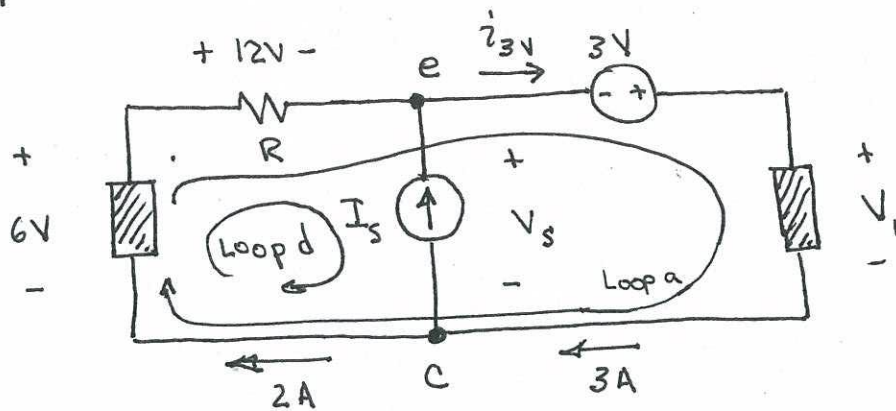


1.1



a) KVL, Loop a: $-6V + 12V - 3V + V_1 = 0 \Rightarrow \boxed{V_1 = -3V}$

b) current through R is 2A (from KCL), so from Ohm's Law:

$$(2A)(R) = 12V \Rightarrow \boxed{R = 6\Omega}$$

c) KCL at node "c"

$$2A + I_s = 3A \Rightarrow \boxed{I_s = 1A}$$

d) KVL, Loop d:

$$-6V + 12V + V_s = 0 \Rightarrow V_s = -6V$$

Note: current enters positive voltage node.

$$P = i v \Rightarrow P = (6V)(1A)$$

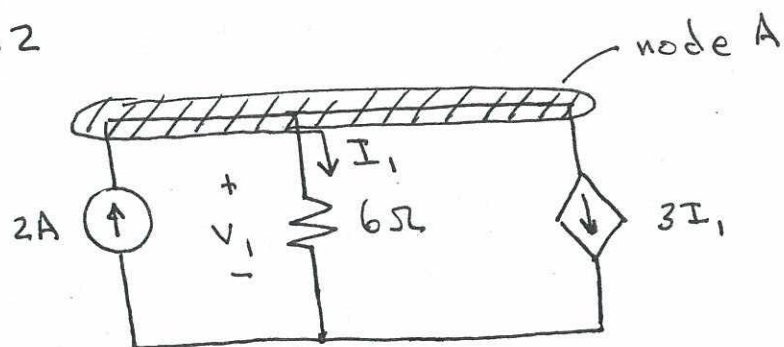
$$\boxed{P = 6W \text{ (absorbed)}}$$

e) KCL, node e

$$i_{3V} = 2A + 1A = 3A \text{ (enters negative voltage node)}$$

$$P = i v = (-3A)(3V) \Rightarrow \boxed{P = -9W \text{ (generated)}}$$

1.2



KCL, node A:

$$2A = I_1 + 3I_1$$

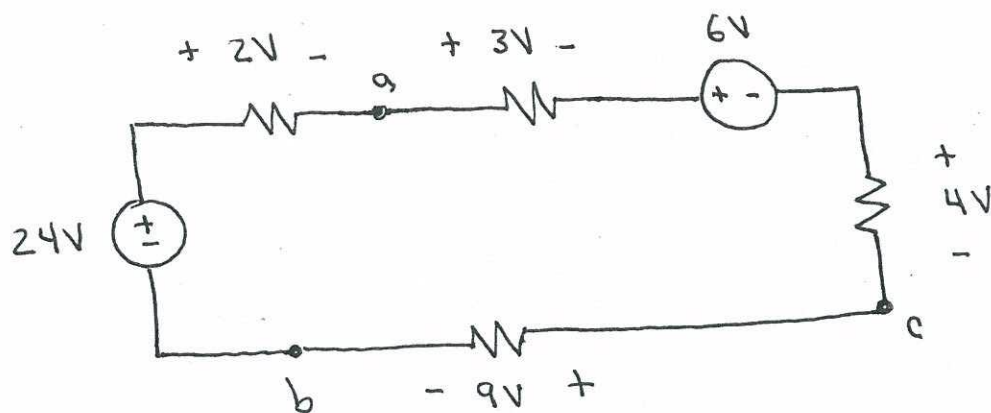
$$2A = 4I_1 \Rightarrow I_1 = 0.5A$$

Ohm's Law:

$$V_1 = (6\Omega) I_1 = (6\Omega)(0.5A)$$

$$V_1 = 3V$$

1.3



$$\text{KVL: } -24\text{V} + 2\text{V} + v_{ab} = 0$$

$$v_{ab} = 22\text{V}$$

$$\text{KVL: } -24\text{V} + 2\text{V} + v_{ac} + 9\text{V} = 0$$

$$v_{ac} = 13\text{V}$$

By inspection,

$$v_{cb} = 9\text{V}$$

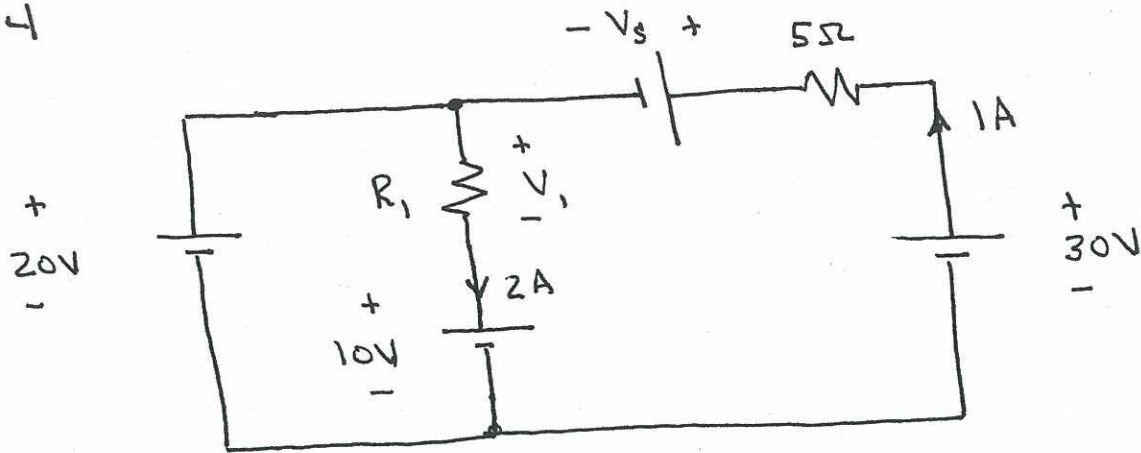
basically given,
though KVL gives

$$v_{bc} + 9\text{V} = 0;$$

$$\text{since } v_{cb} = -v_{bc}$$

$$\Rightarrow \underline{\underline{v_{cb} = 9\text{V}}}$$

1.4



(a) KVL around outer loop:

$$-20V - V_s - \underbrace{(5\Omega)(1A)}_{\text{Ohm's Law}} + 30V = 0 \Rightarrow \boxed{V_s = 5V}$$

(b) KVL around leftmost loop:

$$-20V + V_1 + 10V = 0 \Rightarrow V_1 = 10V$$

Ohm's Law:

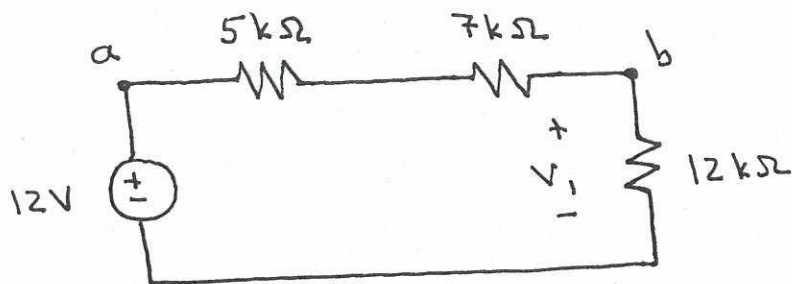
$$V_1 = R_1(2A)$$

$$\rightarrow R_1 = \frac{10V}{2A} \Rightarrow \boxed{R_1 = 5\Omega}$$

(c) $P = i v = (2A)(10V) \Rightarrow \boxed{P = 20W \text{ (absorbed)}}$

current enters positive voltage node $\Rightarrow$ power absorbed

1.5



Note: KCL, applied at any node in the circuit, tells us that the current (i) is the same in all circuit elements.

KVL: $-12V + v_{ab} + V_1 = 0$

ohm's law:

$$v_{ab} = 5k\Omega \cdot i + 7k\Omega \cdot i = (12 \times 10^3 \Omega) i$$

$$V_1 = 12k\Omega \cdot i = (12 \times 10^3 \Omega) i$$

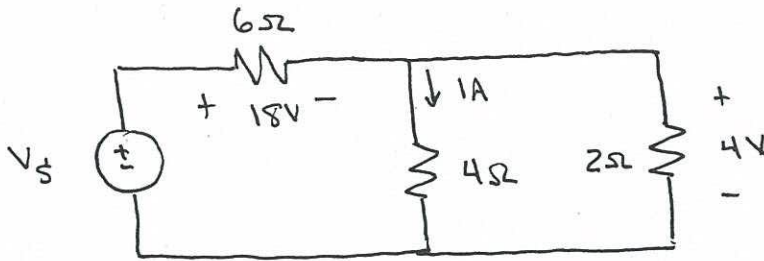
$$0 = -12V + (12 \times 10^3) i + (12 \times 10^3) i$$

$$12V = (24 \times 10^3) i \Rightarrow i = 0.5 \text{ mA}$$

$$v_{ab} = (12 \times 10^3) i = (12 \times 10^3)(0.5 \times 10^{-3}) \Rightarrow \boxed{v_{ab} = 6V}$$

$$V_1 = (12 \times 10^3) i \Rightarrow \boxed{V_1 = 6V}$$

1.6



$$(a) \quad P = \frac{v^2}{R} = \frac{(18V)^2}{6\Omega} = \boxed{54W}$$

$$(b) \quad v = iR = (1A)(4\Omega) = \boxed{4V}$$

$$(c) \quad i = \frac{v}{R} = \frac{4V}{2\Omega} = \boxed{2A}$$

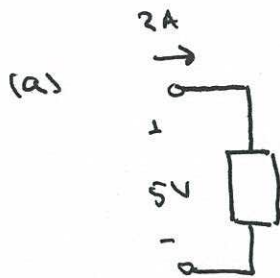
$$(a) \quad P = i v = (2A)(5V) = \boxed{10W} \text{ (absorbed - current enters positive voltage node)}$$

$$(b) \quad P = (-1A)(3V) = \boxed{-3W} \text{ (generated - current enters negative voltage node)}$$

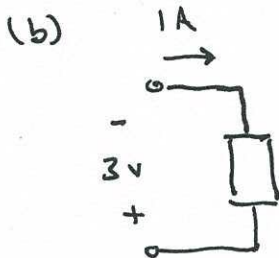
$$(c) \quad P = (3A)(10V) = \boxed{30W} \text{ (absorbed - current leaves negative voltage node; same as entering positive voltage node)}$$

$$(d) \quad P = (-2A)(2V) = \boxed{-4W} \text{ (generated; negative current enters positive voltage node)}$$

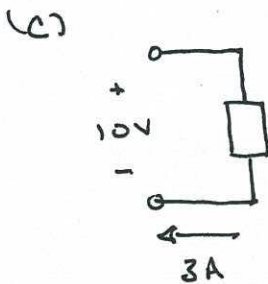
1.7



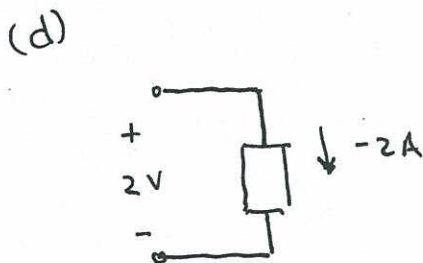
$$P = \underbrace{(2A)(5V)}_{\substack{\uparrow \text{ Agree with passive} \\ \text{sign convention}}} = \underline{\underline{10W}} \quad (\text{absorbed})$$



$$P = \underbrace{(-1A)(4V)}_{\substack{\uparrow \text{ voltage, current disagree} \\ \text{with passive sign convention}}} = -\underline{\underline{4W}} \quad (\text{generated})$$

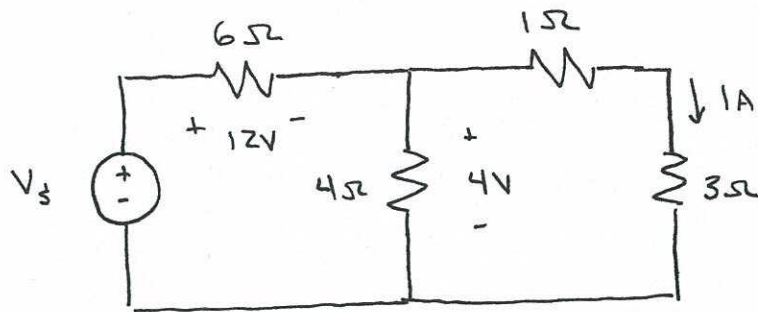


$$P = \underbrace{(10V)(3A)}_{\substack{\uparrow \text{ Agree with passive} \\ \text{sign convention}}} = \underline{\underline{30W}} \quad (\text{absorbed})$$



$$P = \underbrace{(2V)(-2A)}_{\substack{\uparrow \text{ voltage, current directions} \\ \text{agree with passive sign} \\ \text{convention, BUT current} \\ \text{is negative}}} = -\underline{\underline{4W}} \quad (\text{generated})$$

1.8



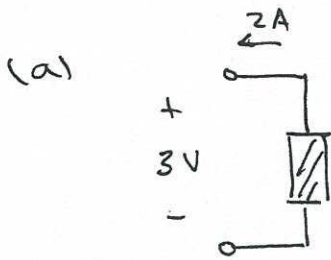
$$(a) \quad P = \frac{V^2}{R} = \frac{(12)^2}{6} = \underline{\underline{72 \text{ W}}}$$

$$(b) \quad i = 1 \text{ A} \Rightarrow v = iR \Rightarrow v = (1 \text{ A})(3 \Omega)$$

$$\underline{\underline{v = 3 \text{ V}}}$$

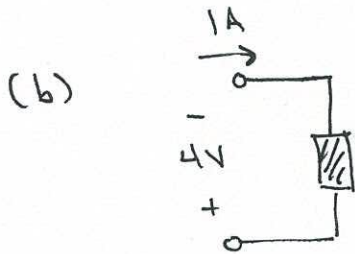
$$(c) \quad i = \frac{v}{R} \Rightarrow i = \frac{4 \text{ V}}{4 \Omega} \Rightarrow \underline{\underline{i = 1 \text{ A}}}$$

1.9



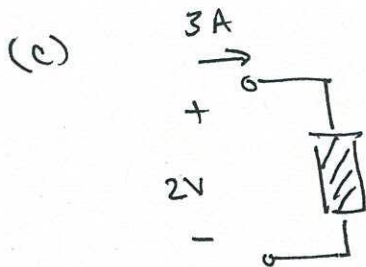
$$P = \underbrace{(-2A)(3V)}_{\uparrow} = -\underline{6W} \text{ (generated)}$$

current, voltage don't agree with passive sign convention



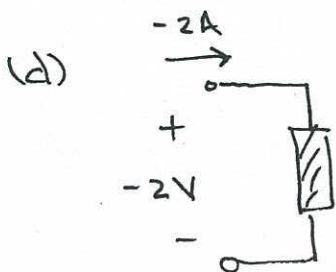
$$P = \underbrace{(-1A)(4V)}_{\uparrow} = -\underline{4W} \text{ (generated)}$$

passive sign convention opposite to given voltage, current directions



$$P = \underbrace{(2V)(3A)}_{\uparrow} = \underline{6W} \text{ (absorbed)}$$

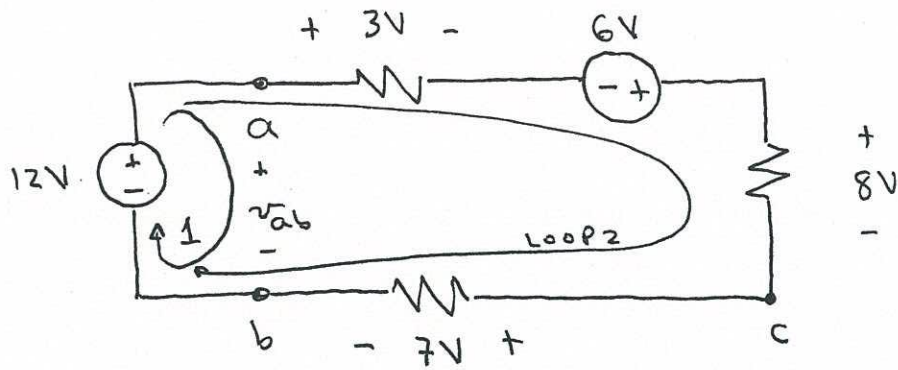
↑ voltage, current agree with passive sign convention



$$P = \underbrace{(-2A)(-2V)}_{\uparrow} = 4W \text{ (absorbed)}$$

↑ given voltage, current directions consistent with passive sign convention. But both are negative.

1.10



(a) KVL, Loop 1:

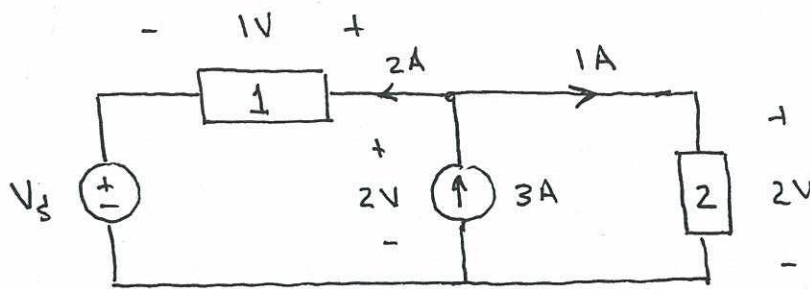
$$-12V + v_{ab} = 0 \Rightarrow \underline{\underline{v_{ab} = 12V}}$$

(b) KVL, Loop 2:

$$-12V + v_{ac} + 7V = 0 \Rightarrow \underline{\underline{v_{ac} = 5V}}$$

(c) By inspection, $\underline{\underline{v_{cb} = 9V}}$

1.11



$$P_1 = (2A)(1V) = 2W \quad (\text{note: current enters positive node})$$

$$P_{3A} = (2V)(-3A) = -6W \quad (\text{note: current leaves positive node})$$

$$P_2 = (1A)(2V) = 2W \quad (\text{note: current enters positive node})$$

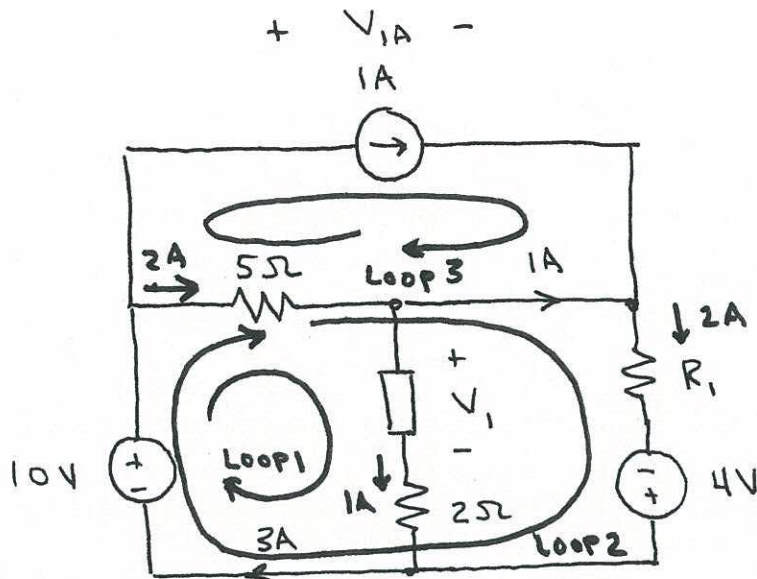
$$\sum P = 0 = P_1 + P_{3A} + P_2 + P_{V_s}$$

$$2W - 6W + 2W + P_{V_s} = 0$$

$$P_{V_s} = 2W$$

(note: the voltage source is absorbing power. This makes sense, since the 2A current enters the positive terminal of V_s .)

1.12



(a) KVL, Loop 1: $-10V + (2A)(5\Omega) + V_1 + (2\Omega)(1A) = 0$

$\uparrow$ $\uparrow$
 From KCL From KCL

$$V_1 = -2V$$

(b) KVL, Loop 2: $-10V + (2A)(5\Omega) + (2A)(R_1) - 4V = 0$

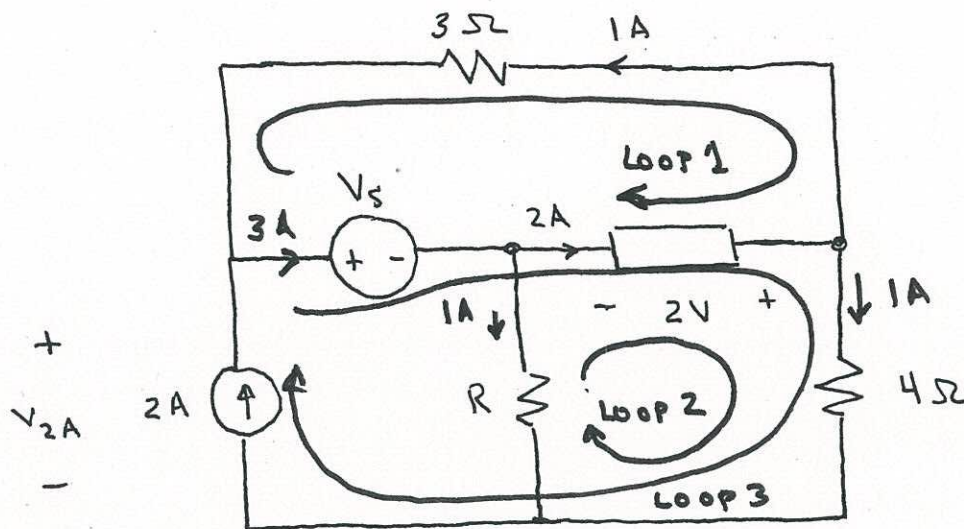
$$R_1 = 1\Omega$$

(c) KVL, Loop 3: $(-2A)(5\Omega) + V_{1A} = 0 \Rightarrow V_{1A} = 10V$

$$P = (I_A)(V_{I_A})$$

$P = 10 \text{ W}$ (absorbed; current enters positive voltage terminal)

1.13



(a) Loop 1: $-V_s - (1A)(3\Omega) + 2V = 0$

$$V_s = -1V$$

(b) Loop 2: $-R(1A) - 2V + (4\Omega)(1A) = 0 \Rightarrow R = 2\Omega$

↑
↑
 From KCL From KCL

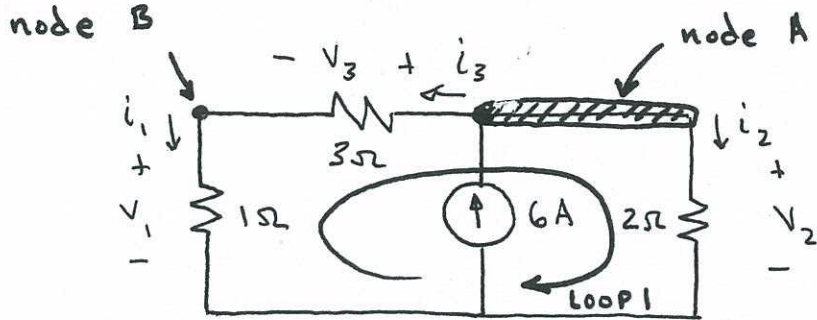
(c) Loop 3: $-V_{2A} + V_s - 2V + (4\Omega)(1A) = 0$

$$V_{2A} = 1V$$

$$P_{2A} = (1V)(-2A) = -2W \quad (\text{generated; current enters negative voltage terminal})$$

1.14

There are many different approaches, but we'll use the exhaustive approach:



$$\text{KCL, node B: } \underline{i_3 = i_1} \quad (1)$$

$$\text{KCL, node A: } \underline{6A = i_2 + i_3} \quad (2)$$

$$\text{KVL, loop 1: } \underline{-V_1 - V_3 + V_2 = 0} \quad (3)$$

Ohm's Law:

$$\underline{V_1 = (1\Omega) i_1} \quad (4)$$

$$\underline{V_2 = (2\Omega) i_2} \quad (5)$$

$$\underline{V_3 = (3\Omega) i_3} \quad (6)$$

Equations (1) - (6) comprise a system of six equations in six unknowns - we're done!

15

Ohm's Law: $V_6 = (6\Omega)(1A) = 6V$

KVL, Loop 1: $-18V - v_L + v_x = 0 \Rightarrow -18V - 6V + v_x = 0$

$$V_x = 24 \text{ V}$$

KCL gives current through 3Ω & 4Ω resistors as shown on diagram.

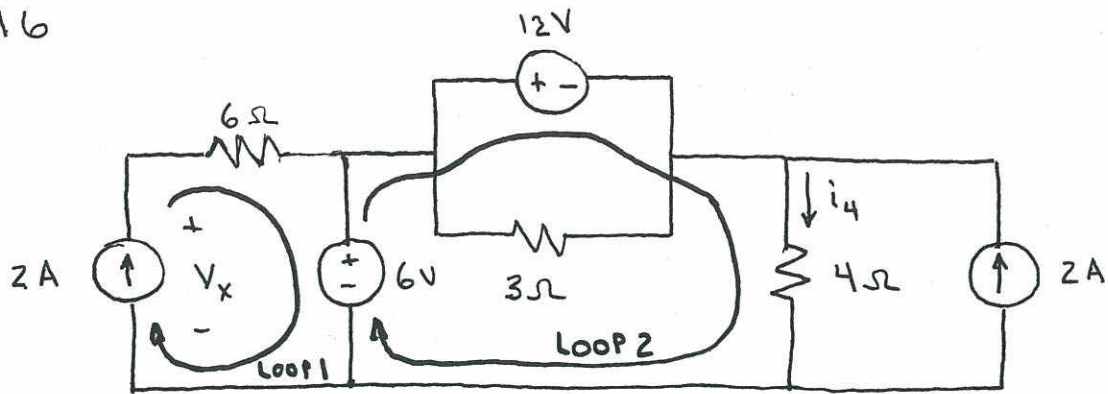
$$\text{KVL, Loop 2: } -v_x - v_3 + v_4 + v_{ba} = 0$$

$$-24V - (2A)(3\Omega) + (4A)(4\Omega) + v_{ba} = 0$$

$$V_{ba} = 14V \Rightarrow \boxed{V_{ab} = -14V}$$

$$P = iv = (-3A)(24V) = \boxed{-72W \text{ (generated)}}$$

1.16



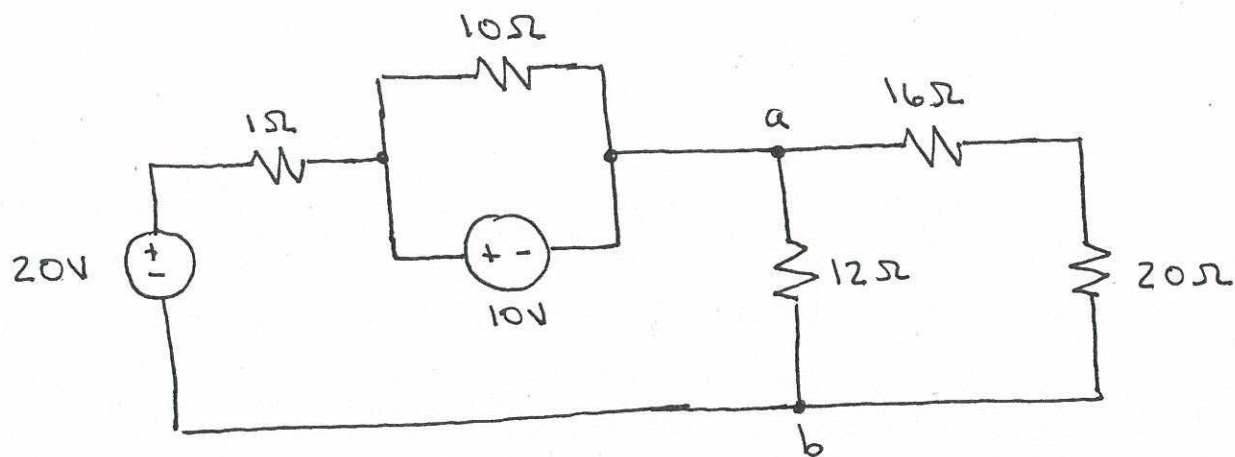
$$\text{KVL, Loop 1: } -V_x + (6\Omega)(2\text{A}) + 6\text{V} = 0$$

$$\underline{\underline{V_x = 18\text{V}}}$$

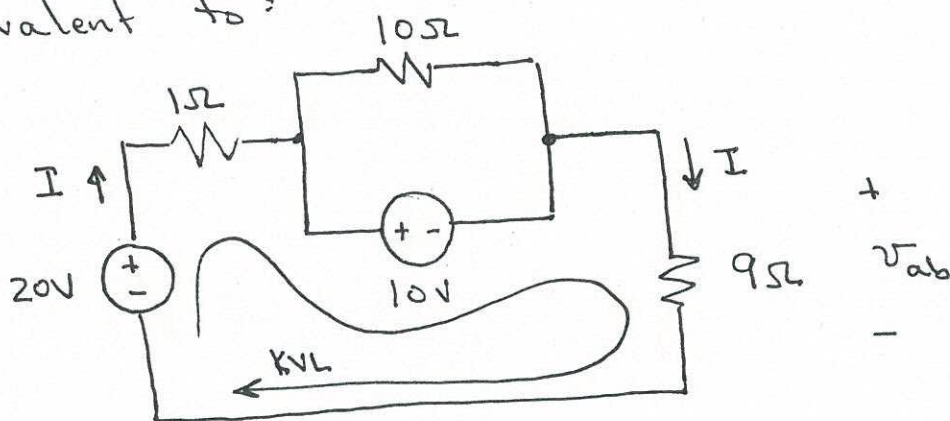
$$\text{KVL, Loop 2: } -6\text{V} + 12\text{V} + (4\Omega)i_4 = 0$$

$$i_4 = \frac{-6\text{V}}{4\Omega} \Rightarrow \underline{\underline{i_4 = -1.67\text{A}}}$$

2.1



Using circuit reduction, the above circuit is equivalent to:



KVL, Loop shown: $-20V + 1\Omega(I) + 10V + 9\Omega(I) = 0$

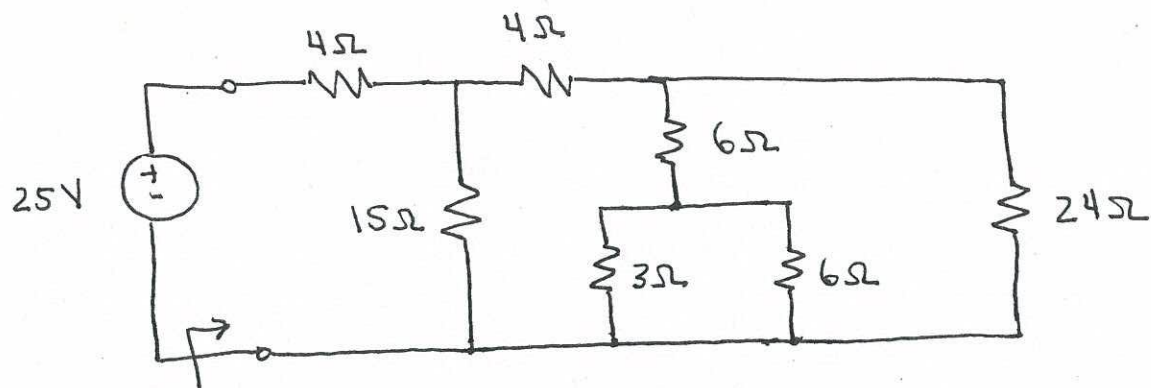
$$I = 1A$$

Ohm's Law:

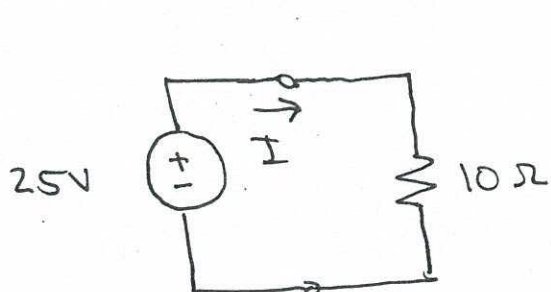
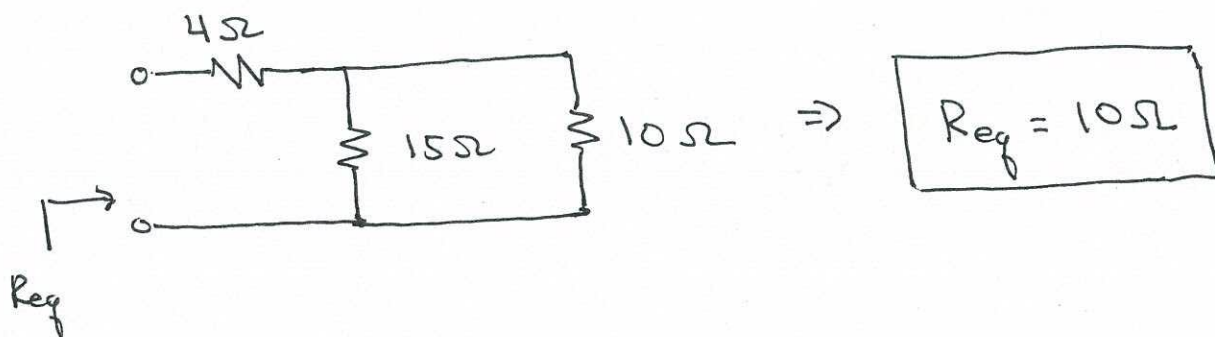
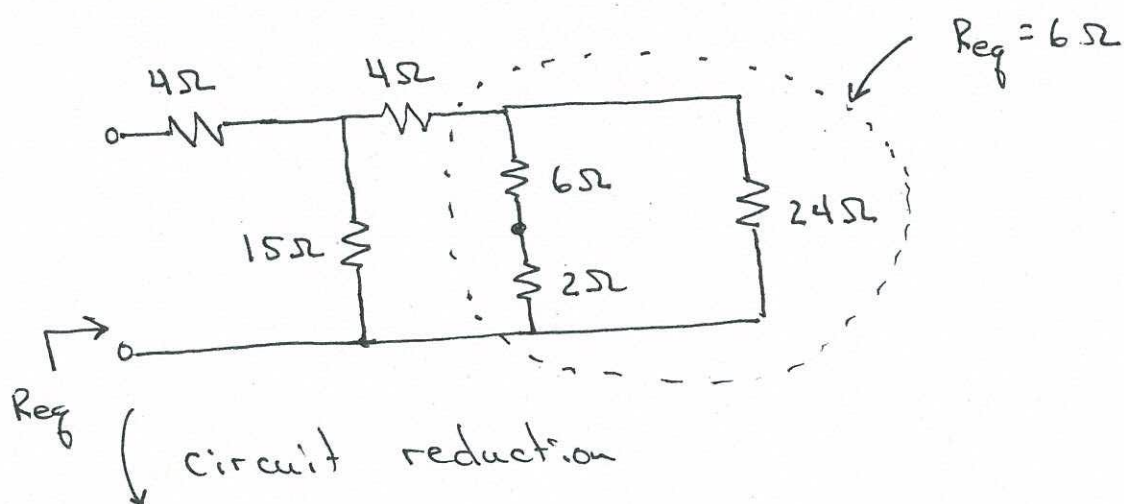
$$v_{ab} = (9\Omega)(1A) = \boxed{9V = v_{ab}}$$

$$P_{20V} = I(20V) = (20V)(1A) = \boxed{20W \text{ (generated)}}$$

2.2



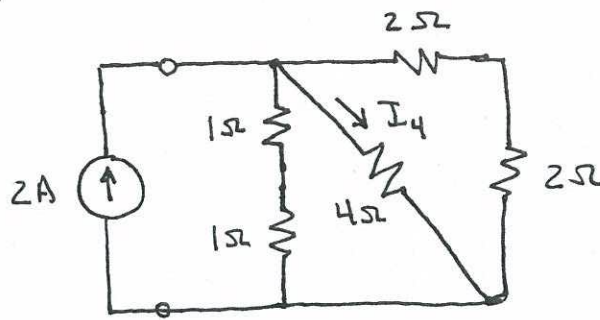
R_{eq} $\downarrow$ circuit reduction



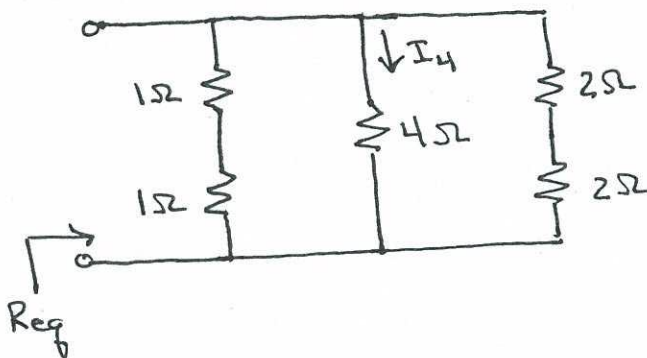
$$I = \frac{25V}{10\Omega} \text{ (ohm's law)}$$

$$I = 2.5A$$

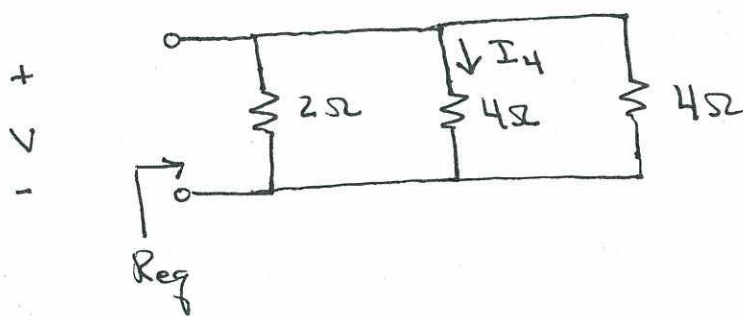
2.3



Re-draw circuit to emphasize parallel, series combinations

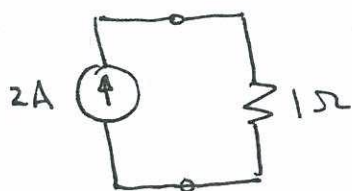


Combine series resistors:



$$\Rightarrow R_{eq} = \frac{1}{\frac{1}{2} + \frac{1}{4} + \frac{1}{4}} = 1\Omega$$

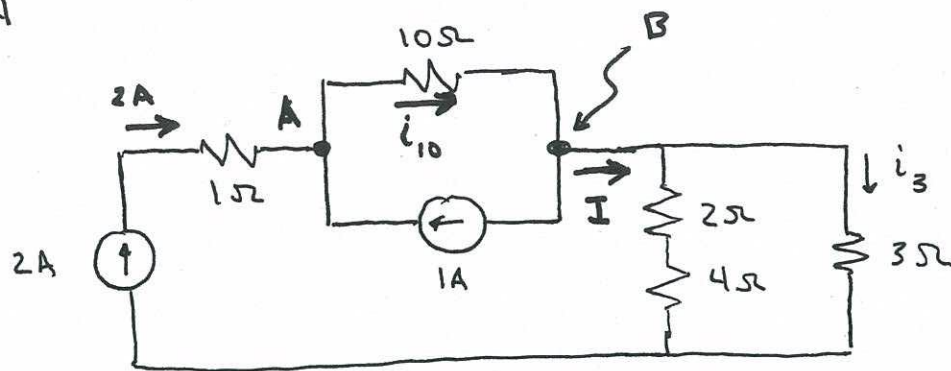
Reduced circuit:



$$+ V = 2V = (2A)(1\Omega) \Rightarrow I_4 = \frac{2V}{4\Omega}$$

$$I_4 = 0.5A$$

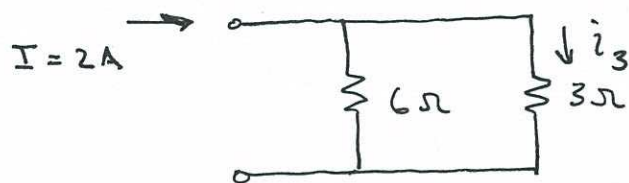
2.4



$$\text{KCL at A: } 2\text{A} + 1\text{A} - i_{10} = 0 \Rightarrow i_{10} = 3\text{A}$$

$$\text{KCL at B: } i_{10} - 1\text{A} - I = 0 \Rightarrow I = 2\text{A}$$

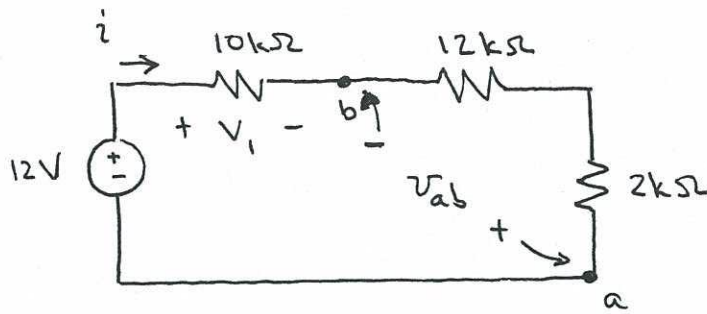
Combine the series combination of the 2Ω and 4Ω resistors to obtain the current divider below:



$$i_3 = 2\text{A} \left(\frac{6\Omega}{3\Omega + 6\Omega} \right)$$

$$\underline{\underline{i_3 = \frac{4}{3}\text{A}}}$$

2.5



$$\text{KVL: } -12\text{V} + (10\text{k}\Omega)i + (12\text{k}\Omega)i + (2\text{k}\Omega)i = 0$$

Note: currents through all elements are the same, by KCL.

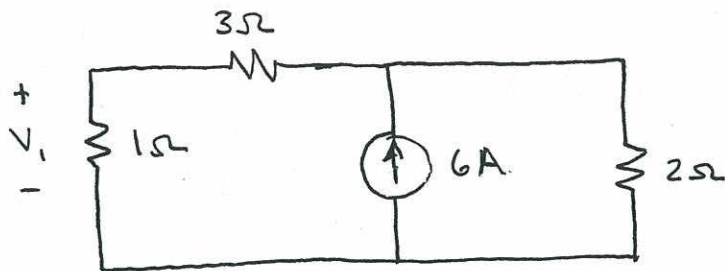
$$i = \frac{12\text{V}}{10\text{k}\Omega + 12\text{k}\Omega + 2\text{k}\Omega} \Rightarrow i = 5\text{mA}$$

$$V_1 = (10\text{k}\Omega)i = (10\text{k}\Omega)(5\text{mA}) \Rightarrow \underline{\underline{V_1 = 5\text{V}}}$$

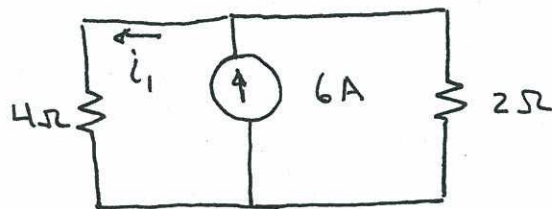
$$\text{KVL: } -12\text{V} + V_1 - v_{ab} = 0$$

$$-12\text{V} + 5\text{V} - v_{ab} = 0 \Rightarrow \underline{\underline{v_{ab} = -7\text{V}}}$$

2.6



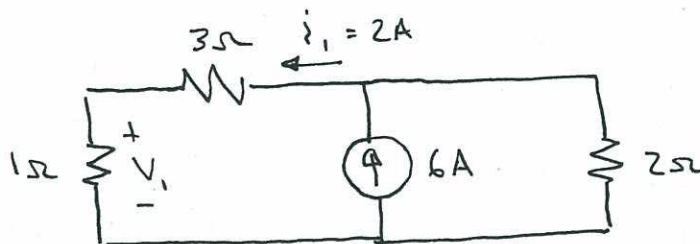
Combine 1Ω & 3Ω resistors:



The above is a current divider, so:

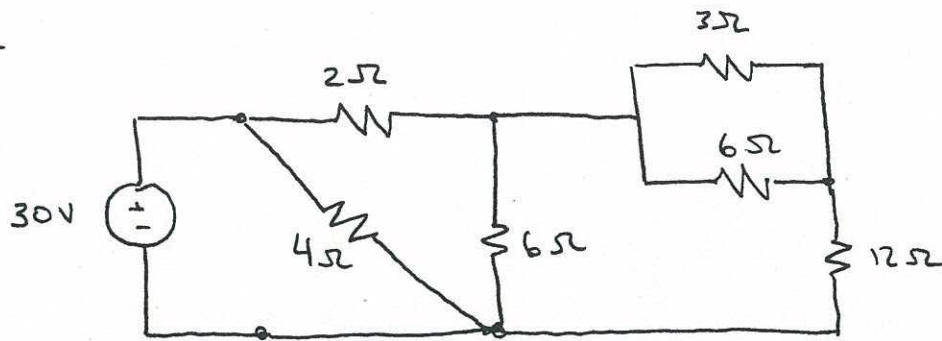
$$i_1 = 6A \left(\frac{2\Omega}{2\Omega + 4\Omega} \right) = 2A$$

Now, from original circuit:

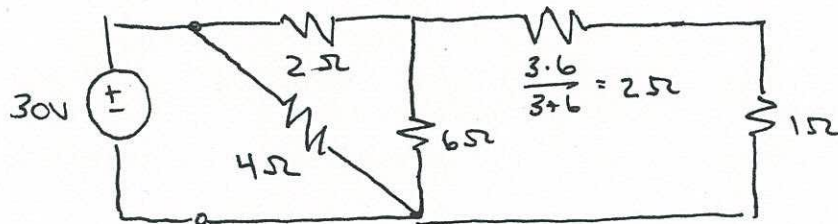


$$V_1 = (1\Omega)(2A) \Rightarrow \underline{\underline{V_1 = 2V}}$$

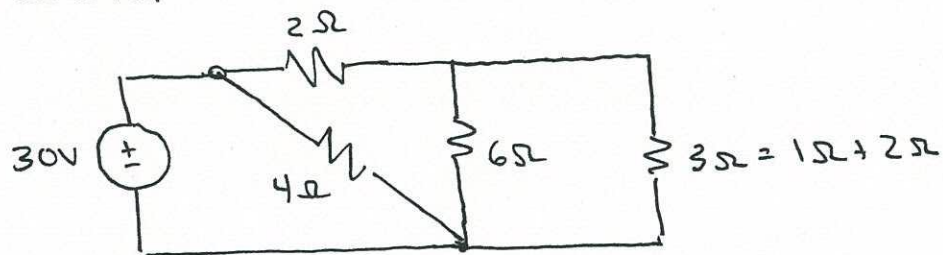
2.7



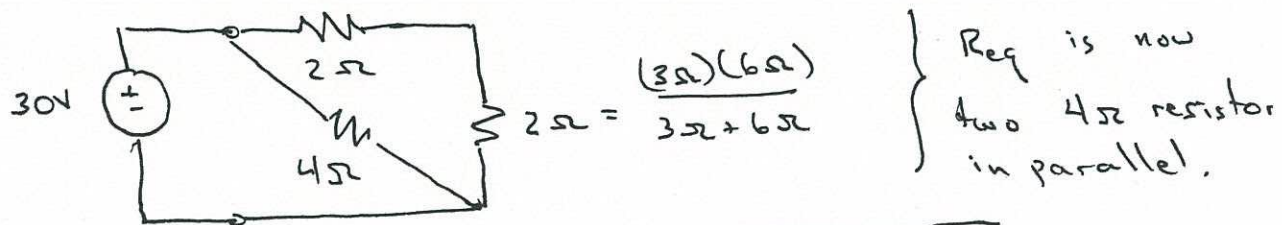
(a) First, combine parallel 3Ω & 6Ω resistors:



The resulting 2Ω resistor is in series with 1Ω resistor:



The 3Ω & 6Ω resistors are in parallel:



The equivalent resistance, $\boxed{R_{eq} = 2\Omega}$

(b) $I = \frac{30V}{R_{eq}} = 15A \Rightarrow P_s = (-30V)(15A)$

$\boxed{P_s = -450W}$ (generated)

2.7 (cont'd)

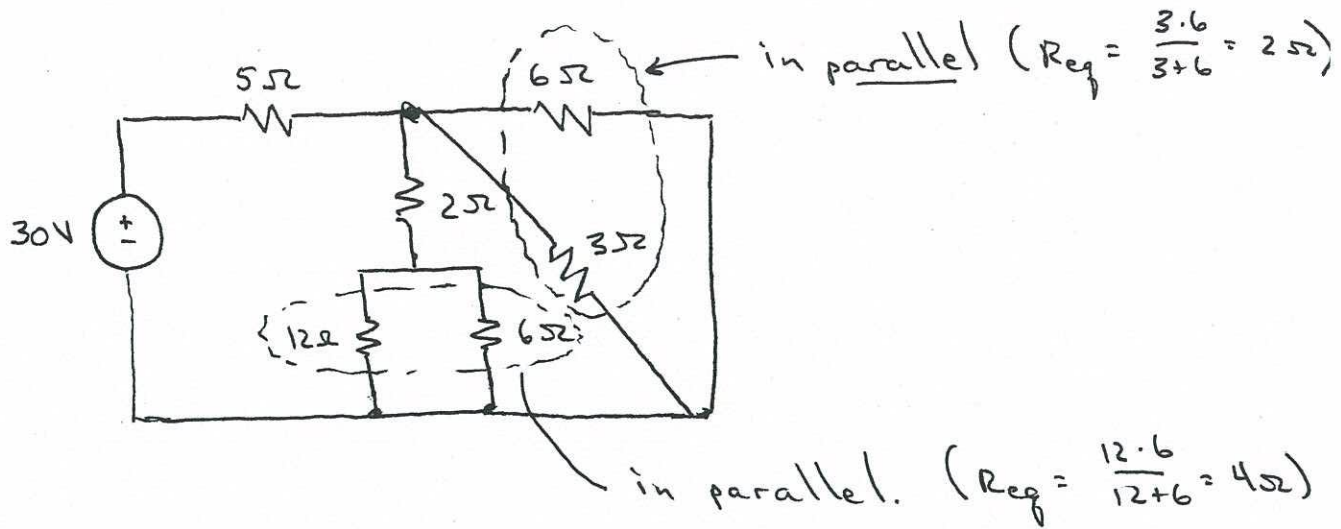
(c) The current through the 4Ω resistor can be found directly from Ohm's Law. The voltage across the 4Ω resistor is

30V. Therefore,

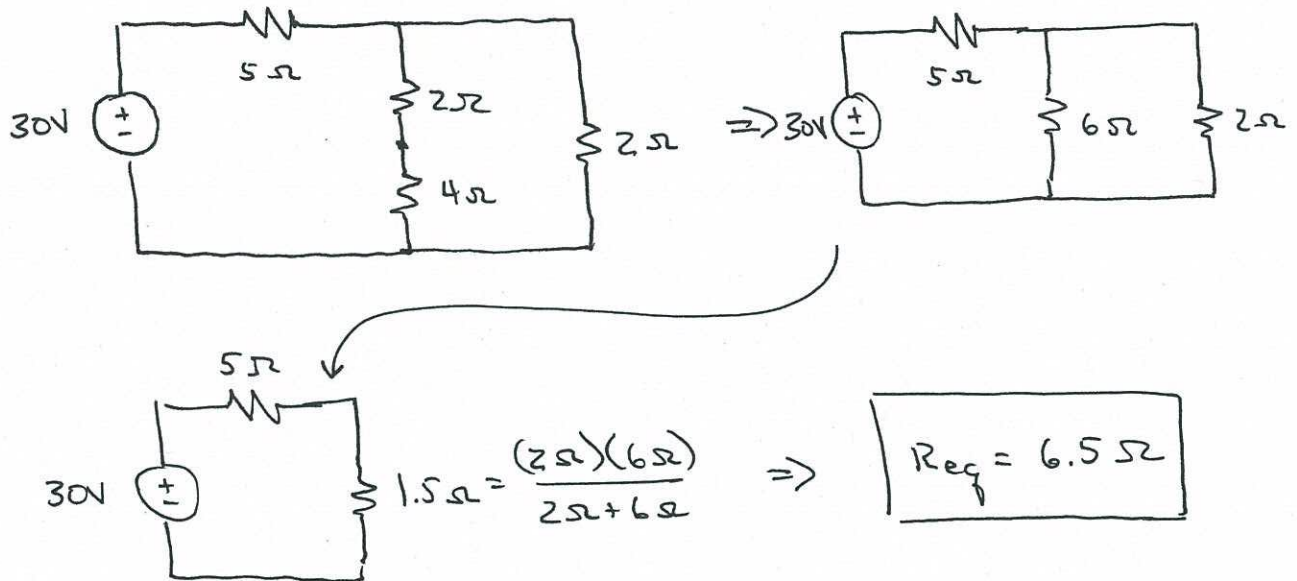
$$I = \frac{30V}{4\Omega} = 7.5A$$

2.8

(a)

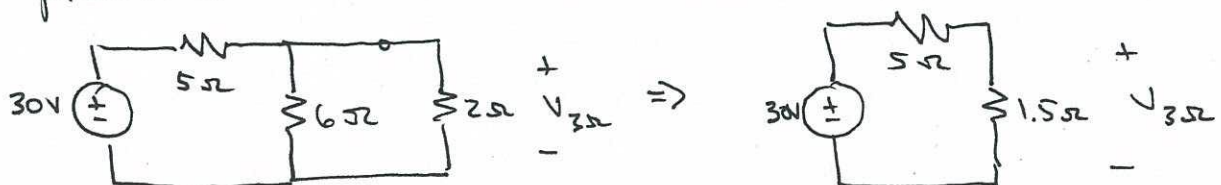


Combining the two sets of parallel resistors:



(b) $I_s = \frac{V_s}{R_{eq}} \Rightarrow I_s = \frac{30V}{6.5\Omega} \Rightarrow I_s = 4.615 A$

(c) The voltage across the 3Ω resistor is the same as the voltage across the parallel combination of the 3Ω & 6Ω resistors. We can use the previous circuit reductions:



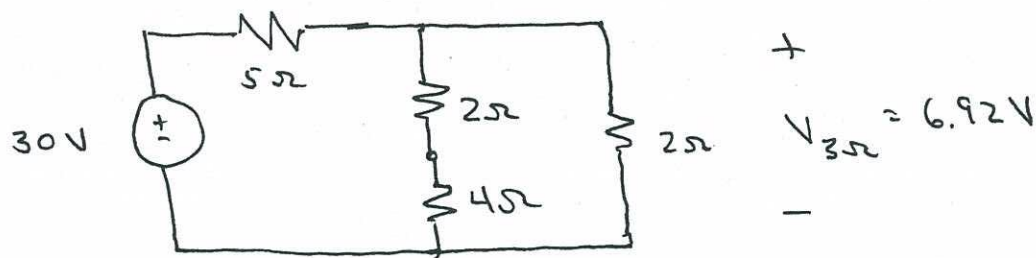
2.8 (cont'd)

The voltage across the 3Ω resistor can now be found from a voltage divide formula:

$$V_{3\Omega} = 30V \left[\frac{1.5\Omega}{5\Omega + 1.5\Omega} \right]$$

$$V_{3\Omega} = 6.92V$$

(d) We can again take advantage of an earlier circuit reduction.

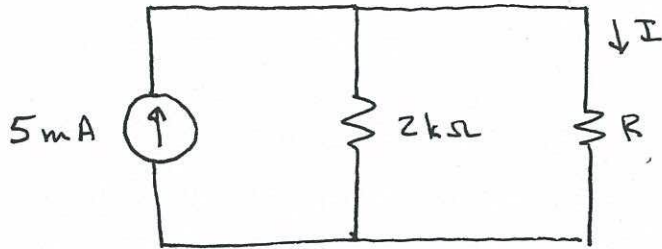


The 2Ω and 4Ω resistors now form a voltage divider. The total voltage across the series combination is $6.92V$. Therefore,

$$V_{2\Omega} = 6.92V \left[\frac{2\Omega}{2\Omega + 4\Omega} \right]$$

$$V_{2\Omega} = 2.31V$$

2.9



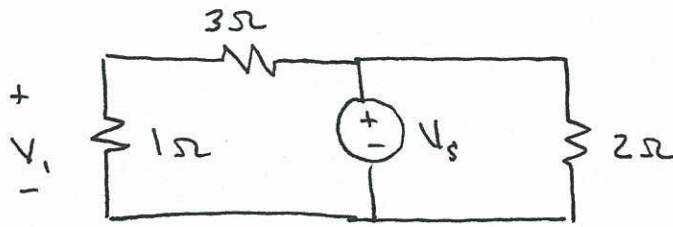
current divider: $I = 5\text{mA} \left[\frac{2\text{k}\Omega}{2\text{k}\Omega + R} \right] = 2\text{mA}$

$$5\text{mA} [2\text{k}\Omega] = 2\text{mA} (2\text{k}\Omega + R)$$

$$\frac{5}{2} (2\text{k}\Omega) = 2\text{k}\Omega + R$$

$$\underline{\underline{R = 3\text{k}\Omega}}$$

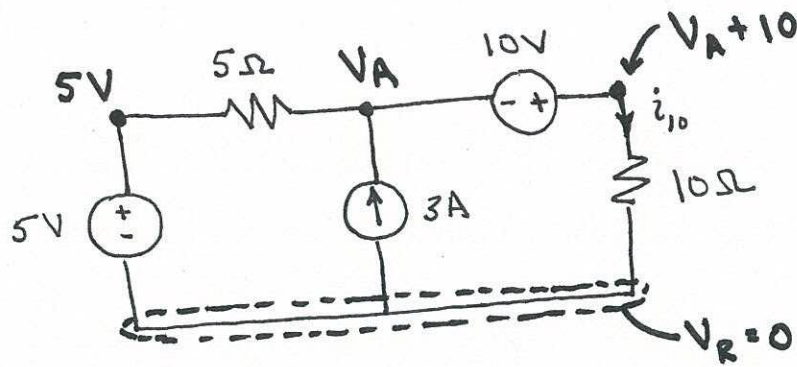
2.10



Voltage divider : $V_1 = V_s \left[\frac{1\Omega}{1\Omega + 3\Omega} \right] = 2V$

$$\underline{\underline{V_s = 8V}}$$

3.1



One independent voltage - V_A

KCL at "A":

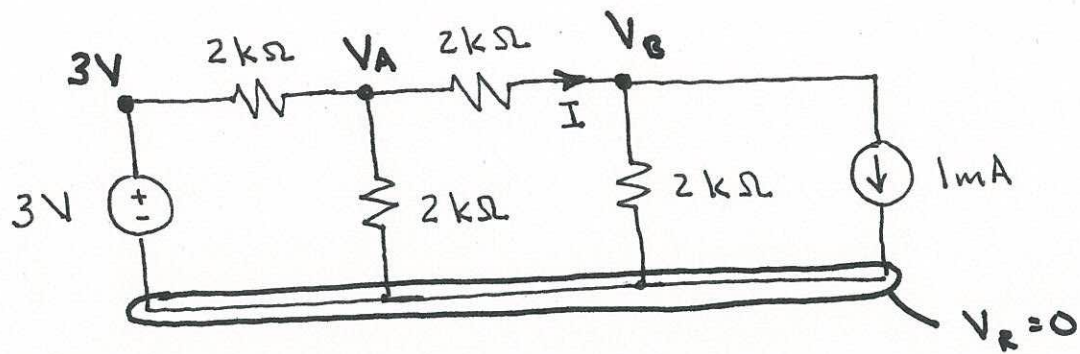
$$\frac{V_A - 5V}{5\Omega} - 3A + \frac{V_A + 10}{10\Omega} = 0$$

$$2V_A - 10 - 30 + V_A + 10 = 0 \Rightarrow 3V_A = 30V \Rightarrow V_A = 10V$$

Ohm's Law: $i_{10} = \frac{V_A + 10}{10\Omega} = \frac{20}{10} \Rightarrow \boxed{i_{10} = 2A}$

In the direction shown.

3.2



Two independent voltages - $V_A \neq V_B$

$$\text{Node A: } \frac{V_A - 3V}{2k\Omega} + \frac{V_A}{2k\Omega} + \frac{V_A - V_B}{2k\Omega} = 0 \Rightarrow 3V_A - V_B = 3$$

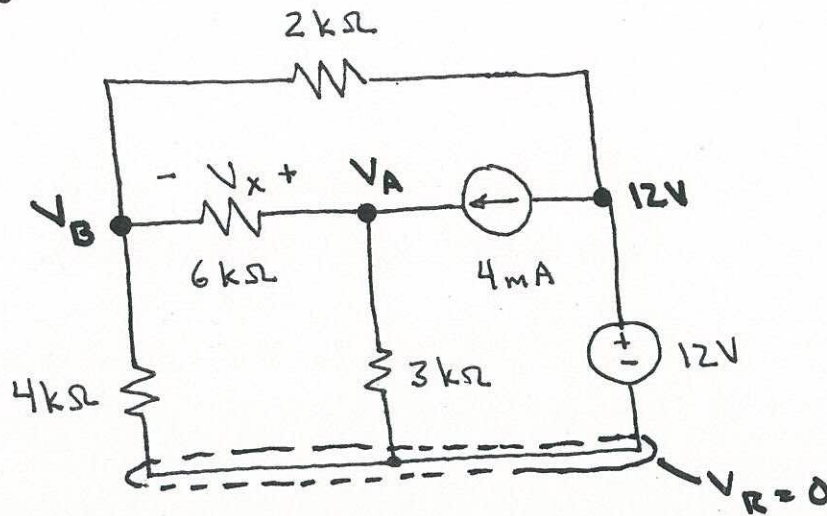
$$\text{Node B: } \frac{V_B - V_A}{2k\Omega} + \frac{V_B}{2k\Omega} + 1mA = 0 \Rightarrow 2V_B - V_A = -2$$

$$V_A = 0.8V$$

$$V_B = -0.6V$$

$$I = \frac{V_A - V_B}{2k\Omega} = \frac{0.8 - (-0.6)}{2k\Omega} \Rightarrow I = 0.7mA$$

3.3



Two independent
voltages, V_A & V_B

Node A: $\frac{V_A - V_B}{6k\Omega} + \frac{V_A}{3k\Omega} - 4mA = 0 \Rightarrow 3V_A - V_B = 24$

Node B: $\frac{V_B}{4k\Omega} + \frac{V_B - V_A}{6k\Omega} + \frac{V_B - 12}{2k\Omega} = 0 \Rightarrow 11V_B - 2V_A = 72$

$$V_A = 10.84V$$

$$V_B = 8.52V$$

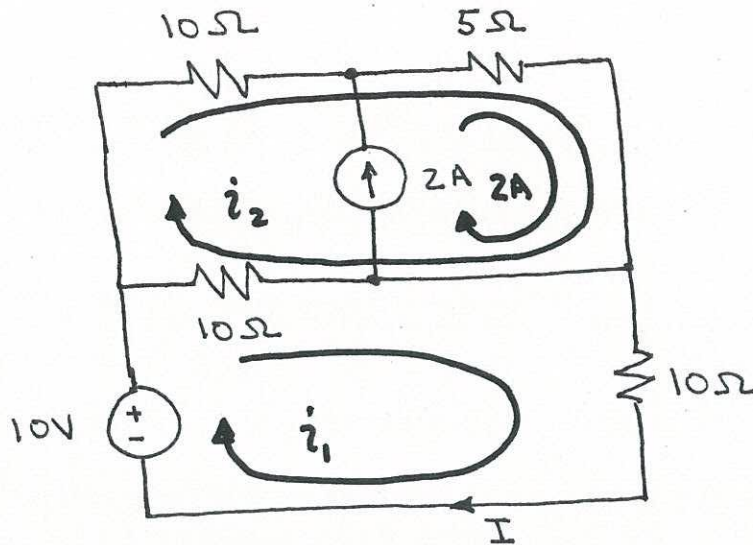
solving

$$V_X = V_A - V_B$$

$$= 10.84 - 8.52$$

$$V_X = 2.32V$$

3.4



2 mesh loops,
1 constrained
loop, as
shown

$$\text{Loop 1: } -10 + 10(i_1 - i_2) + 10i_1 = 0 \Rightarrow i_2 = 2i_1 - 1$$

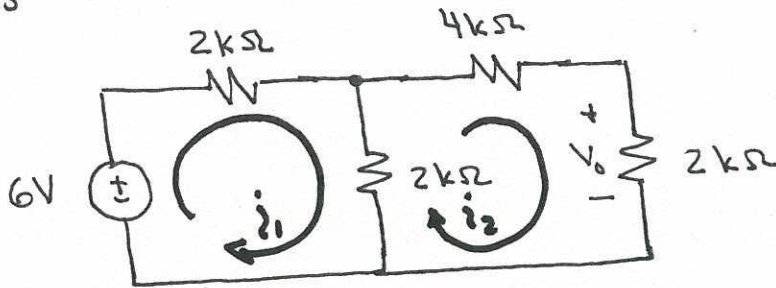
$$\text{Loop 2: } 10i_2 + 5(i_2 + 2) + 10(i_2 - i_1) = 0 \Rightarrow 25i_2 - 10i_1 = -10$$

solving for i_i

$$i_1 = \frac{3}{8} \text{ A}$$

$$I = i_1 \Rightarrow \boxed{I = \frac{3}{8} \text{ A}}$$

3.5



2 mesh currents;
no constraint loops

$$\text{Loop 1: } -6 + 2\text{k}\Omega (i_1) + 2\text{k}\Omega (i_1 - i_2) = 0$$

$$4000 i_1 - 2000 i_2 = 6$$

$$\text{Loop 2: } 2\text{k}\Omega (i_2 - i_1) + 4\text{k}\Omega (i_2) + 2\text{k}\Omega (i_2) = 0$$

$$8000 i_2 - 2000 i_1 = 0$$

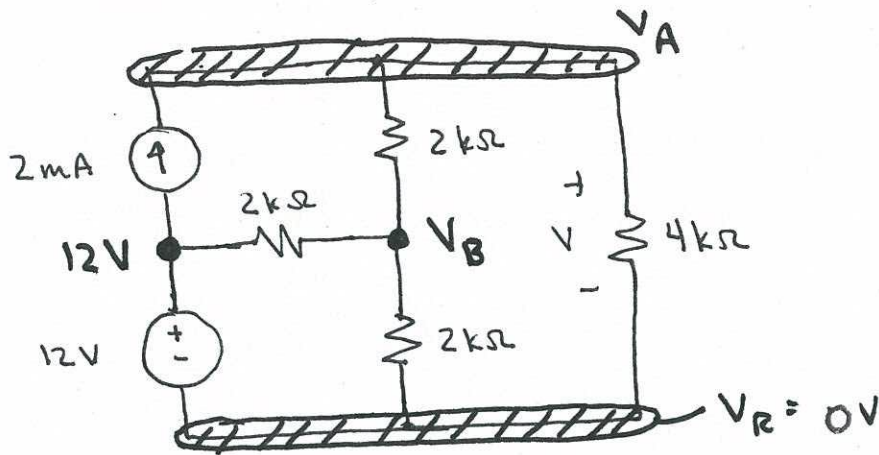
$$i_1 = 1.714 \text{ mA}$$

$$i_2 = 0.429 \text{ mA}$$

SOLVING

$$V_o = 2000 (i_2) \Rightarrow \boxed{V_o = 0.857 \text{ V}}$$

3.6

Nodal Analysis:

2 independent
voltages, $V_A \neq V_B$

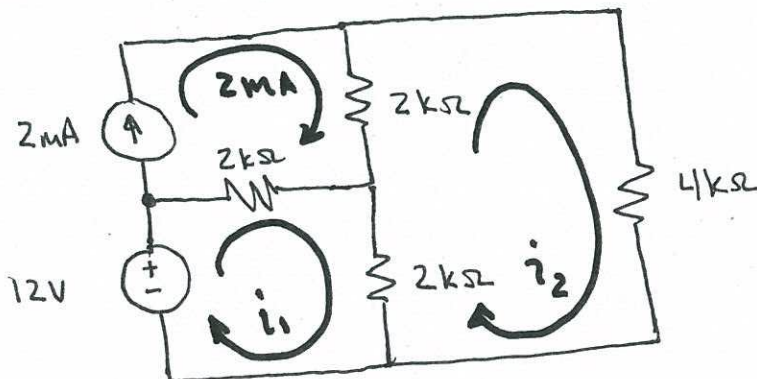
$$\text{Node A: } -2\text{mA} + \frac{V_A - V_B}{2\text{k}\Omega} + \frac{V_A}{4\text{k}\Omega} = 0 \Rightarrow 3V_A - 2V_B = 8$$

$$\text{Node B: } \frac{V_B - 12}{2\text{k}\Omega} + \frac{V_B}{2\text{k}\Omega} + \frac{V_B - V_A}{2\text{k}\Omega} = 0 \Rightarrow 3V_B - V_A = 12$$

$$V_A = 6.857\text{ V}$$

$$V_B = 6.286\text{ V}$$

$$V = V_A = 6.857\text{ V}$$

Mesh AnalysisLoop 1:

$$-12 + (i_1 - 2)2 + 2(i_1 - i_2) = 0$$

$$8 = 2i_1 - i_2$$

Loop 2:

$$2(i_2 - i_1) + 2(i_2 - 2) + 4i_2 = 0$$

$$4i_2 - i_1 = 2$$

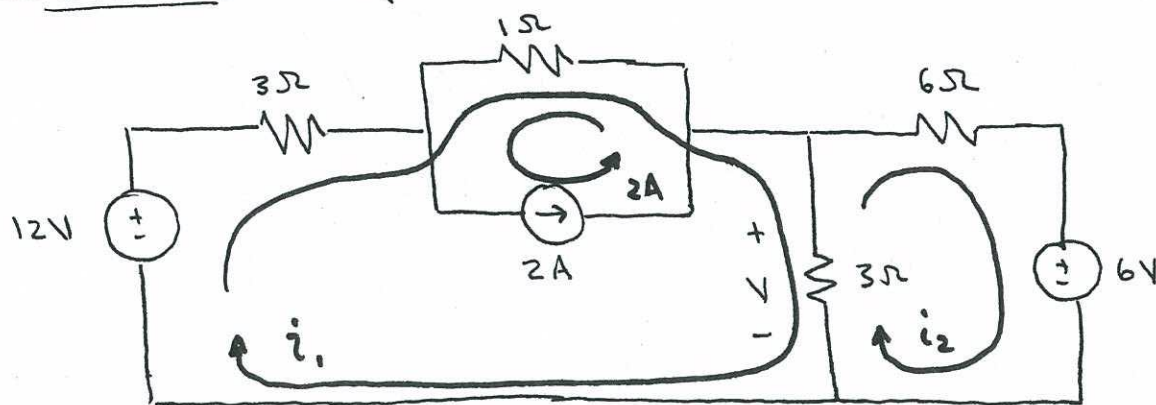
$$i_1 = 4.857\text{ mA}$$

$$i_2 = 1.714\text{ mA}$$

$$V = (4\text{k}\Omega) i_2 \Rightarrow V = 6.857\text{ V}$$

3.7

Two mesh currents, i_1 & i_2 , as shown. One 2A constrained loop:



$$\text{Loop } i_1: -12 + 3\Omega(i_1) + 1\Omega(i_1 - 2A) + 3\Omega(i_1 - i_2) = 0$$

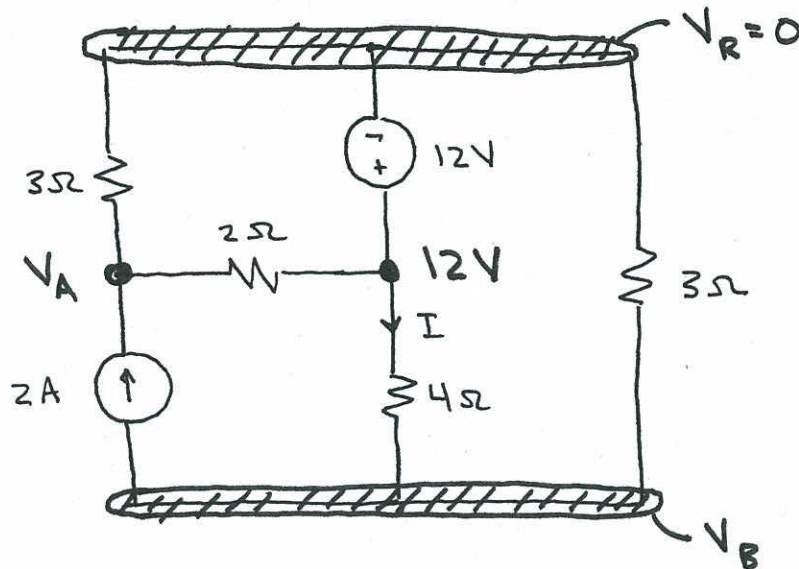
$$\text{Loop } i_2: 3\Omega(i_2 - i_1) + 6\Omega(i_2) + 6V = 0$$

$$V = 3\Omega(i_1 - i_2)$$

Important note: The above solution is not unique! Other valid solutions exist, depending (for one thing) on how you define your constrained loop. You can also define three mesh currents, and write constraint equations explicitly...

3.8

Define reference voltage, two node voltages, and one dependent voltage as shown:



Node A: $\frac{V_A - 0V}{3\Omega} + \frac{V_A - 12V}{2\Omega} - 2A = 0$

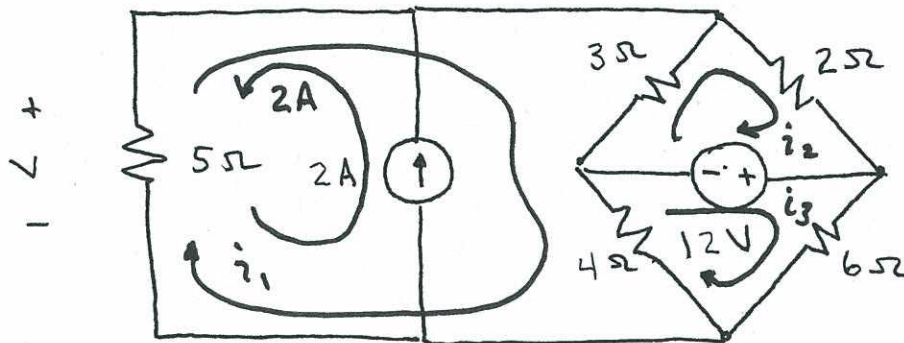
Node B: $2A + \frac{V_B - 12V}{4\Omega} + \frac{V_B - 0V}{3\Omega} = 0$

$$I = \frac{12V - V_B}{4\Omega}$$

The above equations, of course, are not the only possible solution. Other valid solutions exist, depending on (for example) your definitions of reference node and independent nodes.

3.9

Define three mesh currents and one constrained current, as shown below.



$$\text{Loop 1: } (i_1 - 2A)5\Omega + 3\Omega(i_1 - i_2) + 4\Omega(i_1 - i_3) = 0$$

$$\text{Loop 2: } 3\Omega(i_2 - i_1) + 2\Omega(i_2) + 12V = 0$$

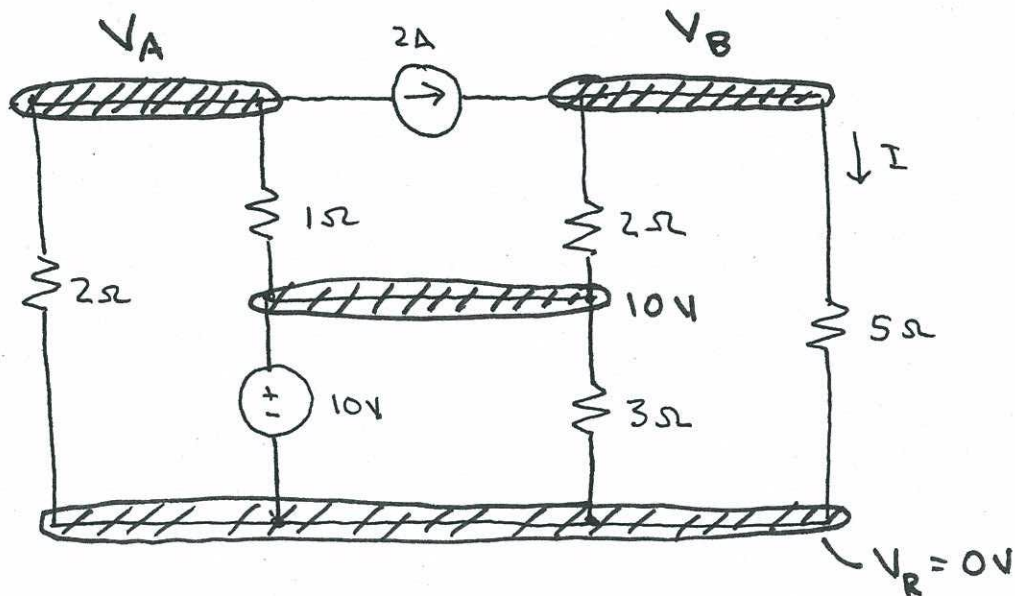
$$\text{Loop 3: } 4\Omega(i_3 - i_1) - 12V + 6\Omega(i_3) = 0$$

$$V = 5\Omega [2A - i_1]$$

Note: Other valid solutions exist, depending upon your definition of mesh loops and constrained loops.

3.10

Define reference voltage, two node voltages, and one dependent voltage, as shown:



Node A: $\frac{V_A - 0V}{2\Omega} + \frac{V_A - 10V}{1\Omega} + 2A = 0$

Node B: $-2A + \frac{V_B - 10V}{2\Omega} + \frac{V_B - 0V}{5\Omega} = 0$

$$I = \frac{V_B}{5\Omega}$$

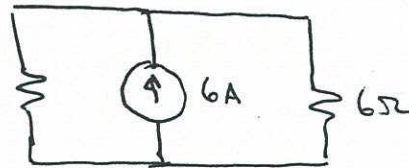
Note: The above solution depends upon my choice of reference node, independent nodes, and dependent voltages. Alternate, valid solutions are possible!

3.11

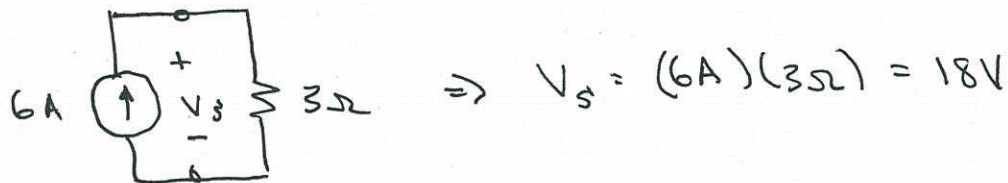
I chose to do the problem using circuit reduction techniques.

The resistances to the left of the 6A source combine to a single equivalent 6Ω resistor:

$$\frac{6\Omega \cdot 12\Omega}{6\Omega + 12\Omega} + 2\Omega = 6\Omega$$

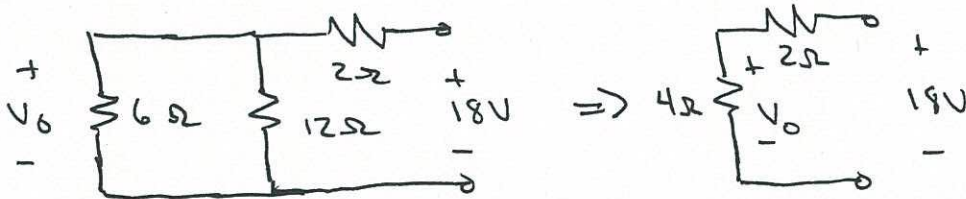


The two parallel 6Ω resistors combine to a single 3Ω equivalent resistance:



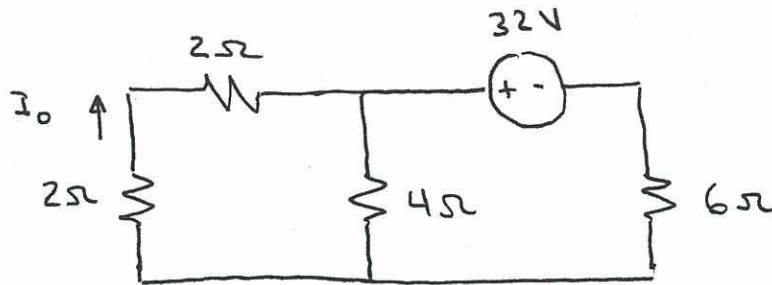
The voltage across the current source is 18V.

we can find V_o from a

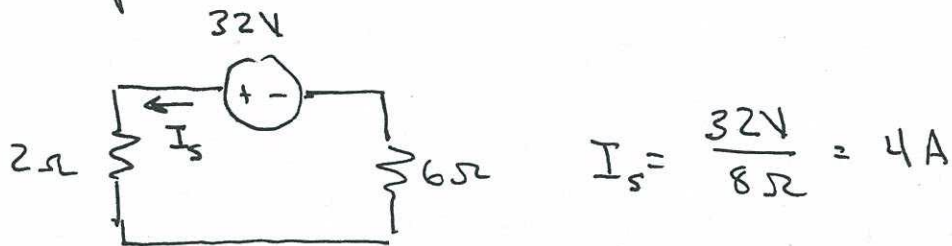


$$V_o = \left(\frac{4\Omega}{4\Omega + 2\Omega} \right) 18V \Rightarrow \boxed{V_o = 12V}$$

3.12



I used circuit reduction. The above circuit is equivalent to:



We can find I_0 from I_s and a current divider formula:

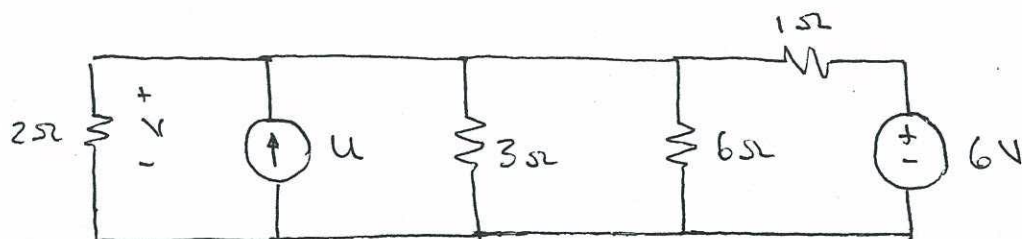


$$I_1 = \left(\frac{4\Omega}{4\Omega + 4\Omega} \right) 4A$$

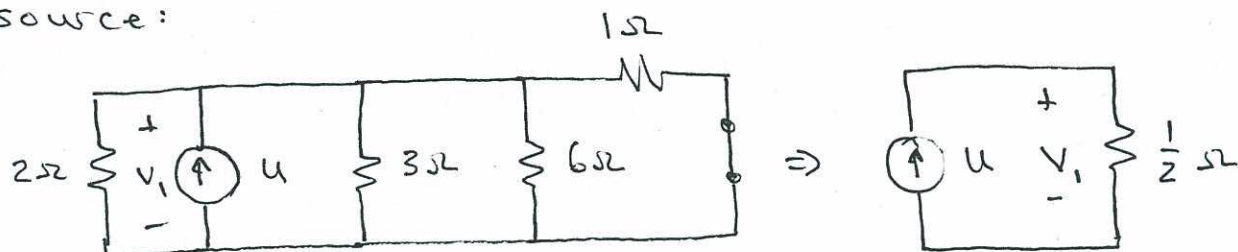
$$I_1 = 2A$$

$$I_0 = -I_1 \Rightarrow \boxed{I_0 = -2A}$$

4.1.

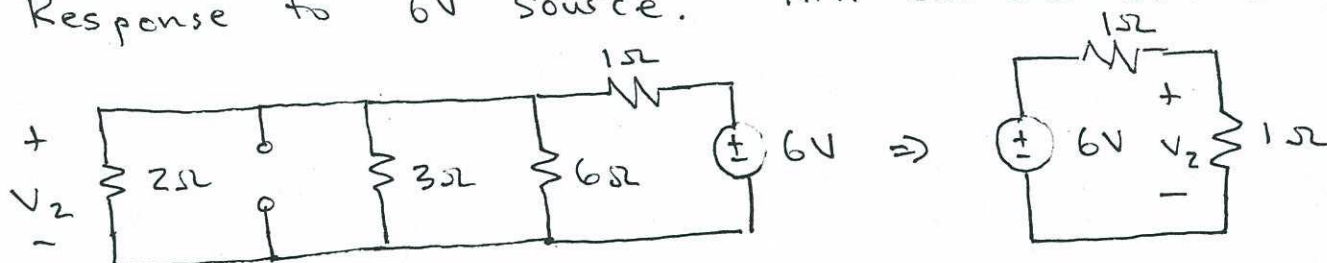


Response to current source, u . Kill voltage source:



$$V_1 = u \left(\frac{1}{2} \Omega \right) = \frac{u}{2} V$$

Response to 6V source. Kill current source:



$$V_2 = 6V \left(\frac{1\Omega}{1\Omega + 1\Omega} \right) = 3V$$

Superimpose:

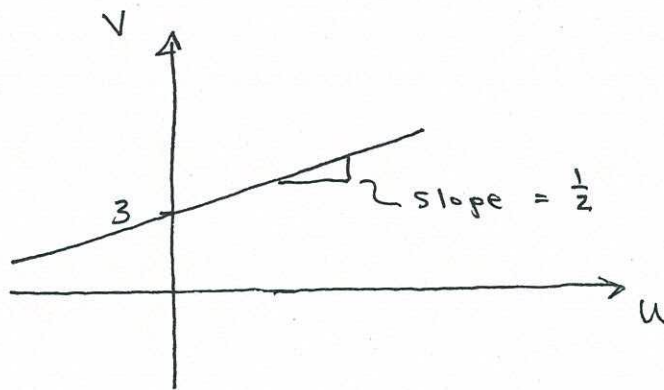
$$V = V_1 + V_2 \Rightarrow \underline{\underline{V = 3V + \frac{u}{2}}}$$

4.2.

From problem 1:

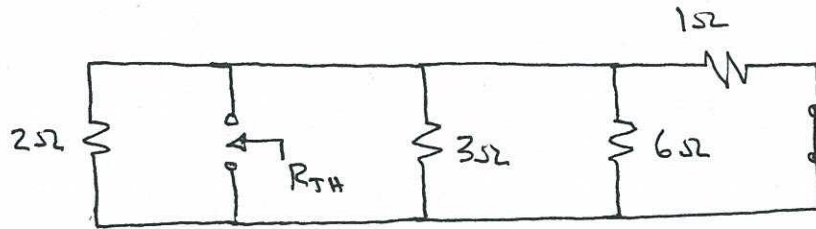
$$V = \frac{u}{2} + 3V$$

This is the input-output relation
for the circuit. u is the input, V
is the output.



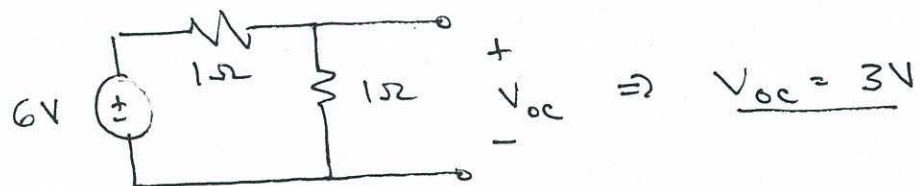
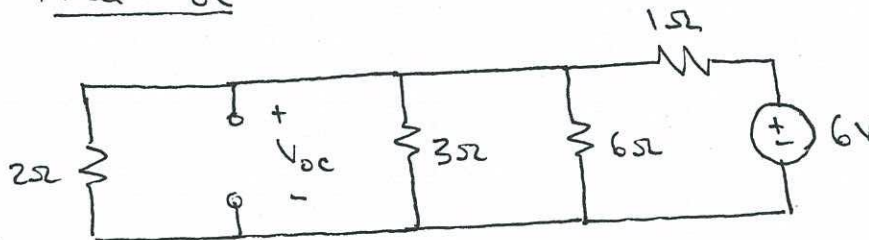
4.3

Find R_{TH} :

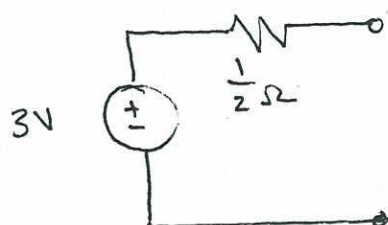


$$R_{TH} = \frac{1}{\frac{1}{2\Omega} + \frac{1}{3\Omega} + \frac{1}{6\Omega} + \frac{1}{1\Omega}} \Rightarrow \underline{R_{TH} = \frac{1}{2}\Omega}$$

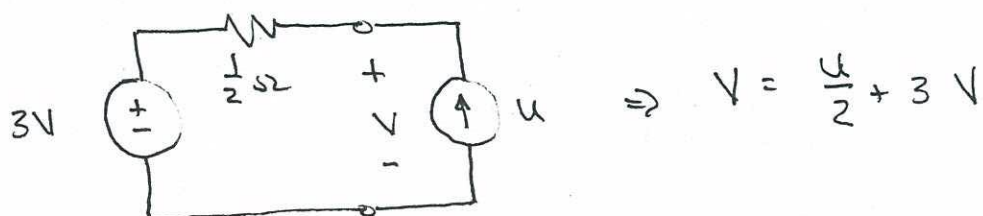
Find V_{oc} :



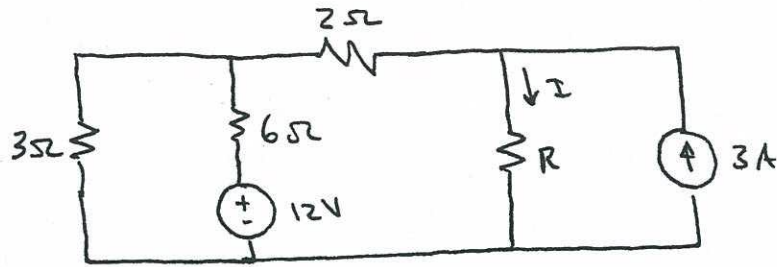
Thevenin circuit:



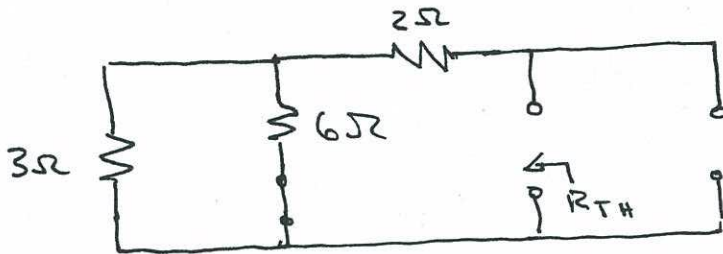
Loaded circuit:



4.4



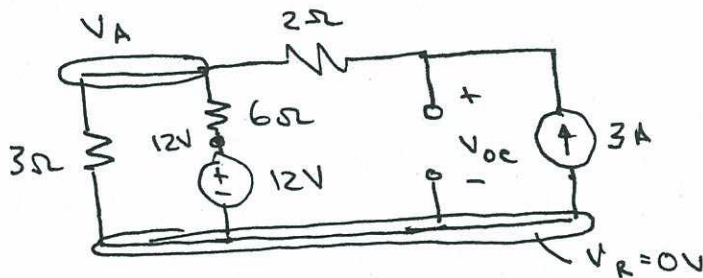
(a) Remove resistor R , kill sources and determine Thevenin resistance:



$$R_{TH} = \frac{(3\Omega)(6\Omega)}{3\Omega + 6\Omega} + 2\Omega$$

$$\underline{R_{TH} = 4\Omega}$$

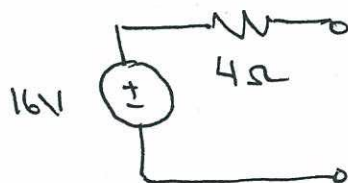
Replace sources & Find V_{oc} :



$$\frac{V_A}{3\Omega} + \frac{V_A - 12V}{6\Omega} - 3A = 0$$

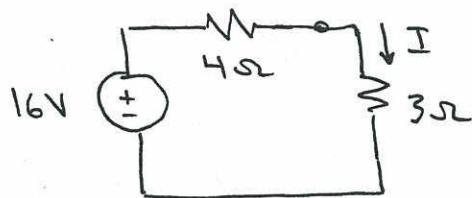
$$V_A = 10V \Rightarrow \underline{V_{oc} = 16V}$$

Thevenin Circuit:



4.4 (cont'd)

(b) Replace resistor & set $R = 3\Omega$:

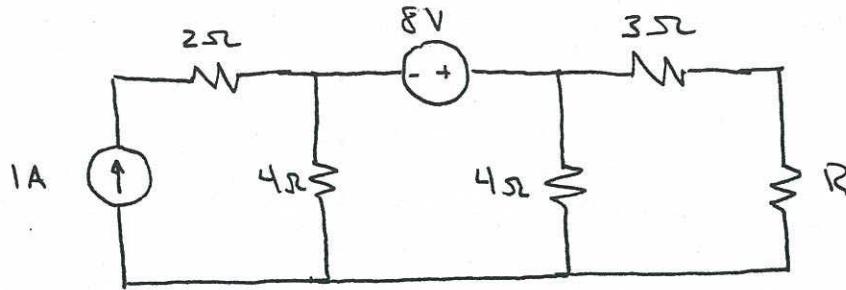


$$I = \frac{16V}{7\Omega} \Rightarrow \boxed{I = \frac{16}{7} A}$$

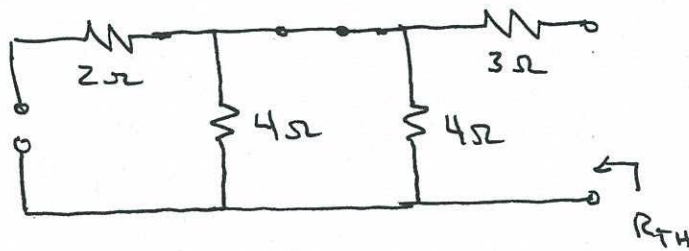
(c) Maximum power when $R = R_{TH}$:

$$\boxed{R = 4\Omega}$$

4.5



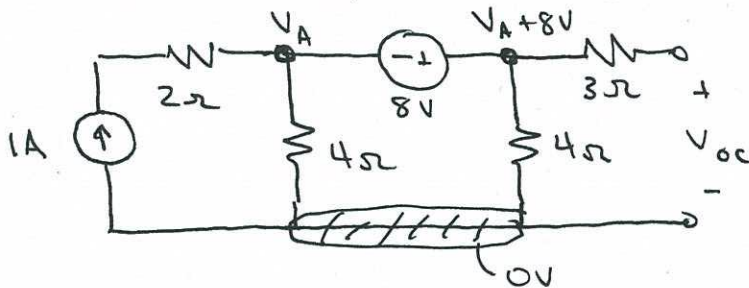
(a) Kill sources, find resistance "seen" by R:



$$R_{TH} = 3\Omega + \frac{(4\Omega)(4\Omega)}{4\Omega + 4\Omega}$$

$$\underline{R_{TH} = 5\Omega}$$

Return sources, find V_{oc} :

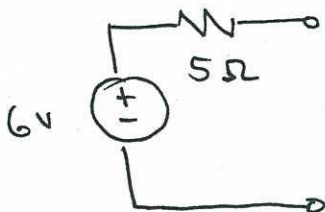


$$-1A + \frac{V_A}{4\Omega} + \frac{V_A + 8V}{4\Omega} = 0$$

$$V_A = -2V$$

$$V_{oc} = V_A + 8V = \underline{\underline{6V}}$$

Thevenin circuit:

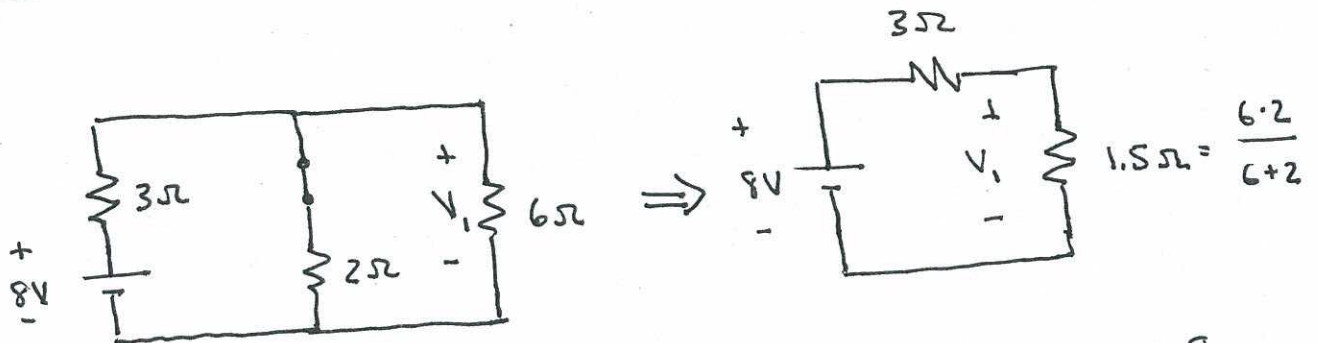


(b) Maximum power transfer when $R = R_{TH}$:

$R = 5\Omega$

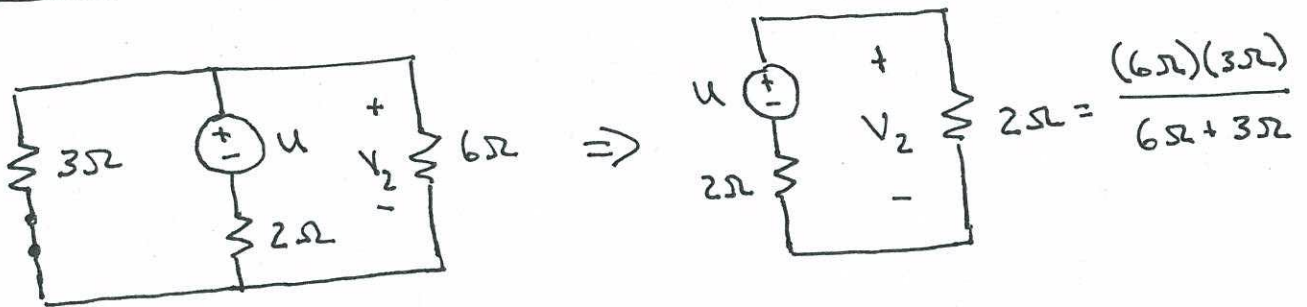
4.6

Contribution from 8V source:



Voltage divider: $V_1 = \frac{1.5\Omega}{3\Omega + 1.5\Omega} \cdot 8V = \left(\frac{1}{3}\right) 8V \Rightarrow \underline{\underline{V_1 = \frac{8}{3} V}}$

Contribution from "u" Volt input:



Voltage divider: $V_2 = \left(\frac{2\Omega}{2\Omega + 2\Omega}\right) u \Rightarrow \underline{\underline{V_2 = \frac{u}{2} V}}$

Superimpose:

$V = V_1 + V_2 = \left(\frac{8}{3} V\right) + \left(\frac{u}{2}\right) V \Rightarrow \boxed{V = \frac{8}{3} + \frac{u}{2}}$

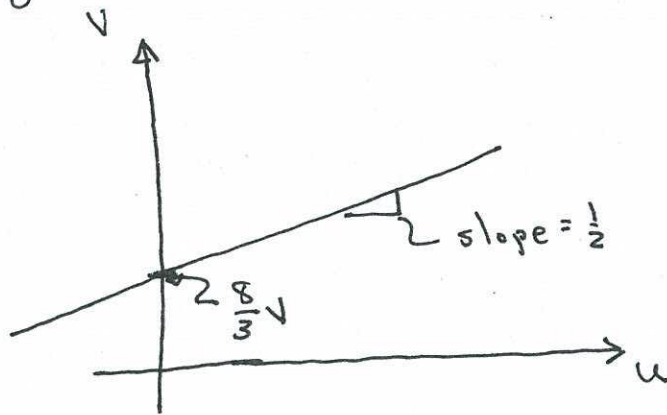
4.7

Using superposition (in previous problem), we found that:

$$V = \frac{8}{3} + \frac{1}{2} u$$

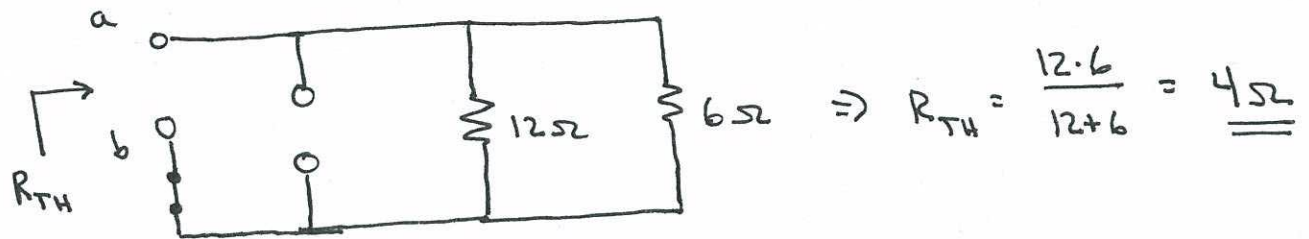
This is a valid input-output relationship!

Plotting:

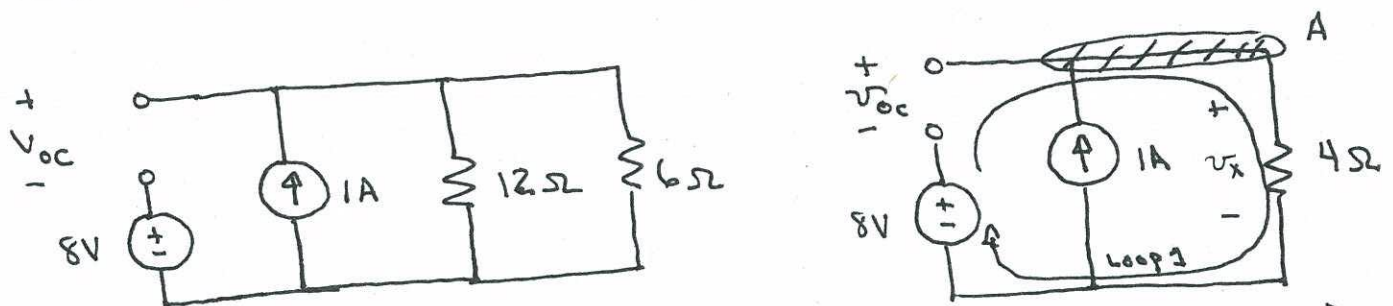


4.8

Find R_{TH} :



Find V_{oc} :

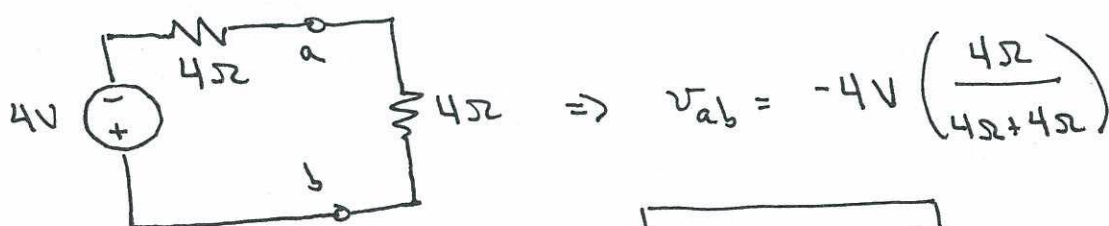


equivalent circuit.

KCL at "A" + Ohm's Law $\Rightarrow v_x = (4\Omega)(1A) = 4V$

KVL, Loop 1: $v_{oc} = v_x - 8V = -4V$

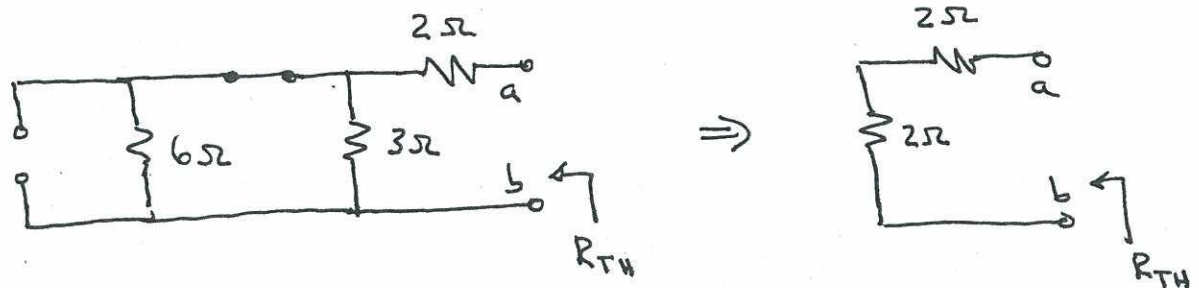
Thevenin Circuit:



$v_{ab} = -2V$

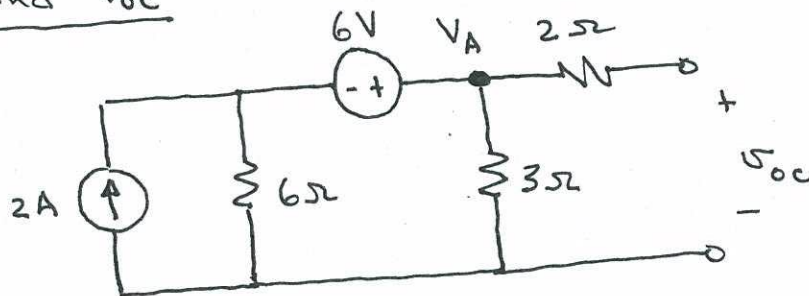
4.9

Find R_{TH} :



$$R_{TH} = 2\Omega + 2\Omega = \underline{\underline{4\Omega}}$$

Find V_{oc} :

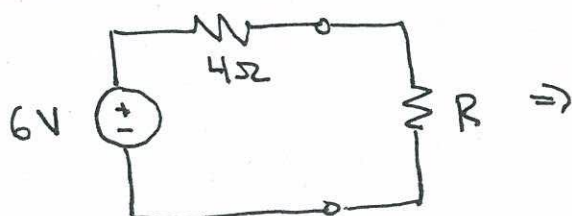


Nodal analysis to find V_A :

$$-2 + \frac{V_A - 6}{6\Omega} + \frac{V_A}{3\Omega} = 0 \Rightarrow 3V_A = 18 \Rightarrow V_A = 6V$$

$$V_{oc} = V_A = \underline{\underline{6V}}$$

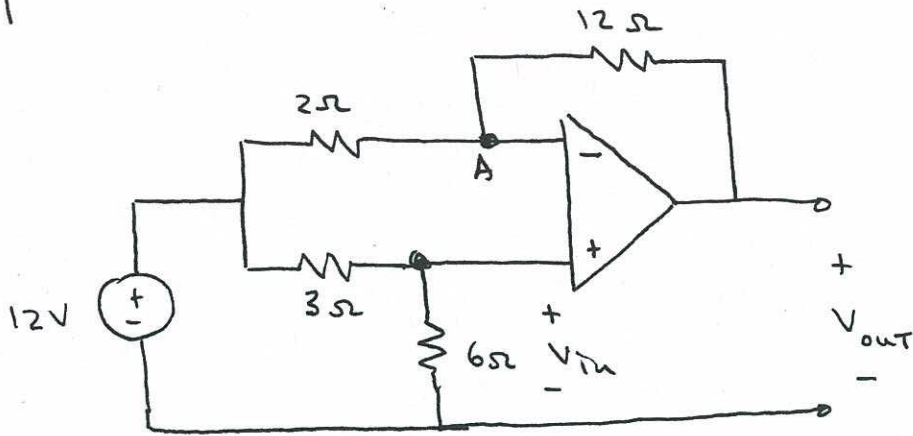
Thevenin Circuit



Maximum power when

$$R = R_{TH} = 4\Omega$$

5.1



$$V_{in} = \left(\frac{6\Omega}{3\Omega + 6\Omega} \right) 12V = 8V$$

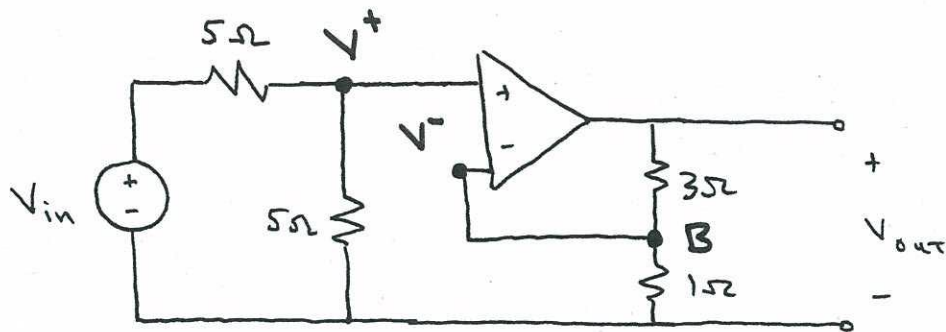
Op-amp rules: $V_A = V_{in} = 8V$

KCL at node A:

$$\frac{12V - V_{in}}{2\Omega} = \frac{V_{in} - V_{out}}{12\Omega} \Rightarrow \frac{12V - 8V}{2\Omega} = \frac{8V - V_{out}}{12\Omega}$$

$$V_{out} = -16V$$

5.2



$$V^+ = \left(\frac{5\Omega}{5\Omega + 5\Omega} \right) V_{in} = \frac{V_{in}}{2}$$

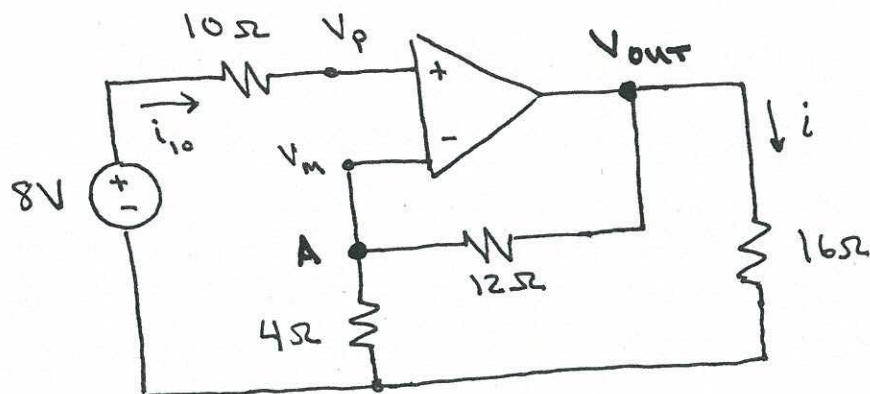
Op-amp rules: $V^- = V^+ = \frac{V_{in}}{2}$

KCL at B:

$$\frac{V_B}{1\Omega} = \frac{V_{out} - V_B}{3\Omega} \Rightarrow \frac{V_{in}}{2} = \frac{V_{out} - \frac{V_{in}}{2}}{3}$$

$$V_{out} = 2 V_{in}$$

5.3



Apply op-amp rules:

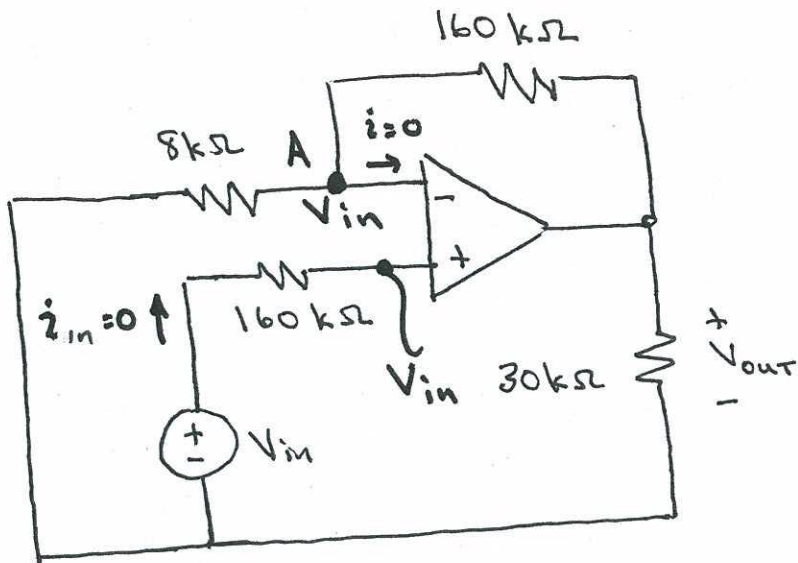
$$i_{10} = 0 \Rightarrow V_p = 8V$$

$$V_p = V_m \Rightarrow V_m = 8V$$

$$\text{KCL at "A": } \frac{8V}{4\Omega} = \frac{V_{OUT} - 8V}{12\Omega} \Rightarrow V_{OUT} = 32V$$

$$\text{Ohm's Law: } i = \frac{V_{OUT}}{16\Omega} = \frac{32V}{16\Omega} \Rightarrow \boxed{i = 2A}$$

5.4



Op-Amp rules:

$$i_{in} = 0 \Rightarrow V_{in} = V_p$$

$$V_p = V_n = V_{in}$$

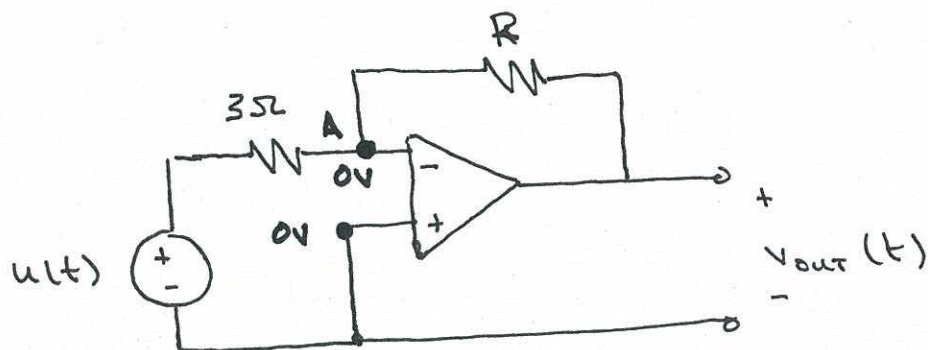
KCL at A:

$$\frac{0 - V_{in}}{8 \text{ k}\Omega} = \frac{V_{in} - V_{out}}{160 \text{ k}\Omega} \Rightarrow -20 V_{in} = V_{in} - V_{out}$$

$$V_{out} = 21 V_{in}$$

5.5

Probably easiest to find an input-output relation, and apply both inputs to that:



$$R = \frac{(8\Omega)(24\Omega)}{8\Omega + 24\Omega} = 6\Omega$$

Equivalent resistance of parallel 8Ω & 24Ω resistors

KCL at A: $\frac{u(t) - 0}{3\Omega} = \frac{0 - v_{out}(t)}{R} \Rightarrow 2u(t) = -v_{out}(t)$

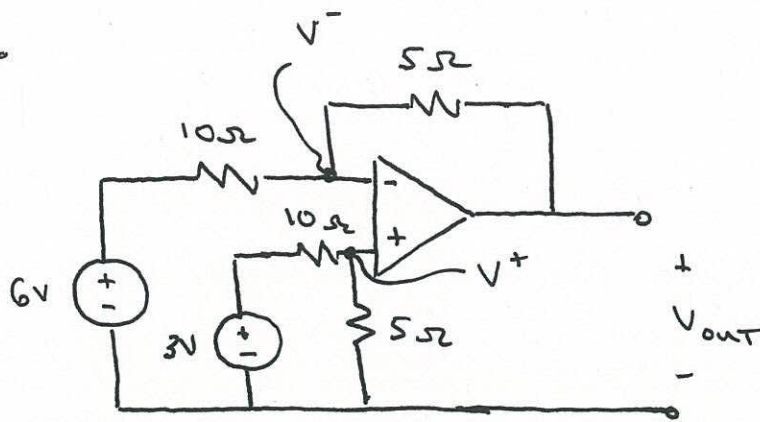
$v_{out}(t) = -2u(t)$

(a)

$$v_{out}(t) = -2[4V] \Rightarrow \boxed{v_{out}(t) = -8V}$$

(b) $v_{out}(t) = -2[2\cos(5t)] \Rightarrow \boxed{v_{out}(t) = -4\cos(5t) V}$

5.6



Non-inverting input: $\frac{3V - V^+}{10\Omega} = \frac{V^+ - 0V}{5\Omega} \Rightarrow V^+ = 1V$

Opamp rules $\Rightarrow V^- = V^+ \Rightarrow V^- = 1V$

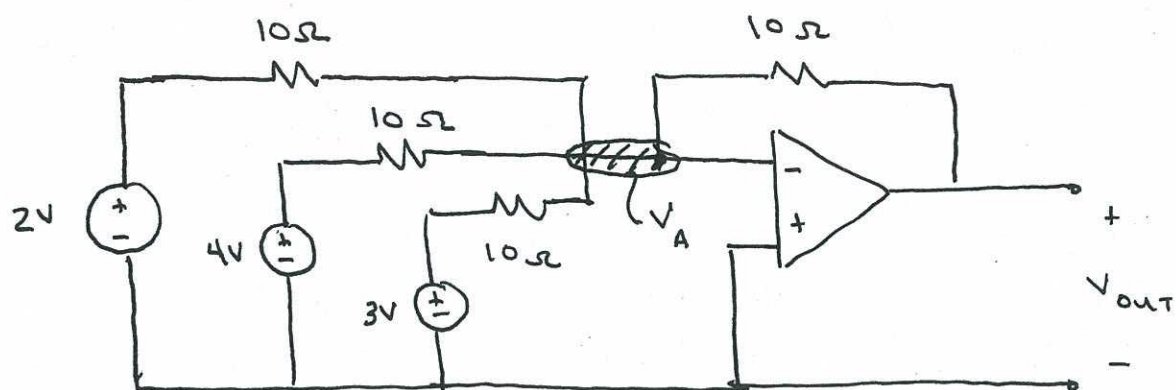
KCL at inverting input:

$$\frac{6V - V^-}{10\Omega} = \frac{V^- - V_{out}}{5\Omega} \quad \rightarrow \quad \frac{6V - 1V}{10\Omega} = \frac{1V - V_{out}}{5\Omega}$$

$$5V = 2V - 2V_{out}$$

$$\underline{\underline{V_{out} = -\frac{3}{2} V}}$$

5.7



Op-amp rules: $V_A = 0V$

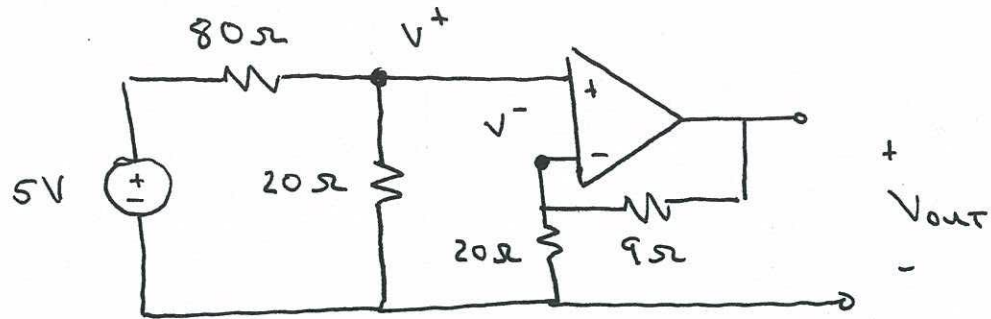
KCL at node A:

$$\frac{2V - V_A}{10\Omega} + \frac{4V - V_A}{10\Omega} + \frac{3V - V_A}{10\Omega} = \frac{V_A - V_{out}}{10\Omega}$$

$$\frac{2V}{10\Omega} + \frac{4V}{10\Omega} + \frac{3V}{10\Omega} = -\frac{V_{out}}{10\Omega}$$

$$\underline{\underline{V_{out} = -9V}}$$

5.8



KCL at non-inverting input:

$$\frac{5V - V^+}{80\Omega} = \frac{V^+ - 0V}{20\Omega} \Rightarrow V^+ = 1V$$

Op-amp rules: $V^- = V^+ \Rightarrow V^- = 1V$

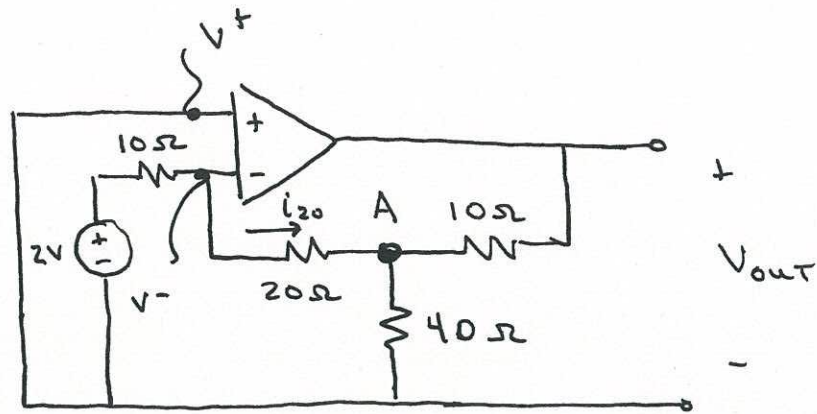
KCL at inverting input:

$$\frac{V^- - 0V}{20\Omega} + \frac{V^- - V_{out}}{9\Omega} = 0$$

$$\frac{1V}{20\Omega} + \frac{1V - V_{out}}{9\Omega} = 0$$

$$\underline{\underline{V_{out} = \frac{29}{20} V}}$$

5.9



$$V^+ = 0 \Rightarrow V^- = 0 \quad (\text{op-amp rules})$$

KCL at A:

$$\frac{V_A - V^-}{20\Omega} + \frac{V_A - 0V}{40\Omega} + \frac{V_A - V_{out}}{10\Omega} = 0$$

$$\frac{V_A}{20\Omega} + \frac{V_A}{40\Omega} + \frac{V_A}{10\Omega} = \frac{V_{out}}{10\Omega} \Rightarrow V_{out} = \frac{7}{4} V_A$$

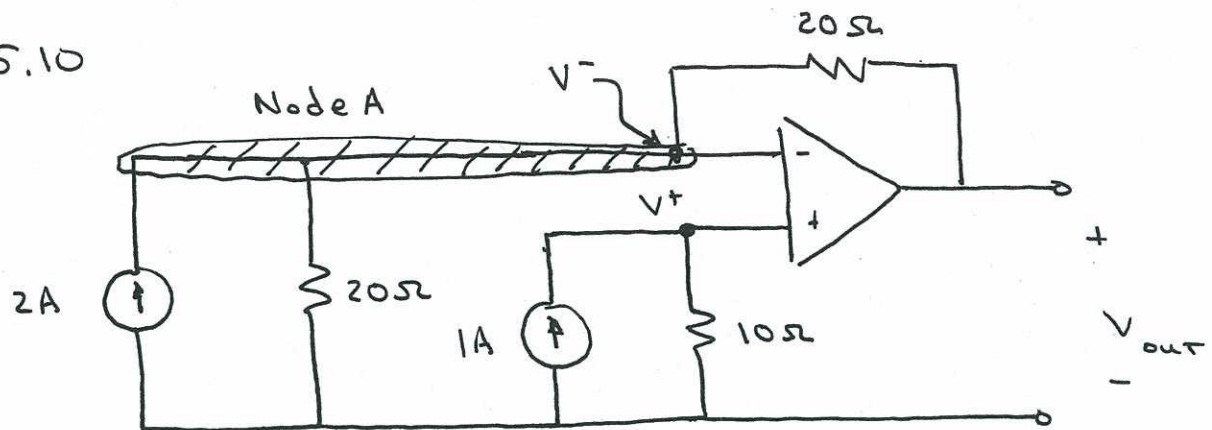
KCL at inverting input:

$$\frac{2V - V^-}{10\Omega} = i_{20} \Rightarrow i_{20} = \frac{1}{5} A$$

$$\text{Ohm's Law: } V^- - V_A = (20\Omega) i_{20} \Rightarrow V_A = -4V$$

$$\underline{\underline{V_{out} = 7V}}$$

5.10



$$V^+ = (10\Omega)(1A) = 10V$$

Op-amp rule: $V^+ = V^- = 10V$

KCL, node A:

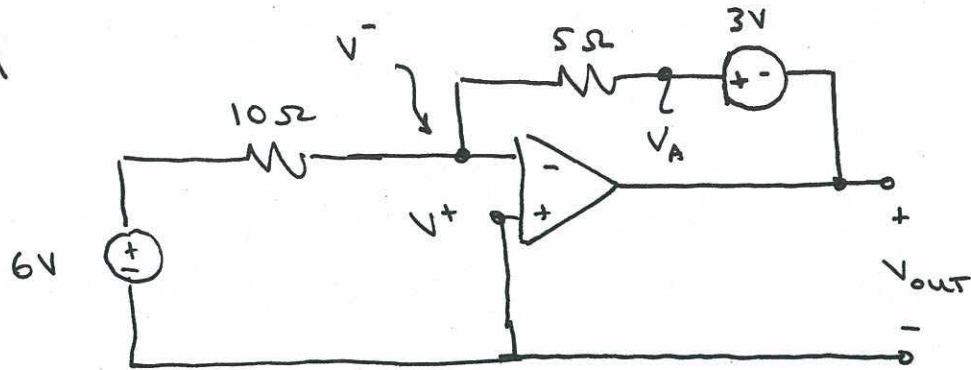
$$2A = \frac{V^-}{20\Omega} + \frac{V^- - V_{out}}{20\Omega}$$

$$40V = V^- + V^- - V_{out}$$

$$40V = 10V + 10V - V_{out}$$

$$\underline{\underline{V_{out} = -20V}}$$

5.11



$$V^- = V^+ = 0V \quad (\text{op-amp rules})$$

KCL at inverting input:

$$\frac{6V - V^-}{10\Omega} = \frac{V^- - V_A}{5\Omega} \Rightarrow 6V - V^- = 2V^- - 2V_A$$

$$6V = -2V_A$$

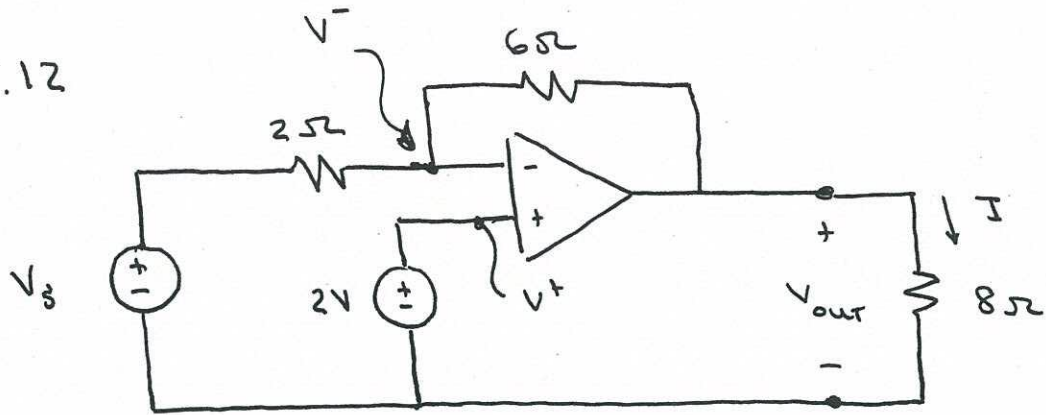
$$V_A = -3V$$

$$V_A = V_{out} + 3V$$

$$V_{out} = V_A - 3V$$

$$\underline{\underline{V_{out} = -6V}}$$

5.12



$$V^+ = 2V \Rightarrow V^- = 2V \quad (\text{op-amp rules})$$

KCL at inverting input:

$$\frac{V_s - V^-}{2\Omega} = \frac{V^- - V_{out}}{6\Omega} \Rightarrow 3V_s - 4V^- = V_{out}$$

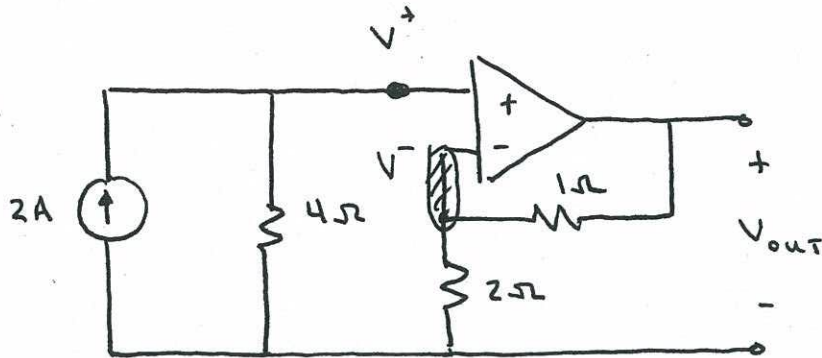
$$V_{out} = 8V - 3V_s$$

Ohm's Law:

$$I = \frac{V_{out}}{8\Omega}$$

$$I = \underline{\underline{\frac{8V - 3V_s}{8\Omega}}}$$

5.13



$$V^+ = (2A)(4\Omega) = 8V$$

KCL at inverting input
+ Ohm's law

$$V^- = V^+ = 8V \quad (\text{op-amp rules})$$

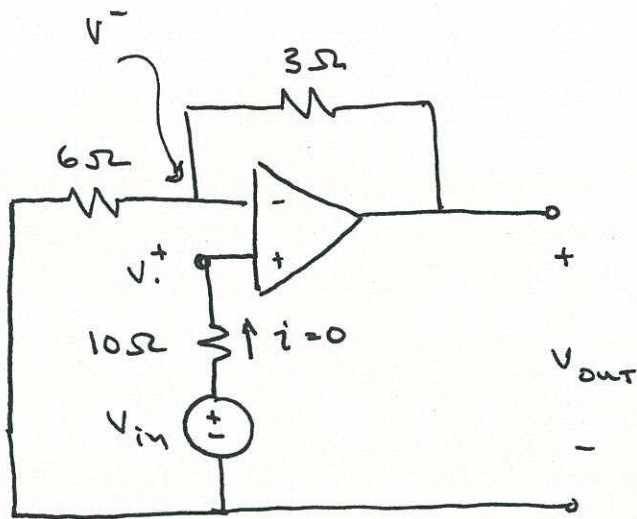
KCL at inverting input:

$$\frac{V^- - 0V}{2\Omega} + \frac{V^- - V_{out}}{1\Omega} = 0 \Rightarrow 3V^- = 2V_{out}$$

$$2V_{out} = (3)(8V)$$

$$\underline{\underline{V_{out} = 12V}}$$

5.14



$$V^+ = V_{in}$$

$$V^- = V^+ = V_{in}$$

no current into input terminals, so no voltage drop across 10Ω resistor

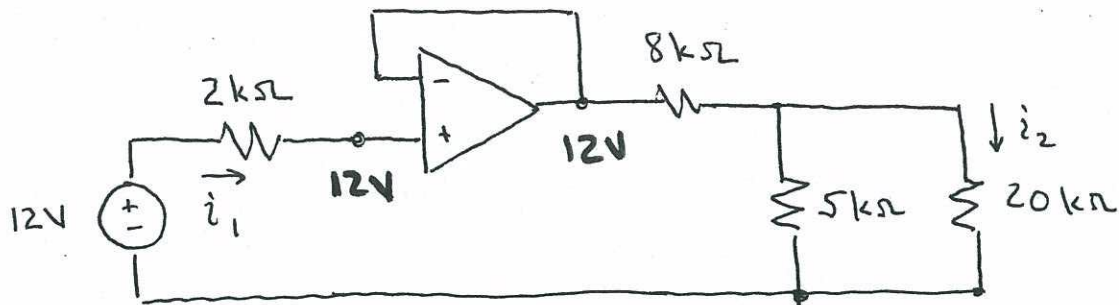
KCL at inverting input:

$$\frac{V^- - 0V}{6\Omega} = \frac{V_{out} - V^-}{3\Omega} \Rightarrow 2V_{out} = 3V^-$$

$$2V_{out} = 3(V_{in})$$

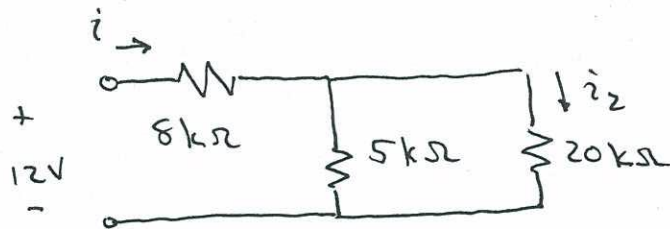
$$\underline{\underline{V_{out} = \frac{3}{2} V_{in}}}$$

S.15

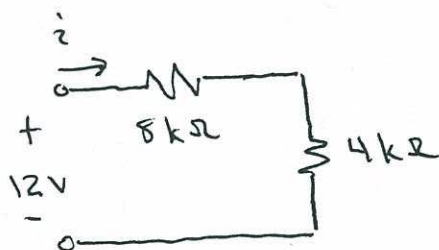


From op-amp rules, $i_1 = 0A$

From op-amp rules, op-amp output voltage = 12V,
so circuit to determine i_2 is:

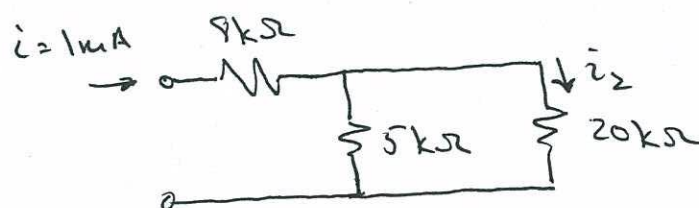


↓ Equivalent circuit



$$\Rightarrow i = \frac{12V}{8k\Omega + 4k\Omega} = 1mA$$

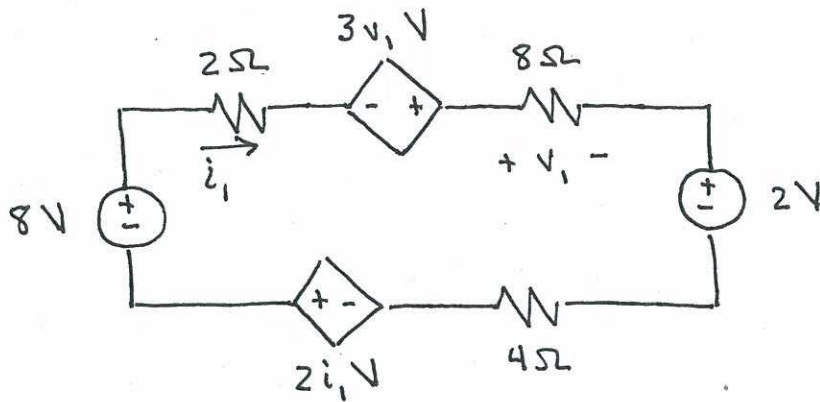
current divider:



$$i_2 = \frac{5k\Omega}{5k\Omega + 20k\Omega} (1mA)$$

$$i_2 = 0.2mA$$

5.16



KVL :

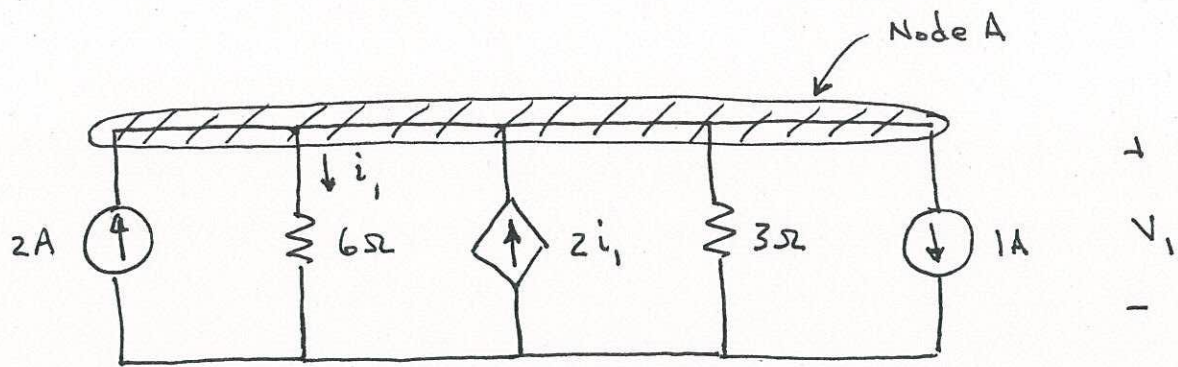
$$-8V + (2\Omega) i_1 - 3v_1 + v_1 + 2V + (4\Omega) i_1 - 2i_1 = 0$$

$$v_1 = (8\Omega) i_1$$

$$-6 - 2v_1 + \frac{v_1}{2} = 0 \Rightarrow -3v_1 = 12$$

$$v_1 = -4V$$

5.17



KCL, Node A:

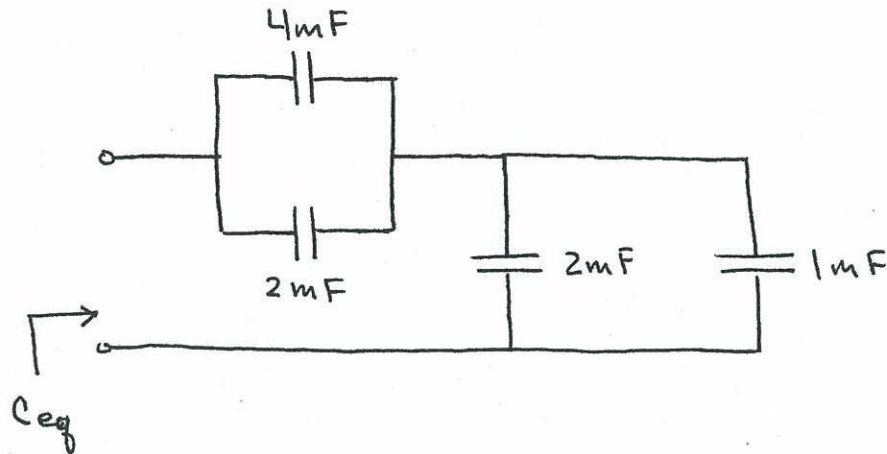
$$-2A + i_1 - 2i_1 + \frac{V_1}{3\Omega} + 1A = 0$$

Ohm's Law:

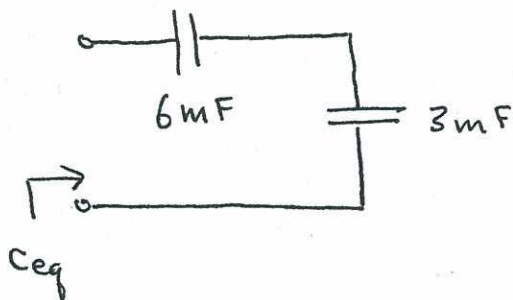
$$i_1 = \frac{V_1}{6\Omega} \Rightarrow V_1 = 6i_1$$

$$\rightarrow -2 + i_1 - 2i_1 + \frac{6i_1}{3\Omega} + 1 = 0 \Rightarrow \boxed{i_1 = 1A}$$

6.1



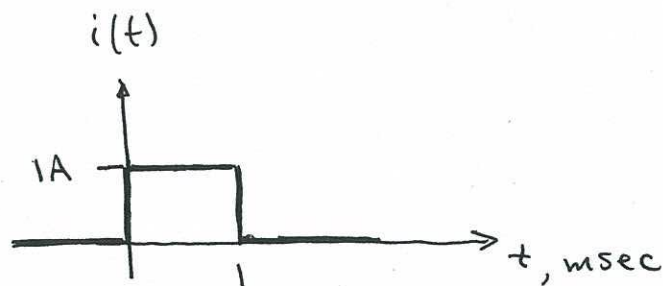
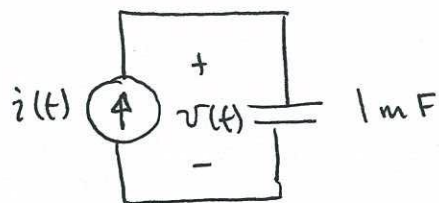
Capacitors in parallel add directly, so the above is equivalent to:



Capacitors in series add like resistors in parallel:

$$C_{eq} = \frac{(3\text{mF})(6\text{mF})}{3\text{mF} + 6\text{mF}} \Rightarrow \boxed{C_{eq} = 2\text{mF}}$$

6.2

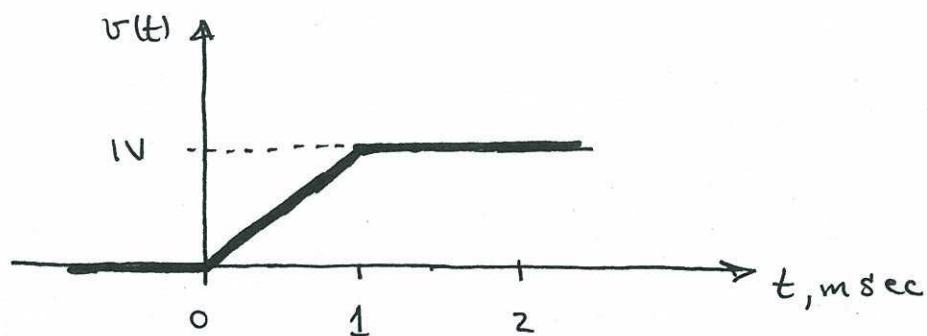


voltage - current relation for capacitors:

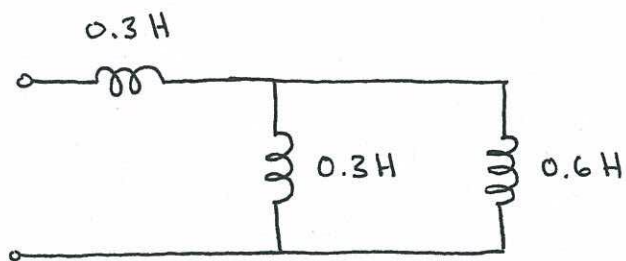
$$v(t) = \frac{1}{C} \int_0^t i(t) dt + v(0)$$

0 (capacitor initially uncharged)

$$v(t) = 1000 \int_0^t 1 dt, \quad 0 < t < 1 \times 10^{-3} \text{ sec}$$



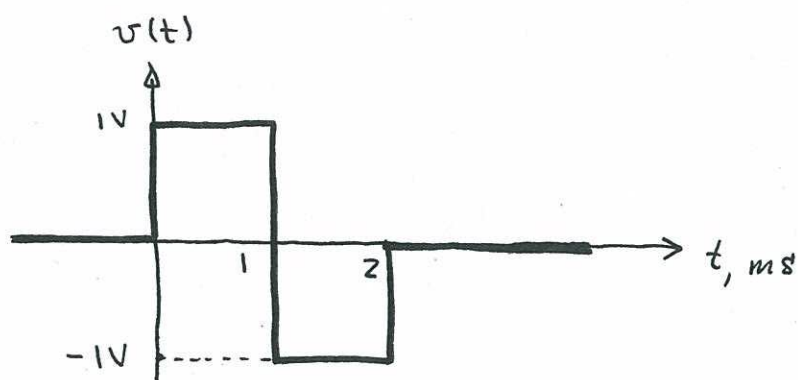
6.3



$$L_{eq} = 0.3 \text{ H} + \frac{(0.3 \text{ H})(0.6 \text{ H})}{0.3 \text{ H} + 0.6 \text{ H}}$$

$$L_{eq} = 0.5 \text{ H}$$

6.4

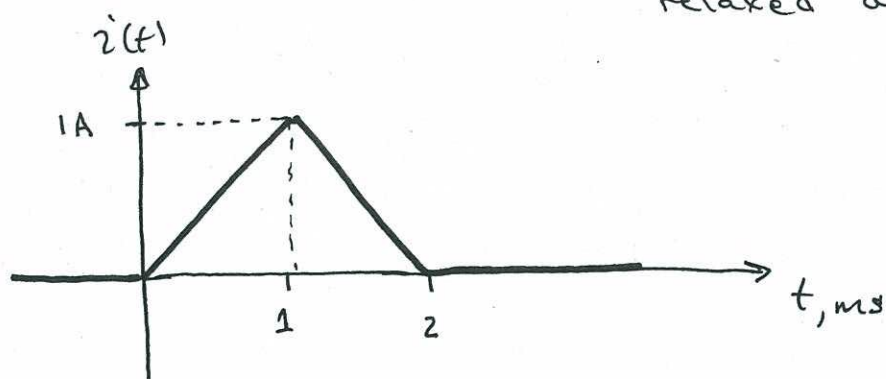


$$L = 1 \text{ mH}$$

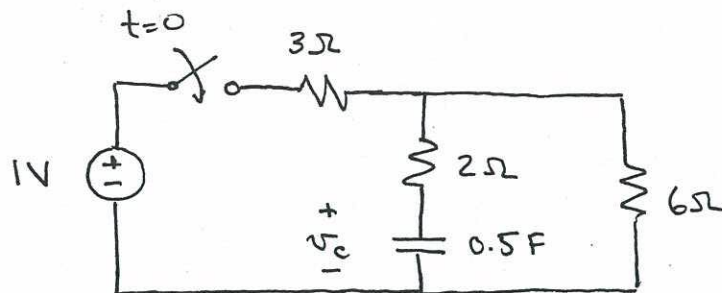
Voltage-current relationship:

$$i(t) = \frac{1}{L} \int_0^t v(t) dt + i(t=0)$$

← assume initial current is zero (inductor relaxed at time $t=0$).

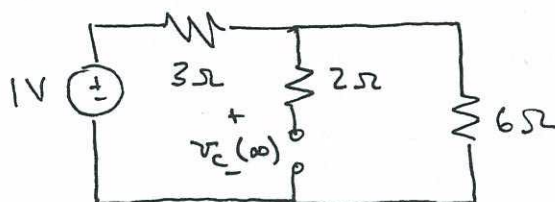


7.1



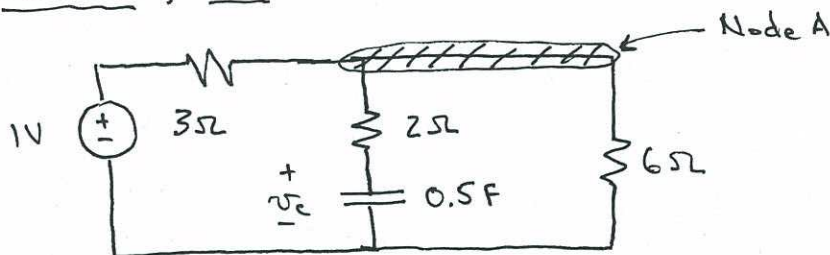
Initial condition : $v_c(t=0^+) = v_c(t=0^-) = \underline{\underline{0V}} = v_c(0)$

Final condition :



$$v_c(t \rightarrow \infty) = \left(\frac{6\Omega}{9\Omega} \right) 1V = \underline{\underline{\frac{2}{3}V}}$$

Circuit, $t > 0$:



Nodal analysis (KCL at A):

$$\frac{1}{3\Omega} \left[1 - (v_c + 2(0.5F) \frac{dv_c}{dt}) \right] = (0.5F) \frac{dv_c}{dt} + \left[v_c + 2(0.5F) \frac{dv_c}{dt} \right] \frac{1}{6\Omega}$$

$$2 - 2v_c - 2 \frac{dv_c}{dt} = 3 \frac{dv_c}{dt} + v_c + \frac{dv_c}{dt}$$

$$2 = 6 \frac{dv_c}{dt} + 3v_c$$

$$\underline{\underline{\frac{dv_c}{dt} + \frac{1}{2} v_c = \frac{1}{3}}}$$

Differential equation governing $v_c(t)$, $t > 0$

↓ (continued)

7.1 (cont'd)

Governing differential equation: $\frac{dv_c}{dt} + \frac{1}{2} v_c = \frac{1}{3}$

$$\tau = 2 \text{ seconds}$$

Form of solution:

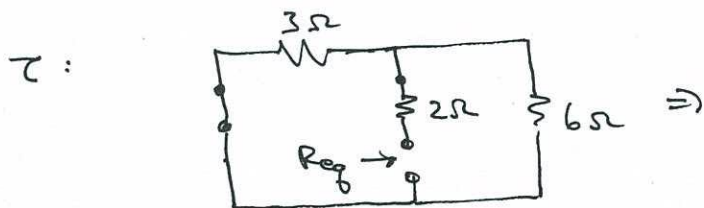
$$v_c(t) = K_1 + K_2 e^{-t/\tau}$$

Initial, final conditions:

$$\left. \begin{array}{l} v_c(0) = 0 = K_1 + K_2 \\ v_c(\infty) = \frac{2}{3} = K_1 \end{array} \right\} \Rightarrow K_1 = \frac{2}{3}, K_2 = -\frac{2}{3}$$

$$v_c(t) = \frac{2}{3} [1 - e^{-t/2}] \text{ V}, t > 0$$

check τ , steady-state response:



$$R_{eq} = 2\Omega + \frac{(6\Omega)(3\Omega)}{6\Omega + 3\Omega} = 4\Omega$$

$$\tau = R_{eq}C = (4)(0.5) = 2 \text{ sec} \checkmark$$

steady-state response:

$$\frac{dv_c(\infty)}{dt} + \frac{1}{2} v_c(\infty) = \frac{1}{3} \Rightarrow v_c(\infty) = \frac{2}{3} \text{ V} \quad \left(\begin{array}{l} \text{matches} \\ \text{final} \\ \text{condition} \end{array} \right)$$

continued

7.1 (cont'd)

(b) capacitance = 1F

changing capacitance only changes the time constant. Equivalent resistance is same as in part (a), so:

$$\tau = R_{eq}C = (4\Omega)(1F) = 4\text{ sec}$$

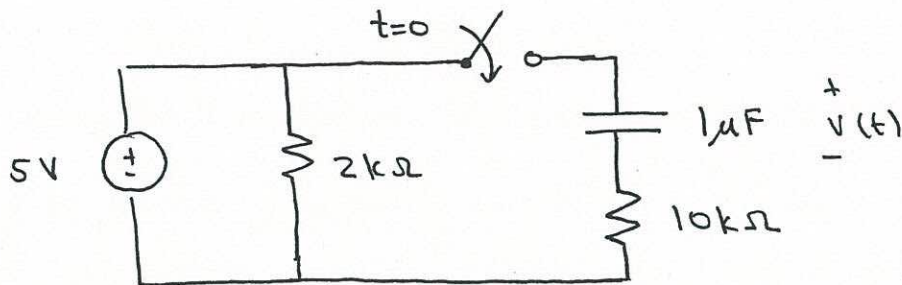
$$v_c(t) = \frac{2}{3} [1 - e^{-t/4}], \quad t > 0$$

(c) capacitance = 0.25F

Same approach as (b):

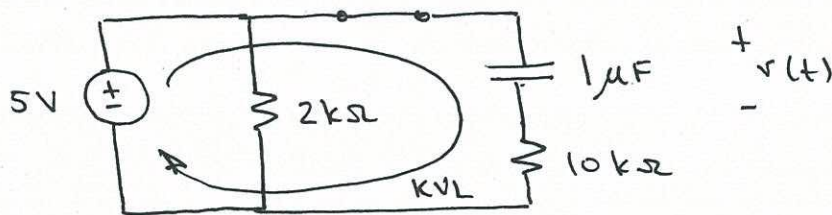
$$v_c(t) = \frac{2}{3} [1 - e^{-t}], \quad t > 0$$

7.2



Initial condition: $v(t=0^+) = v(t=0^-) = \underline{\underline{0V}}$

Circuit, for $t > 0$:

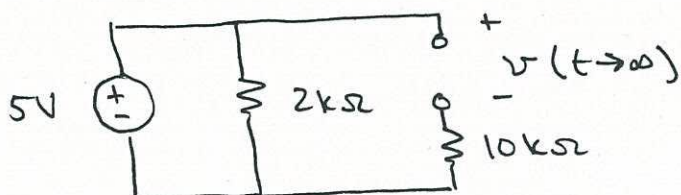


$$\text{KVL: } 5V = v(t) + (10k\Omega) \left[1\mu F \frac{dv(t)}{dt} \right]$$

$$5V = v + 10^{-2} \frac{dv}{dt} \Rightarrow \frac{dv}{dt} + 100v = 500$$

$$\tau = \underline{\underline{\frac{1}{100} \text{ sec}}}$$

Final condition:



$$\Rightarrow \underline{\underline{v(t \rightarrow \infty) = 5V}}$$

Form of solution:

$$v(t) = K_1 + K_2 e^{-t/\tau}$$

$$\xrightarrow{\tau = \frac{1}{100}}$$

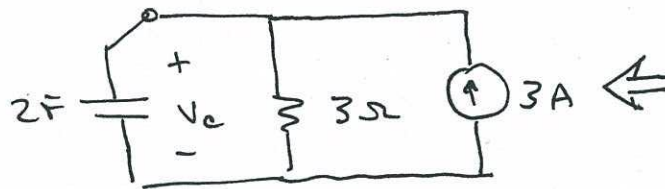
$$\boxed{v(t) = 5(1 - e^{-100t}), t > 0}$$

$$v(0) = 0, v(\infty) = 5$$

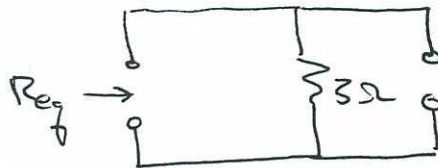
7.3

- Solution is of form: $v(t) = K_1 + K_2 e^{-t/\tau}$
- Find τ from RC time constant

Circuit for $t > 0$:



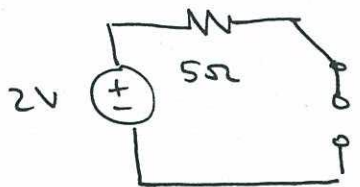
Kill sources, find R_{eq} :



$$\Rightarrow R_{eq} = 3\Omega \Rightarrow R_{eq}C = \underline{\underline{6 \text{ sec}}}$$

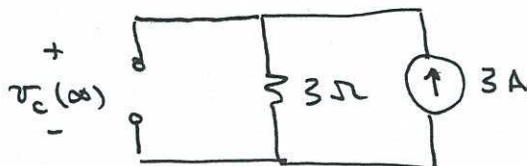
$$\tau = \underline{\underline{6 \text{ sec.}}}$$

- Initial condition (circuit at $t < 0$):



$$v_c(t=0^-) = \underline{\underline{2V}} \Rightarrow \underline{\underline{2V = K_1 + K_2}}$$

- Final Condition, (circuit as $t \rightarrow \infty$):



$$v_c(t \rightarrow \infty) = (3\Omega)(3A) = \underline{\underline{9V}}$$

$$\underline{\underline{9V = K_1}}$$

$$K_1 = 9V$$

$$K_2 = -7V$$

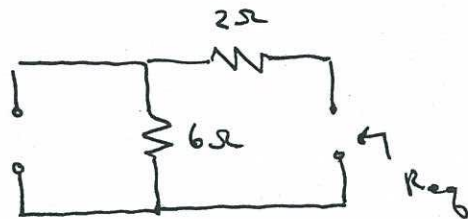
$$\tau = 6 \text{ sec}$$

$$\boxed{v(t) = 9 - 7e^{-t/6} \text{ V}}$$

7.4

Solution is of the form $y(t) = K_1 + K_2 e^{-t/\tau}$.

Finding τ from RC:



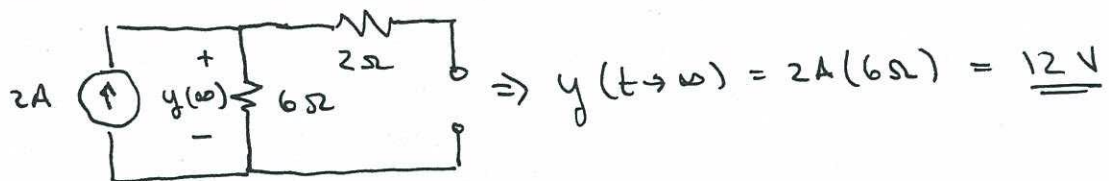
$$\Rightarrow R_{eq} = 2\Omega + 6\Omega = 8\Omega$$

$$\tau = (8\Omega)(\frac{1}{4}F) = \underline{2\text{sec}}$$

Initial condition

$$y(t=0) = 0V \quad (\text{no current applied before } t=0)$$

Final condition:



Solving for K_1 & K_2 :

$$\left. \begin{array}{l} y(0) = K_1 + K_2 = 0V \\ y(\infty) = K_1 = 12V \end{array} \right\} \Rightarrow \begin{array}{l} K_1 = 12V \\ K_2 = -12V \end{array}$$

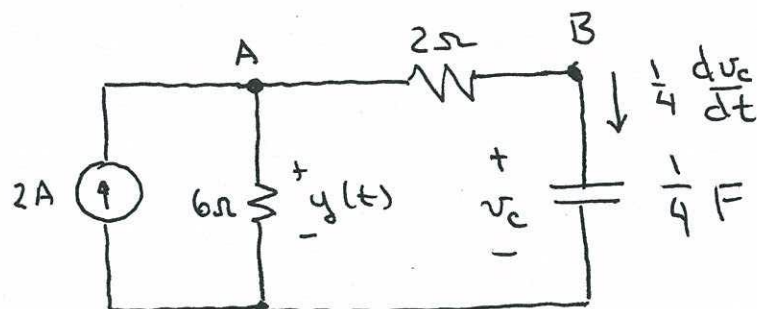
$$y(t) = 12(1 - e^{-t/2})V$$

Let's check this result by writing the diff. eqn. governing the circuit

} continued

7.4 (cont'd)

circuit for $t > 0$:



KVL around right-most loop:

$$y(t) = (2\Omega) \left(\frac{1}{4} \frac{dv_c}{dt} \right) + v_c \Rightarrow \underline{\frac{1}{2} \frac{dv_c}{dt} + v_c = y} \quad (*)$$

KCL at A:

$$2A = \frac{y}{6\Omega} + \frac{1}{4} \frac{dv_c}{dt}$$

This equation is fine, but looks like it will get messy when combined with the above. Let's try something else

Re-do KCL at A:

$$2A = \frac{y}{6\Omega} + \frac{y - v_c}{2\Omega}$$

$$12 = y + 3(y - v_c) \Rightarrow v_c = \frac{4}{3}y - 4 \Rightarrow \underline{\frac{dv_c}{dt} = \frac{4}{3} \frac{dy}{dt}}$$

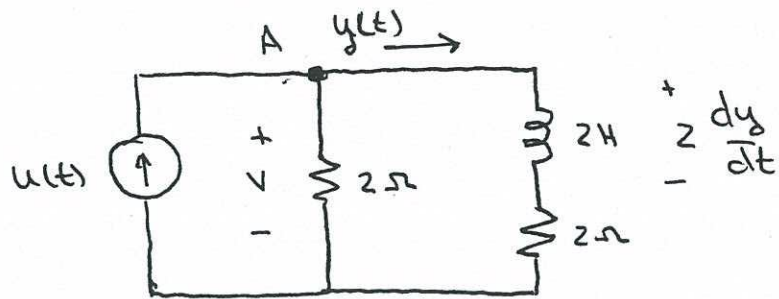
These two are easy to substitute into eqn. (*)

Diff. eqn. becomes:

$$y = \frac{1}{2} \left[\frac{4}{3} \frac{dy}{dt} \right] + \left[\frac{4}{3}y - 4 \right] \Rightarrow \underline{\frac{dy}{dt} + \frac{1}{2}y = 6} \Rightarrow y(\infty) = 12V$$

$\tau = 2 \text{ sec}$
 $\Rightarrow$ AGREES with previous solution!

7.5

(a) circuit for $t > 0$:

$$\text{KCL at A} \Rightarrow u = \frac{v}{2} + y$$

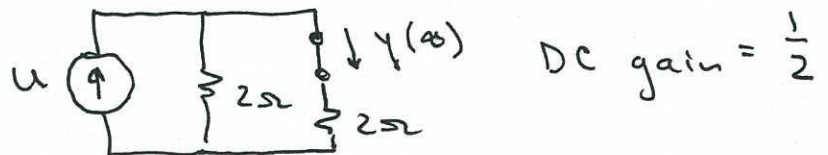
$$\text{KVL, right loop} \Rightarrow v = 2 \frac{dy}{dt} + 2y$$

$$\boxed{\frac{dy}{dt} + 2y = u}$$

- check the diff. equation based on expected circuit response.

$$\text{D.E : } \tau = \frac{1}{2}, \text{ DC gain} = \frac{Y(\omega)}{u(\omega)} = \frac{1}{2}$$

$$\text{circuit : } \tau = \frac{L}{R_{eq}} = \frac{2H}{4\Omega} = \frac{1}{2},$$



$\Rightarrow$ The Diff. eqn. and expected circuit response agree. ✓

↓ continued

7.5 (cont'd)

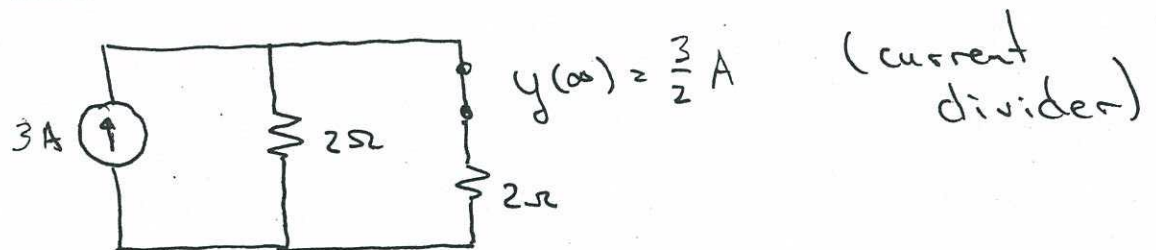
(b) From (a), $\frac{dy}{dt} + 2y = u = 3u_0(t) = \begin{cases} 3A, & t \geq 0 \\ 0A, & t < 0 \end{cases}$

form of solution (if u is a step function) is:

$$y(t) = K_1 + K_2 e^{-t/\tau}$$

$y(0) = 0A$ (no current through inductor before step is applied.)

Final condition:



Solving for K_1 & K_2

$$\left. \begin{aligned} y(0) &= K_1 + K_2 = 0 \\ y(\infty) &= K_1 = \frac{3}{2} \end{aligned} \right\} \Rightarrow K_1 = \frac{3}{2}, \quad K_2 = -\frac{3}{2}$$

$$\boxed{y(t) = \frac{3}{2} (1 - e^{-2t})}$$

(c) In part (a), we

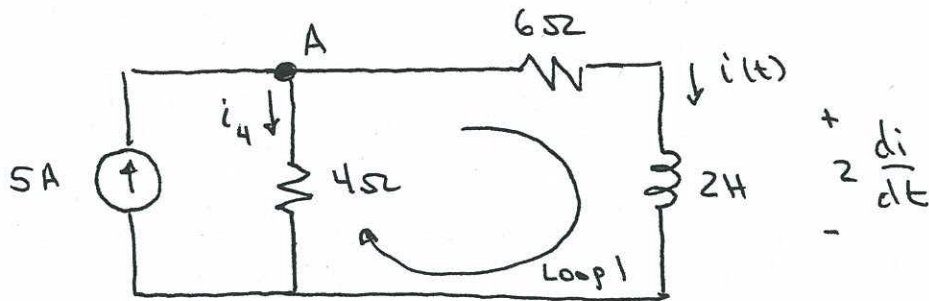
saw that the circuit's time constant

$$\tau = \frac{L}{R_{eq}} = \frac{2H}{4\Omega} = \frac{1}{2} \text{ sec. and the DC gain}$$

is $\frac{1}{2}$. Therefore, the above solution

agrees with what we expect!

7.6

(a) circuit for $t > 0$:

KCL at A: $i_4 = 5 - i(t)$ (*)

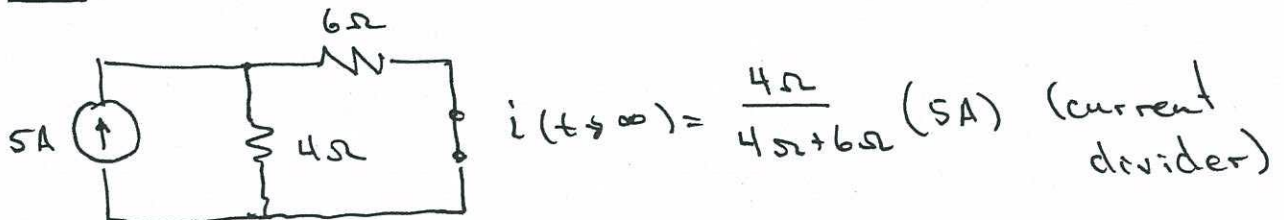
KVL, Loop 1: $(4\Omega) i_4 = (6\Omega) i + (2H) \frac{di}{dt}$ (**)

Substitute (*) into (**):

$$4(5 - i) = 6i + 2 \frac{di}{dt} \Rightarrow \boxed{20 = 10i + 2 \frac{di}{dt}}$$

(b) Initial: $i(t=0) = \underline{\underline{0A}}$ (no current before switch closes)

Final: Inductor looks like short circuit as $t \rightarrow \infty$:



$$i(t \rightarrow \infty) = \frac{4}{10} (5) = \underline{\underline{2A}}$$

↓ continued.

7.6 (cont'd)

(c) Form of solution:

$$i(t) = K_1 + K_2 e^{-t/\tau}$$

From diff. eqn.: $20 = 10i + 2 \frac{di}{dt} \Rightarrow \frac{di}{dt} + 5i = 10$

$$\tau = \frac{1}{5}$$

From initial, final conditions:

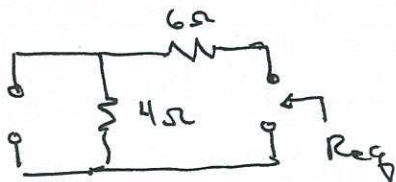
$$\left. \begin{array}{l} y(0) = K_1 + K_2 = 0 \text{ A} \\ y(\infty) = K_1 = 2 \text{ A} \end{array} \right\} \Rightarrow \begin{array}{l} \underline{K_1 = 2 \text{ A}} \\ \underline{K_2 = -2 \text{ A}} \end{array}$$

Solution is:

$$i(t) = 2(1 - e^{-5t}) \text{ A}$$

Check results based on circuit (Not required, but good!)

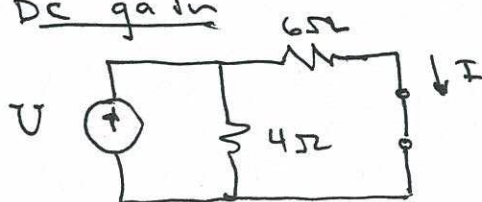
• Time constant: $\tau = \frac{L}{R_{eq}}$



$$\Rightarrow R_{eq} = 10 \Omega$$

$$\tau = \frac{2 \text{ H}}{10 \Omega} = \frac{1}{5} \text{ sec} \checkmark$$

• DC gain

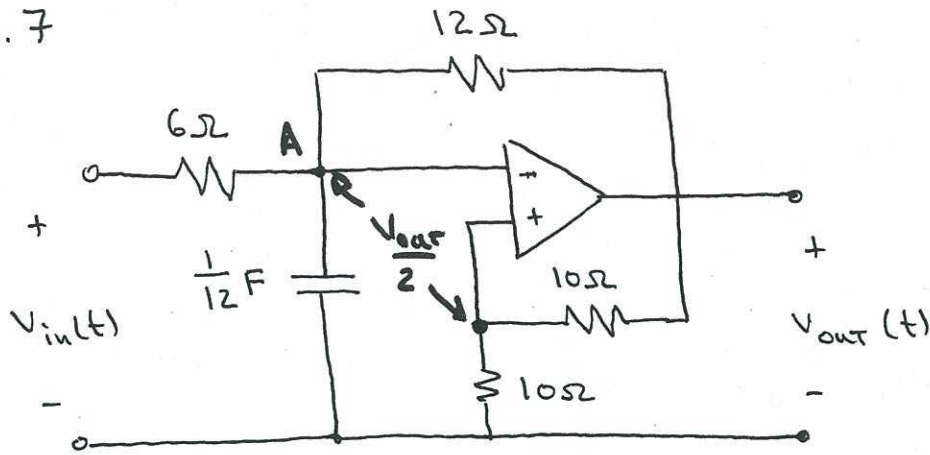


$$I = \frac{4}{6+4} U \Rightarrow \frac{I}{U} = \frac{4}{10} = \frac{2}{5}$$

$$\Rightarrow \text{Apply } 5 \text{ A input, } i(\infty) \rightarrow 2 \text{ A} \checkmark$$

Both consistent with above.

7.7



Voltages at both op-amp input nodes are $\frac{V_{out}}{2}$.

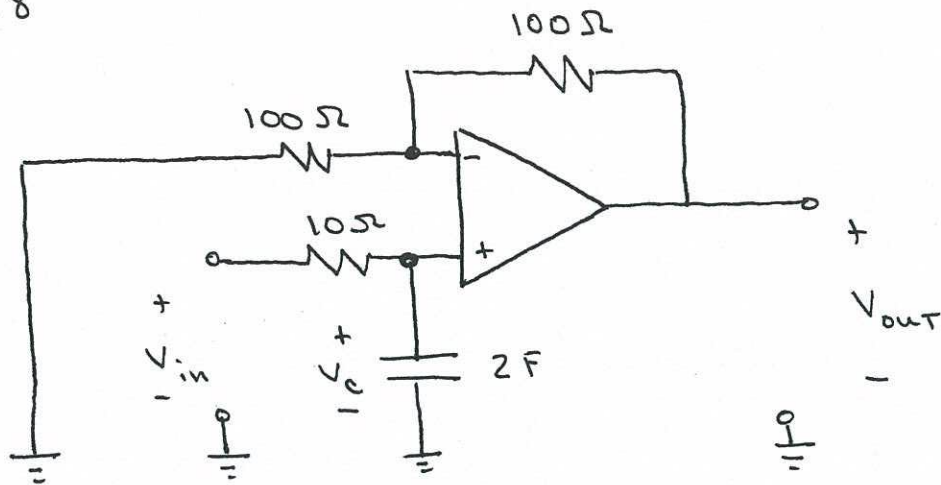
KCL at A:

$$\frac{V_{in} - \frac{V_{out}}{2}}{6\Omega} = \left(\frac{1}{12}F\right) \frac{d}{dt} \left[\frac{V_{out}}{2} \right] + \frac{\frac{V_{out}}{2} - V_{out}}{12\Omega}$$

$$2V_{in} - V_{out} = \frac{1}{2} \frac{dV_{out}}{dt} + \frac{1}{2} V_{out} - V_{out}$$

$$4V_{in} = \frac{dV_{out}}{dt} + V_{out}$$

7.8



KCL at non-inverting input:

$$\frac{V_{in} - V_c}{10\Omega} = (2F) \frac{dV_c}{dt}$$

Op-amp rules:

voltage at inverting input = voltage at non-inverting input = V_c

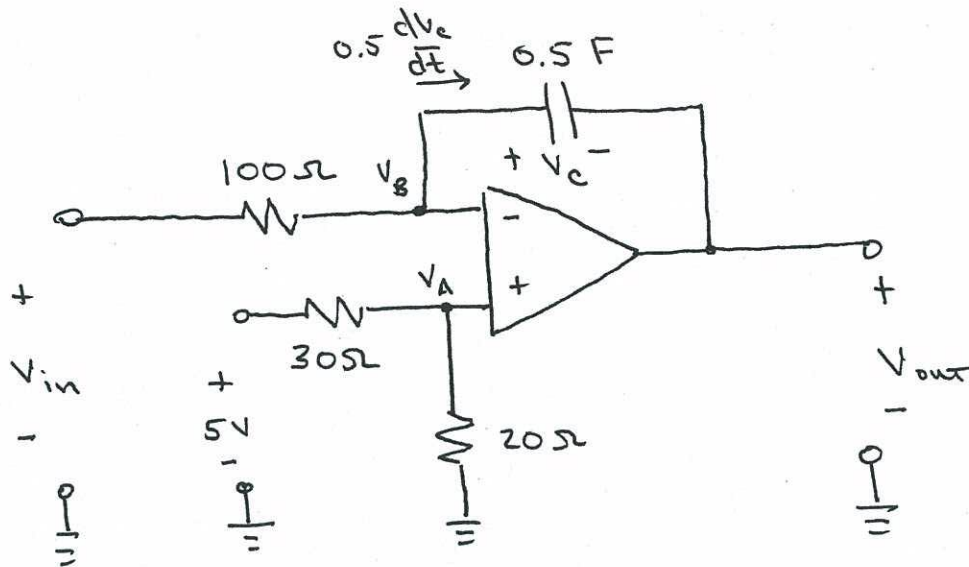
KCL at inverting input:

$$\frac{0 - V_c}{100\Omega} = \frac{V_c - V_{out}}{100\Omega} \Rightarrow V_{out} = 2V_c \Rightarrow V_c = \frac{V_{out}}{2}$$

$$\frac{V_{in} - \frac{V_{out}}{2}}{10\Omega} = (2F) \left[\frac{1}{2} \frac{dV_{out}}{dt} \right]$$

$$\boxed{\frac{V_{in}}{10} = \frac{V_{out}}{20} + \frac{dV_{out}}{dt}}$$

7.9



KCL at non-inverting input:

$$V_A = \frac{2}{5}(5\text{ V}) = 2\text{ V} \quad (\text{voltage divider})$$

Op-amp rules: $V_A = V_B = 2\text{ V}$

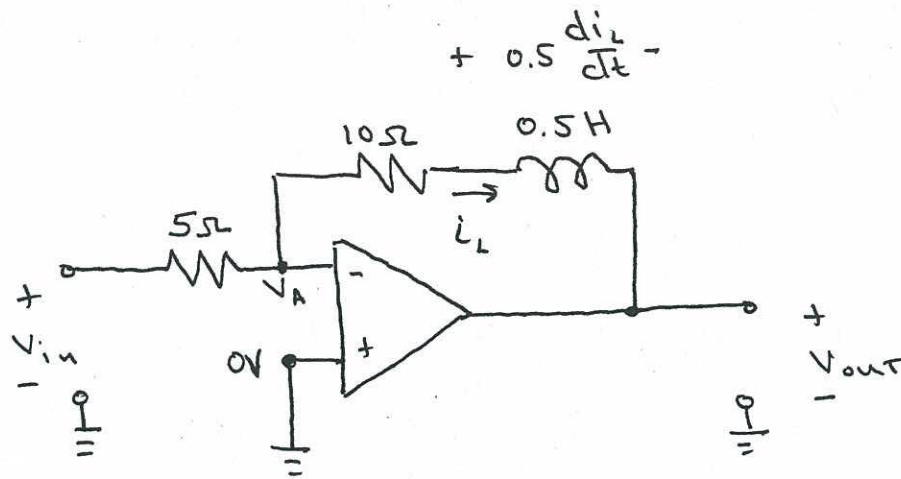
KCL at inverting input

$$\frac{V_{in} - V_B}{100\Omega} = 0.5 \frac{dV_c}{dt} = 0.5 \frac{d}{dt} [V_B - V_{out}]$$

$$V_{in} = V_B - 50 \frac{dV_{out}}{dt}$$

$$V_{in} = 2 - 50 \frac{dV_{out}}{dt}$$

7.10



Op-amp rules: $V_A = 0V$

KCL at inverting input:

$$\frac{V_{in} - V_A}{5\Omega} = i_L \Rightarrow i_L = \frac{V_{in}}{5}$$

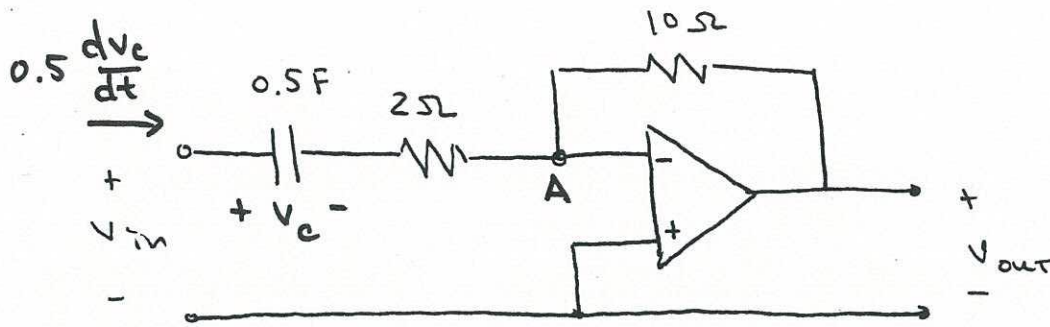
KVL, feedback loop:

$$\underbrace{V_A}_{0V} + (10\Omega)i_L + (0.5H)\frac{di_L}{dt} = -V_{out}$$

$$(10\Omega)\left(\frac{V_{in}}{5}\right) + (0.5H)\left[\frac{1}{5}\frac{dV_{in}}{dt}\right] = -V_{out}$$

$$\boxed{2V_{in} + 0.1\frac{dV_{in}}{dt} = -V_{out}}$$

7.11



Op-amp rules: $V_A = 0V$

KCL at node "A":

$$0.5 \frac{dV_c}{dt} = \frac{0V - V_{out}}{10\Omega} \Rightarrow 5 \frac{dV_c}{dt} = -V_{out} \Rightarrow \frac{dV_c}{dt} = -\frac{1}{5} V_{out}$$

KVL:

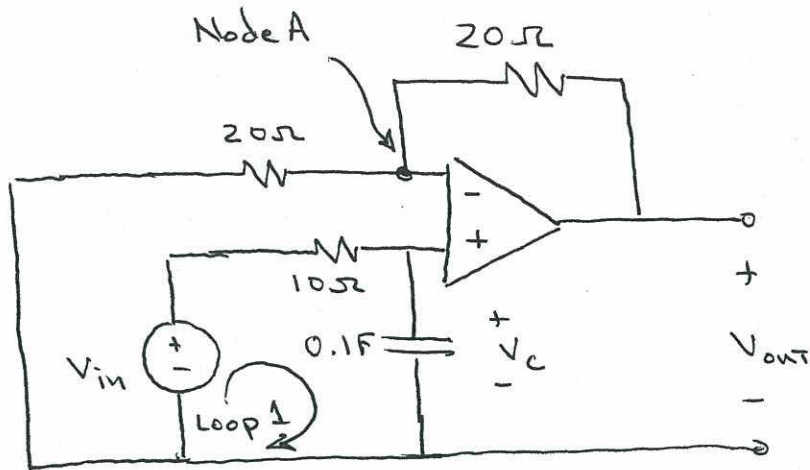
$$V_{in} = V_c + (2\Omega)(0.5 \frac{dV_c}{dt}) \Rightarrow V_{in} = V_c + \frac{dV_c}{dt}$$

$$\frac{dV_{in}}{dt} = \frac{dV_c}{dt} + \frac{d^2V_c}{dt^2}$$

$$\frac{dV_{in}}{dt} = \left(-\frac{1}{5} V_{out}\right) + \left(-\frac{1}{5} \frac{dV_{out}}{dt}\right)$$

$$\boxed{\frac{dV_{in}}{dt} = -\frac{1}{5} \left[V_{out} + \frac{dV_{out}}{dt} \right]}$$

7.12



Voltage at Node A is V_c (from op-amp rules)

KCL at A:

$$\frac{V_c - 0}{20\Omega} + \frac{V_c - V_{out}}{20\Omega} = 0 \Rightarrow 2V_c = V_{out} \Rightarrow V_c = \frac{V_{out}}{2}$$

KVL, Loop 1:

$$V_{in} = 10\Omega \left[(0.1F) \frac{dV_c}{dt} \right] + V_c \Rightarrow V_{in} = \frac{dV_c}{dt} + V_c$$

Combine:

$$\frac{dV_{out}}{dt} + V_{out} = 2V_{in}$$

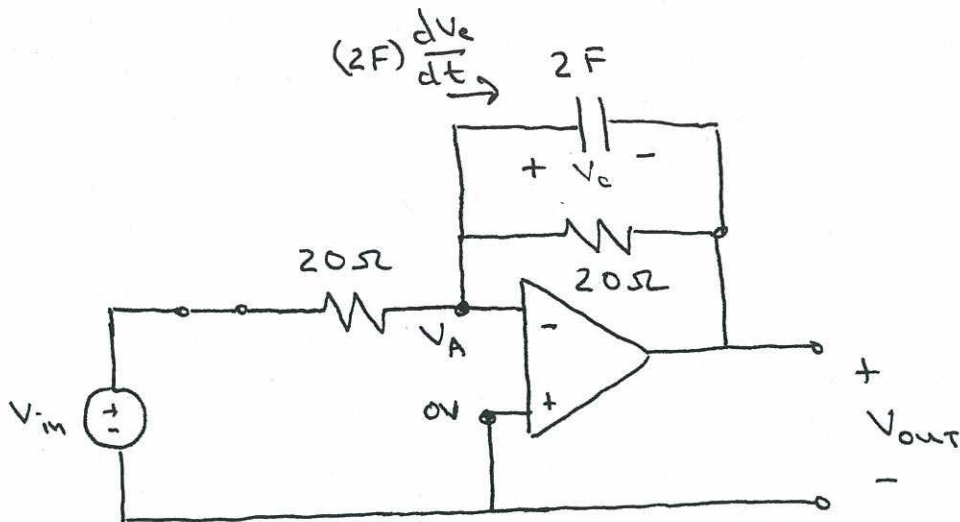
$$\tau = 1 ; V_{out}(t \rightarrow \infty) = 2V_{in} \Rightarrow$$

$$t=0: V_{out} = 0$$

$$\underline{\underline{V_{out} = 2V_{in} [1 - e^{-t}]}}$$

7.13

(a)

Op-amp rules : $V_A = 0V$

KCL at "A" :

$$\left. \begin{aligned} \frac{V_{in} - 0V}{20\Omega} &= \frac{0 - V_{out}}{20\Omega} + (2F) \frac{dV_c}{dt} \\ V_c &= 0 - V_{out} \end{aligned} \right\} \Rightarrow \boxed{V_{in} = -V_{out} - 40 \frac{dV_{out}}{dt}}$$

(b) Governing Diff. eqn. is first order with constant forcing function. Solution is of the form:

$$V_{out}(t) = K_1 + K_2 e^{-t/\tau}$$

Equation from (a) can be written: $-\frac{1}{40} V_{in} = \frac{1}{40} V_{out} + \frac{dV_{out}}{dt}$

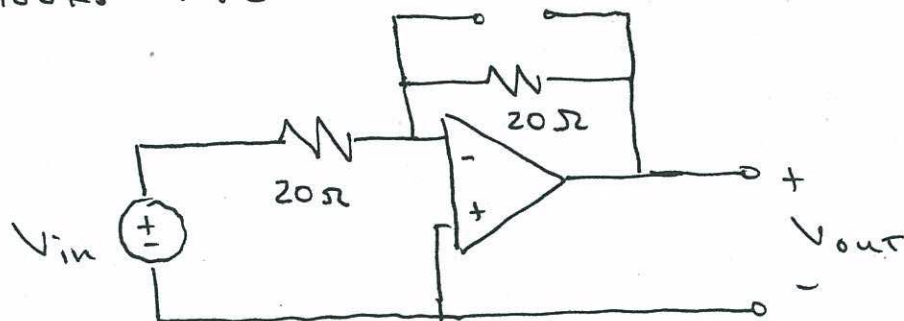
$$\Rightarrow \tau = 40 \text{ seconds}$$

Initial condition: $V_{out}(t=0) = 0V$
 Final condition: $V_{out}(t \rightarrow \infty) = -V_{in}$

$$\boxed{V_{out}(t) = -V_{in} [1 - e^{-t/40}]}$$

7.13 (cont'd)

(c) For constant forcing function, as $t \rightarrow \infty$ the capacitor behaves like an open-circuit. Thus, as $t \rightarrow \infty$, the circuit looks like:



This is an inverting voltage amplifier with gain = 1. Thus,

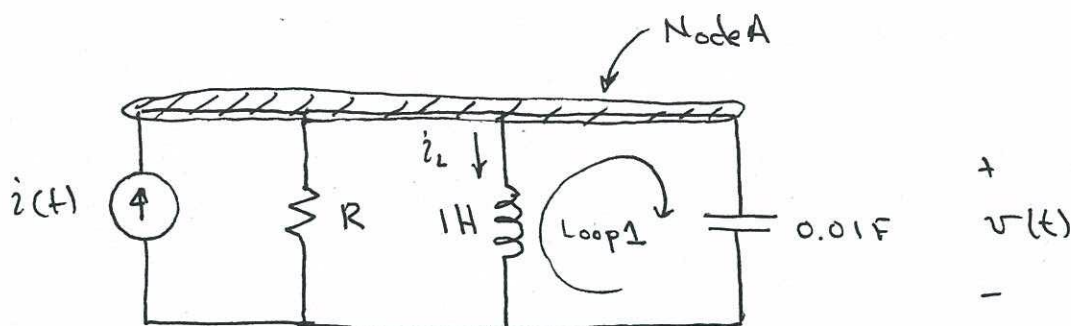
$$V_{out} = -V_{in}$$

From part (b), as $t \rightarrow \infty$:

$$V_{out}(\infty) = -V_{in} [1 - e^{-\infty/40}] = -V_{in}$$

They agree!

8.1



(a) Define the current through the inductor = $i_L(t)$

$$\text{KCL at node A: } i(t) = \frac{v(t)}{R} + i_L(t) + (0.01F) \frac{dv(t)}{dt} \quad (*)$$

$$\text{KVL, loop 1: } v(t) = (1H) \frac{di_L(t)}{dt} \quad (**)$$

Differentiate equation (*) with respect to time:

$$\frac{di(t)}{dt} = \frac{1}{R} \frac{dv(t)}{dt} + \frac{di_L(t)}{dt} + 0.01 \frac{d^2v(t)}{dt^2}$$

substitute equation (**):

$$100 \frac{di(t)}{dt} = \frac{100}{R} \frac{dv(t)}{dt} + 100 v(t) + \frac{d^2v(t)}{dt^2}$$

(b) From above, $\omega_n^2 = 100 \Rightarrow \omega_n = 10 \text{ rad/sec}$

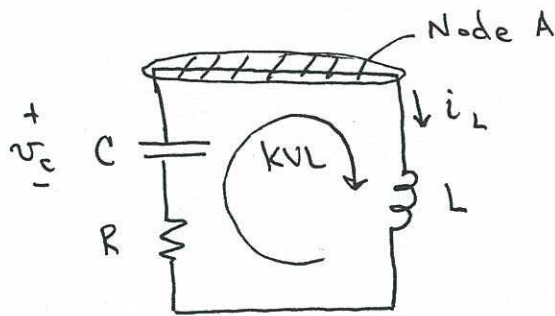
$$2\zeta\omega_n = \frac{100}{R} \Rightarrow 2(1)(10 \text{ rad/s}) = \frac{100}{R} \Rightarrow R = \underline{\underline{5}}$$

↑
 $\zeta=1$, for critical damping

$$R = 5\Omega$$

8.2

After switch opens, circuit becomes:



Define two unknowns,
 $v_c \neq i_L$

(a) KCL, Node A: $i_L = -C \frac{dv_c}{dt}$

KVL: $v_c + RC \frac{dv_c}{dt} = L \frac{di_L}{dt}$

$$\frac{d^2 v_c}{dt^2} + \frac{R}{L} \frac{dv_c}{dt} + \frac{1}{LC} v_c = 0$$

With $R = 2\Omega$, $C = 1\mu F$, $L = 1mH$:

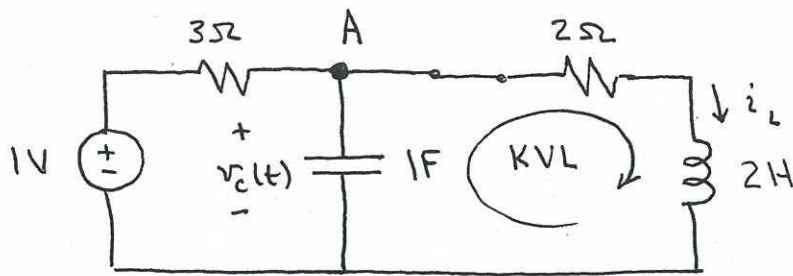
$$\frac{d^2 v_c}{dt^2} + 2000 \frac{dv_c}{dt} + 1 \times 10^9 v_c = 0$$

(b) $\omega_n = \sqrt{1 \times 10^9} = 3.16 \times 10^4 \text{ rad/sec}$

$$\xi = \frac{2000}{2\omega_n} = 0.0316$$

$\xi < 1 \Rightarrow$ System is underdamped

8.3



(a) KCL at A: $\frac{1-v_c}{3\Omega} = (1F) \frac{dv_c}{dt} + i_L \Rightarrow 1-v_c = 3 \frac{dv_c}{dt} + 3i_L \quad (*)$

KVL: $v_c = (2\Omega) i_L + (2H) \frac{di_L}{dt} \quad (**)$

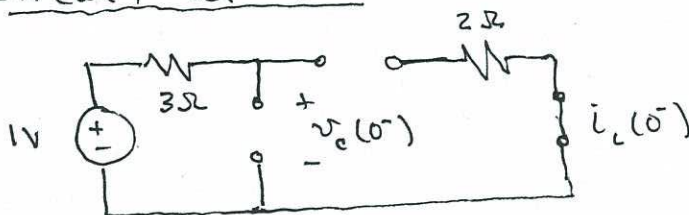
Subst. tute (**) into (*):

$$1 = [2i_L + 2 \frac{di_L}{dt}] + 3 \left[2 \frac{di_L}{dt} + 2 \frac{d^2 i_L}{dt^2} \right] + 3i_L$$

$$1 = 5i_L + 8 \frac{di_L}{dt} + 6 \frac{d^2 i_L}{dt^2}$$

(b) Initial condition

Circuit at $t=0^-$:

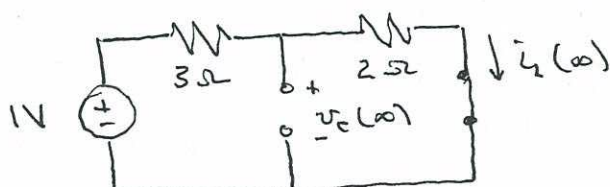


$$v_c(0) = 1V$$

$$i_L(0) = 0A$$

Final Condition:

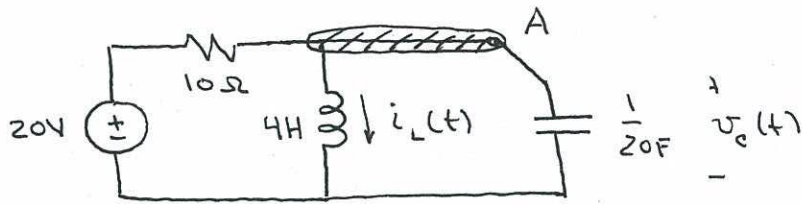
Circuit at $t \Rightarrow \infty$:



$$v_c(\infty) = \frac{2\Omega}{2\Omega + 3\Omega} = \frac{2}{5} V$$

$$i_L(\infty) = \frac{1V}{5\Omega} = \frac{1}{5} A$$

8.4

(a) For $t > 0$, circuit becomes:KVL, right-most loop: $v_c = 4 \frac{di_L}{dt}$ KCL, node A: $\frac{20 - v_c}{10\Omega} = i_L + \frac{1}{20} \frac{dv_c}{dt}$

Combining:

$$2 - 4 \frac{di_L}{dt} = i_L + \frac{1}{20} \left(4 \frac{d^2 i_L}{dt^2} \right)$$

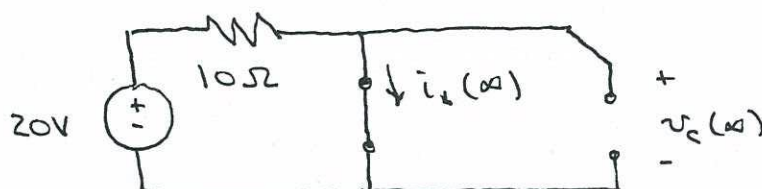
$$2 = \frac{1}{5} \frac{d^2 i_L}{dt^2} + 4 \frac{di_L}{dt} + i_L \Rightarrow \boxed{\frac{d^2 i_L}{dt^2} + 20 \frac{di_L}{dt} + 5 i_L = 10}$$

(b)

Initial condition:

$$i_L(0) = \frac{20V}{10\Omega} = 2A$$

$$v_c(0) = 4V$$

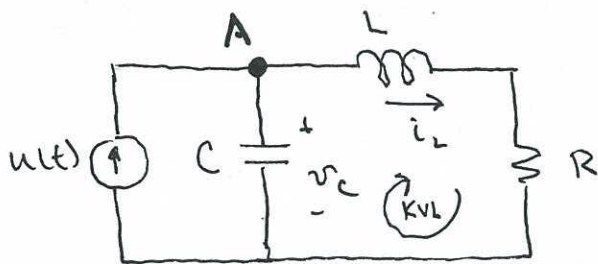
Final condition:

$$v_c(t \rightarrow \infty) = 0V$$

$$i_L(t \rightarrow \infty) = \frac{20V}{10\Omega} = 2A$$

8.5

For generic R, L, C : (we'll plug in specific values later)



- (a) Defining unknowns v_c, i_L as shown & applying KVL & KCL:

$$\text{KCL at A: } u = C \frac{dv_c}{dt} + i_L$$

$$\text{KVL, right loop: } v_c = L \frac{di_L}{dt} + R i_L$$

$$u = \frac{d^2 i_L}{dt^2} + \frac{R}{L} \frac{di_L}{dt} + \frac{1}{LC} i_L$$

with $R = 4\Omega, L = 1\text{H}, C = 0.01\text{F}$ & noting that $i = i_L$:

$$100u = \frac{d^2 i_L}{dt^2} + 4 \frac{di_L}{dt} + 100 i_L$$

- (b) From coefficients of differential equation of part (a):

$$\omega_n = \sqrt{100} = 10 \text{ rad/sec}$$

$$2\zeta\omega_n = 4 \Rightarrow 2\zeta(10) = 4 \Rightarrow \zeta = \frac{4}{20}$$

$$\omega_n = 10 \text{ rad/s}$$

$$\zeta = 0.2$$

↓ continued

8.5 (cont'd)

(c)

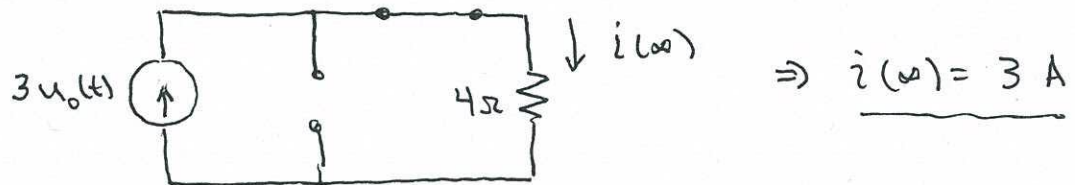
$$M_p \approx 1 - \frac{z}{0.6} = 1 - \frac{0.2}{0.6}$$

$\zeta = 0.2$ From part (a)

$$\underline{M_p = 0.67}$$

steady-state response:

Circuit as $t \rightarrow \infty$, with $u(t) = 3u_0(t)$ A:



Maximum value:

$$i_{\max} = 3A + 0.67(3A)$$

$$\boxed{i_{\max} = 5A}$$

8.6

$$\frac{d^2 i(t)}{dt^2} + 6 \frac{di(t)}{dt} + 100 i(t) = 50 u_0(t)$$

$$(a) \quad \omega_n = \sqrt{100} \Rightarrow \boxed{\omega_n = 10 \text{ rad/sec}}$$

$$2\zeta\omega_n = 6 \Rightarrow \zeta = \frac{6}{2(10)} \Rightarrow \boxed{\zeta = 0.3}$$

$$\omega_d = \omega_n \sqrt{1-\zeta^2} = 10\sqrt{1-0.3^2} \Rightarrow \boxed{\omega_d = 9.54 \text{ rad/s}}$$

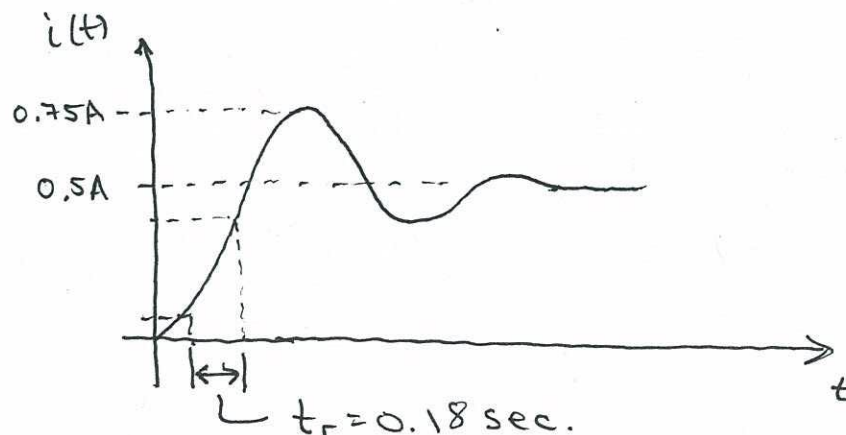
(b) Determine steady-state response, t_r , maximum value:

steady-state response (from diff. eqn):

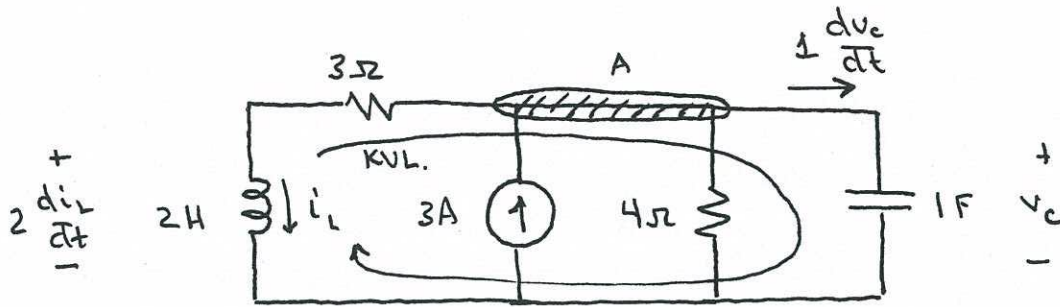
$$\cancel{\frac{d^2 i_{ss}}{dt^2}} + 6 \cancel{\frac{di_{ss}}{dt}} + 100 i_{ss} = 50 \Rightarrow i_{ss} = \frac{50}{100} = 0.5 \text{ A}$$

$$M_p \approx 1 - \frac{\zeta}{0.6} = 1 - \frac{0.3}{0.6} = 0.5 \Rightarrow i_{\text{peak}} = 0.75 \text{ A}$$

$$t_r \approx \frac{1.8}{\omega_n} = 0.18 \text{ sec}$$



8.7

(a) circuit for $t > 0$:

KCL, node A: $3A = i_L + \frac{v_C}{4\Omega} + \frac{dv_C}{dt}$

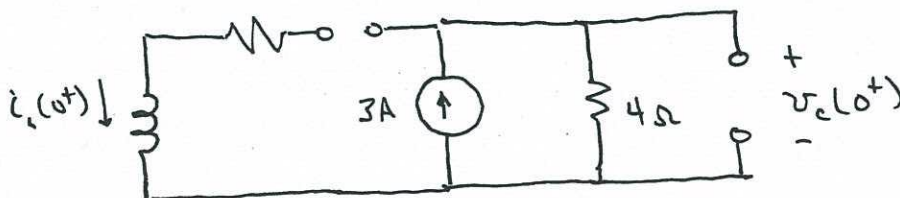
KVL, loop shown: $2 \frac{di_L}{dt} + 3i_L = v_C$

$$\frac{dv_C}{dt} = 2 \frac{d^2 i_L}{dt^2} + 3 \frac{di_L}{dt}$$

Substituting:

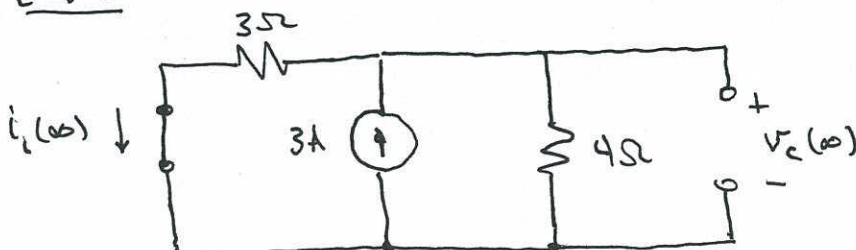
$$3A = i_L + \frac{1}{4} \left[2 \frac{di_L}{dt} + 3i_L \right] + \left[2 \frac{d^2 i_L}{dt^2} + 3 \frac{di_L}{dt} \right]$$

$$2 \frac{d^2 i_L}{dt^2} + \frac{7}{2} \frac{di_L}{dt} + \frac{7}{4} i_L = 3$$

(b) $t = 0^-$:

$$i_L(0) = 0A$$

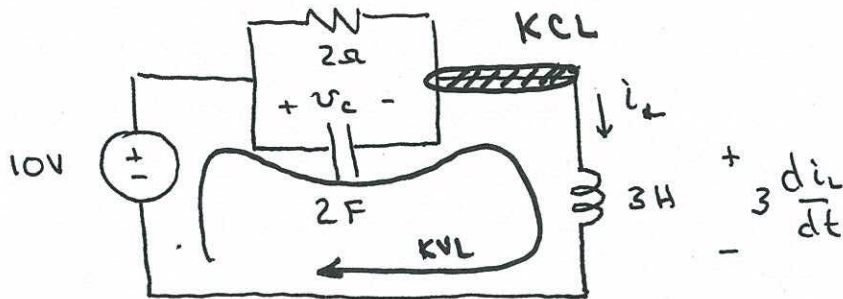
$$v_C(0) = 12V$$

 $t \rightarrow \infty$:

$$i_L(\infty) = \frac{12}{7}A$$

$$v_C(\infty) = \frac{36}{7}V$$

8.8

(a) circuit for $t > 0$ 

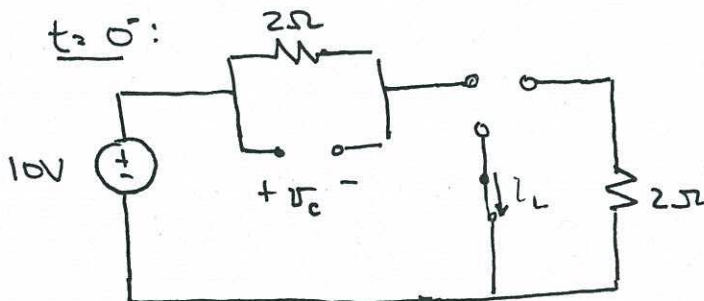
KVL as shown: $10V = v_c + 3 \frac{di_L}{dt}$

KCL as shown: $\frac{v_c}{2\Omega} + (2F) \frac{dv_c}{dt} = i_L \Rightarrow \frac{di_L}{dt} = 2 \frac{d^2 v_c}{dt^2} + \frac{1}{2} \frac{dv_c}{dt}$

Substituting:

$$10V = v_c + 3 \left[2 \frac{d^2 v_c}{dt^2} + \frac{1}{2} \frac{dv_c}{dt} \right]$$

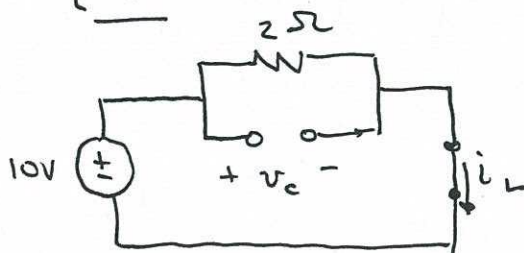
$$10 = v_c + \frac{3}{2} \frac{dv_c}{dt} + 2 \frac{d^2 v_c}{dt^2}$$

(b) $t = 0^-$:

$$v_c(0^+) = v_c(0^-) = 10V \left(\frac{2\Omega}{2\Omega + 2\Omega} \right)$$

$$v_c(0^+) = 5V$$

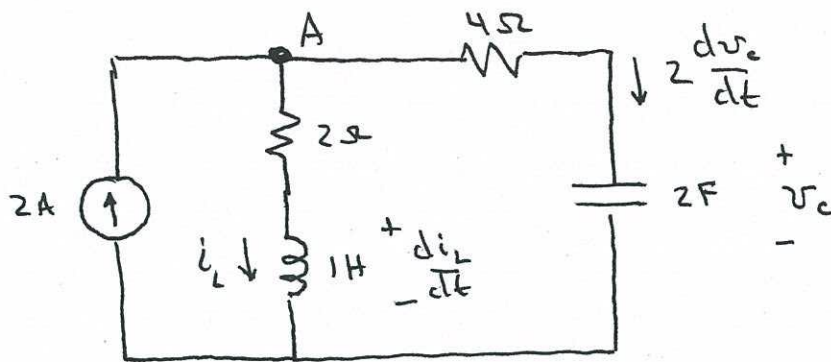
$$i_L(0^+) = 0A$$

 $t \rightarrow \infty$ 

$$v_c(t \rightarrow \infty) = 10V$$

$$i_L(t \rightarrow \infty) = \frac{10V}{2\Omega} = 5A$$

8.9

(a) Circuit for $t > 0$:

KCL at A: $2A = i_L + 2 \frac{dv_c}{dt}$ (*)

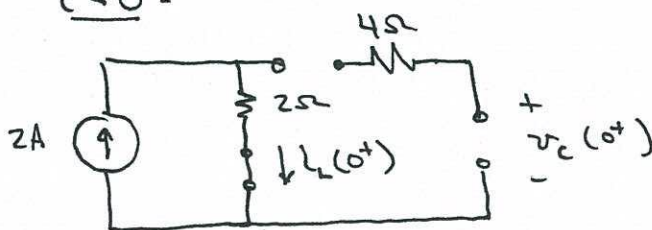
KVL, right loop: $\frac{di_L}{dt} + 2i_L = 4\Omega \left(2 \frac{dv_c}{dt}\right) + v_c$ (**)

From (*), $i_L = 2 - 2 \frac{dv_c}{dt}$ $\Rightarrow \frac{di_L}{dt} = -2 \frac{d^2v_c}{dt^2}$

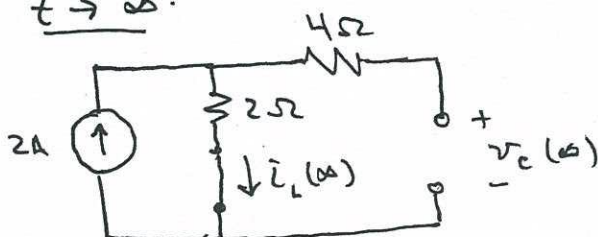
Substituting into (**):

$$-2 \frac{d^2v_c}{dt^2} + 2 \left[2 - 2 \frac{dv_c}{dt}\right] = 8 \frac{dv_c}{dt} + v_c$$

$$4 = 2 \frac{d^2v_c}{dt^2} + 12 \frac{dv_c}{dt} + v_c$$

(b) $t \leq 0$:

$$\begin{aligned} i_L(0) &= 2A \\ v_c(0) &= 0V \end{aligned}$$

 $t \rightarrow \infty$:

$$\begin{aligned} i_L(\infty) &= 2A \\ v_c(\infty) &= 4V \end{aligned}$$

8.10

Put differential equation in standard form (divide by 2):

$$\frac{d^2 v_{out}}{dt^2} + 2 \frac{dv_{out}}{dt} + 25 v_{out} = 50 v_{in}$$

$$\omega_n^2 = 25 \Rightarrow \omega_n = 5 \text{ rad/sec}$$

$$2\zeta\omega_n = 2 \Rightarrow \zeta = \frac{1}{\omega_n} = 0.2$$

$$M_p \approx 1 - \frac{\zeta}{0.6} = 0.67$$

constant input $\Rightarrow \frac{dv_{out}}{dt} = 0, \frac{d^2 v_{out}}{dt^2} = 0$ as $t \rightarrow \infty$

$$25 v_{out}(t \rightarrow \infty) = 50 v_{in}(t \rightarrow \infty)$$

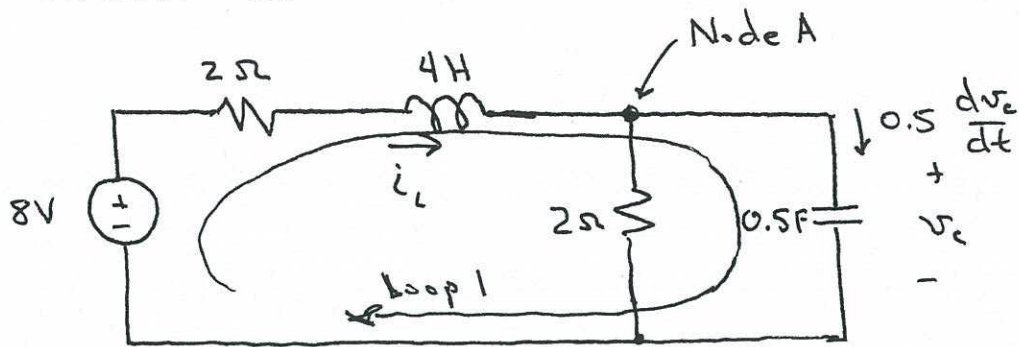
$$v_{out}(t \rightarrow \infty) = \frac{50}{25} v_{in}(t \rightarrow \infty) = \frac{50}{25} (2V)$$

$$v_{out}(t \rightarrow \infty) = 4V$$

$$v_{out} \Big|_{\max} = v_{out}(t \rightarrow \infty) [1 + M_p] = 4V [1 + 0.67]$$

$$\boxed{v_{out} \Big|_{\max} = 6.67V}$$

8.11

Circuit for $t > 0$:

$$\text{KVL, loop 1: } -8V + (2\Omega)i_L + (4H)\frac{di_L}{dt} + v_c = 0$$

$$\text{KCL, Node A: } i_L = \frac{v_c}{2\Omega} + \frac{1}{2} \frac{dv_c}{dt}$$

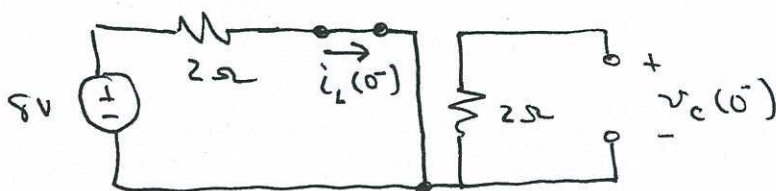
$$\downarrow \frac{d}{dt}$$

$$\frac{di_L}{dt} = \frac{1}{2} \frac{dv_c}{dt} + \frac{1}{2} \frac{d^2 v_c}{dt^2}$$

Substituting:

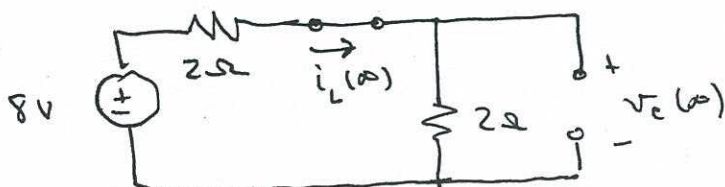
$$8 = 2 \left[\frac{v_c}{2} + \frac{1}{2} \frac{dv_c}{dt} \right] + 4 \left[\frac{1}{2} \frac{dv_c}{dt} + \frac{1}{2} \frac{d^2 v_c}{dt^2} \right]$$

$$\boxed{8 = 2v_c + 3 \frac{dv_c}{dt} + 2 \frac{d^2 v_c}{dt^2}}$$

(b) $t = 0^-$:

$$v_c(t=0^+) = v_c(0^-) = 0V$$

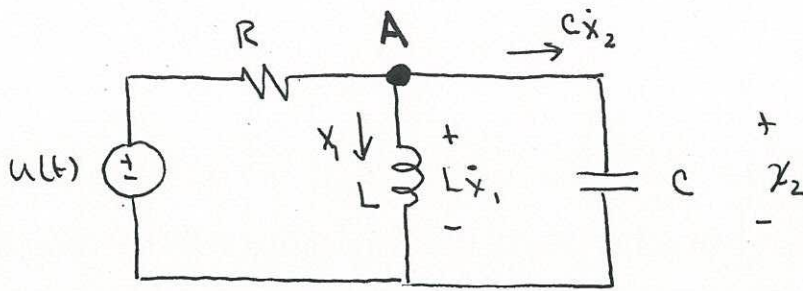
$$i_L(t=0^+) = i_L(0^-) = 4A$$

 $t \rightarrow \infty$:

$$v_c(t \rightarrow \infty) = 4V$$

$$i_L(t \rightarrow \infty) = 2A$$

9.1



- (1) Define state variables $x_1(t)$ & $x_2(t)$ as shown above
- (2) Label voltage across inductor ($L\dot{x}_1$) and current through capacitor ($C\dot{x}_2$)

$$\text{KCL at A: } \frac{u - x_2}{R} = \dot{x}_1 + C\dot{x}_2 \Rightarrow \underline{\dot{x}_2 = \frac{u}{RC} - \frac{1}{RC}x_2 - \frac{1}{C}x_1}$$

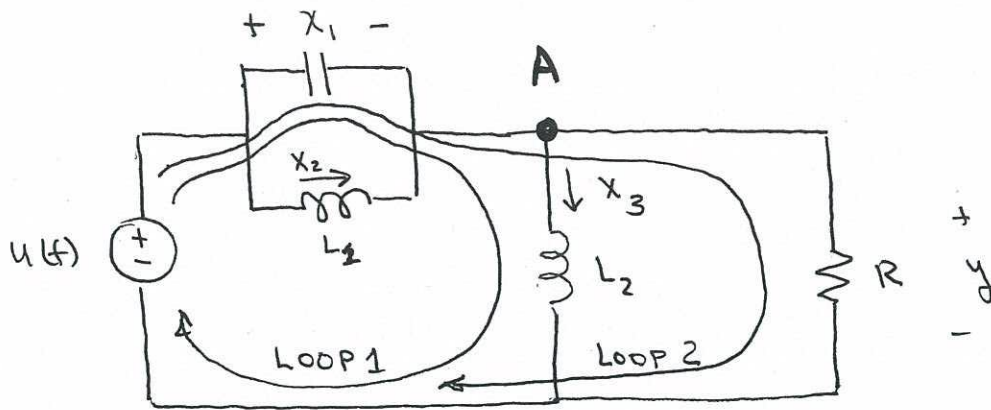
$$\text{KVL, right most loop: } L\dot{x}_1 = x_2 \Rightarrow \underline{\dot{x}_1 = \frac{1}{L}x_2}$$

$$\text{output equation: } y = x_2$$

Matrix form:

$$A = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix}, \quad \underline{b} = \begin{bmatrix} 0 \\ \frac{1}{RC} \end{bmatrix}, \quad \underline{c} = [0 \quad 1], \quad d=0$$

9.2



$$\text{KVL, Loop 1: } u = x_1 + L_2 \dot{x}_3 \Rightarrow \underline{\dot{x}_3 = -\frac{1}{L_2} x_1 + \frac{1}{L_2} u}$$

$$\text{KCL at A: } x_2 + C \dot{x}_1 = x_3 + \frac{y}{R}$$

$$\text{KVL, Loop 2: } u - x_1 - y = 0 \Rightarrow y = u - x_1$$

$$x_2 + C \dot{x}_1 = x_3 + \frac{u}{R} - \frac{x_1}{R}$$

$$\underline{\dot{x}_1 = -\frac{x_1}{RC} - \frac{x_2}{C} + \frac{x_3}{C} + \frac{u}{RC}}$$

Parallel inductor, L_1 , and capacitor:

$$x_1 = L_1 \dot{x}_2 \Rightarrow \underline{\dot{x}_2 = \frac{1}{L_1} x_1}$$

output equation (from KVL, Loop 2 above):

$$\underline{y = u - x_1}$$

↓ continued

9.2 (cont'd)

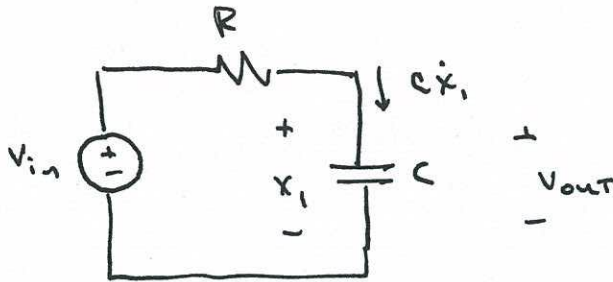
In matrix form:

$$A = \begin{bmatrix} -\frac{1}{RC} & -\frac{1}{C} & \frac{1}{C} \\ \frac{1}{L_1} & 0 & 0 \\ -\frac{1}{L_2} & 0 & 0 \end{bmatrix}, \quad \underline{b} = \begin{bmatrix} \frac{1}{RC} \\ 0 \\ \frac{1}{L_2} \end{bmatrix}$$

$$\underline{c} = [-1 \quad 0 \quad 0], \quad d = 1$$

9.3

(a)



$$\text{KVL: } V_{in}(t) = RC\dot{x}_1 + x_1 \Rightarrow \underline{\underline{\dot{x}_1 = -\frac{1}{RC}x_1 + \frac{1}{RC}V_{in}}}$$

$$\text{Output equation: } V_{out} = x_1$$

$$A = \left[-\frac{1}{RC} \right], \quad b = \frac{1}{RC}, \quad c = 1, \quad d = 0$$

(b) Matlab simulations with $R=470\Omega$, $C=1\mu F$, $V_{in}=4u_0(t)$:

Matlab commands:

```
R = 470;
C = 1e-6;
```

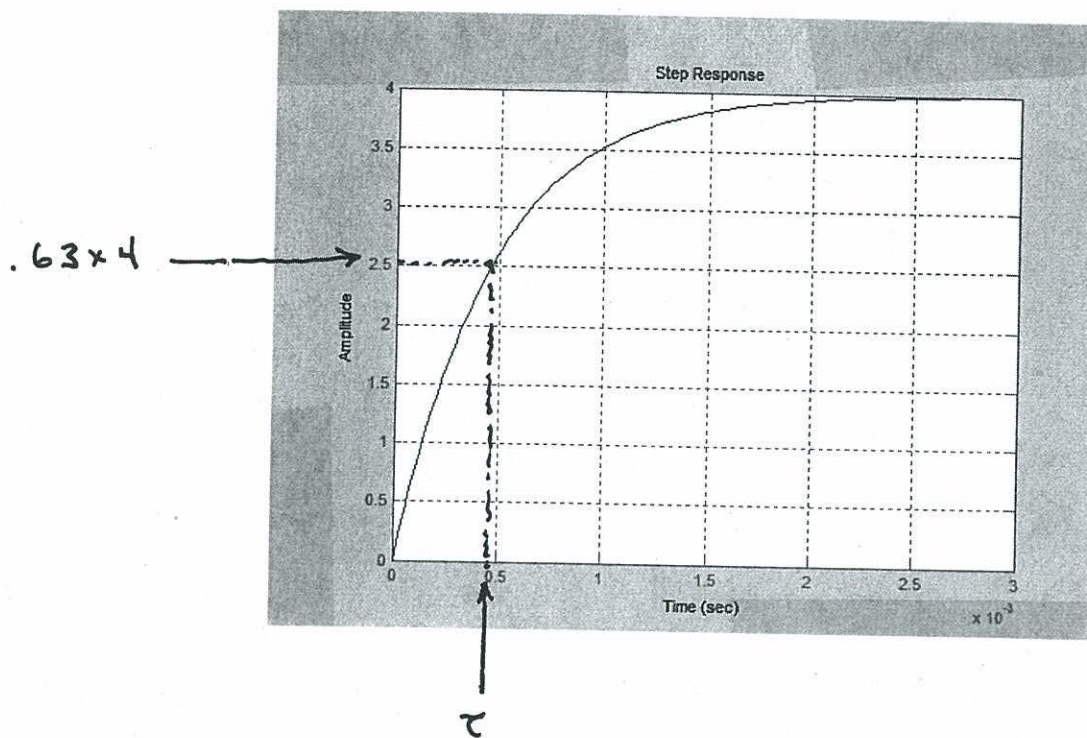
```
A = [-1/(R*C)];
b = [1/(R*C)];
c = 4*[1]; % scale output to correspond to input = 4*u_0(t)
d = 0;
```

```
sys = ss(A,b,c,d);
step(sys)
grid
```

{ continued

9.3 (cont'd)

(b) Simulated response (from MATLAB):



(c) From circuit :

$$\tau = R_{eq} C = (470 \Omega)(1 \times 10^{-6} \text{ F}) \approx 4.7 \text{ msec}$$

From plot above, $\tau \approx 4.8 \text{ msec}$

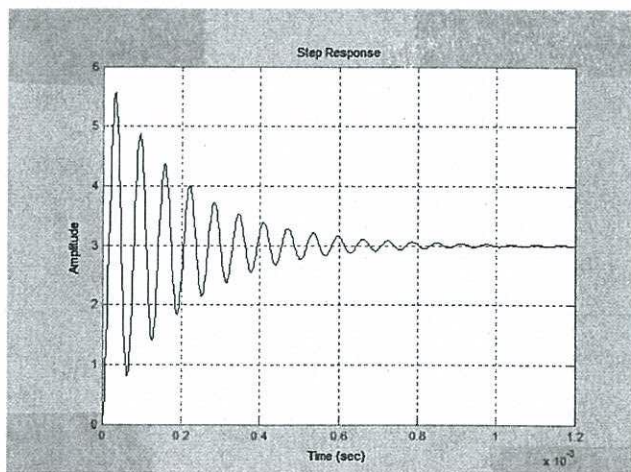
Agreement to within
the accuracy to which
I can read the graph

9.4 The state variable model for this problem was created in exercise 1 of chapter 9.1. We just need to plug in numbers and do the simulation.

Matlab commands:

```
R = 10;  
L = 1e-3;  
C = 100e-9;  
  
A = [-R/L -1/L; 1/C 0];  
b = [1/L; 0];  
c = 3*[0 1]; % scale output to account for Vin = 3*u0(t)  
d = 0;  
  
sys = ss(A,b,c,d);  
step(sys)
```

Resulting Figure:



↖ The scale on the default figure is not too useful for reading rise time and peak value.

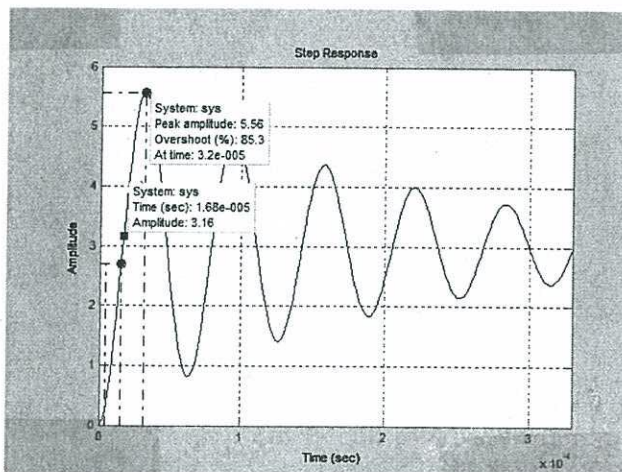
We can, however, modify it...

↓ (cont'd)

9.4 (cont'd)

- MATLAB makes it easy to zoom in on a portion of the figure. click on the magnifying glass on the toolbar, and use your mouse to select the section of the figure to enlarge.
- If the response was generated using MATLAB's "step" command, MATLAB will display certain parameters on the plot. Right-click on the figure, click on "characteristics" and select the value you want displayed.

I labeled the figure as shown below:



$\Leftarrow$ peak value = 5.56
 $t_r = 1.68 \times 10^{-5} \text{ sec}$

↓ (cont'd)

9.4 (st.11 continued)

(c) The input-output relation is:

$$\frac{d^2 V_{out}}{dt^2} + \frac{R}{L} \frac{dV_{out}}{dt} + \frac{1}{LC} V_{out} = \frac{1}{LC} V_{in}$$

$$\omega_n^2 = \frac{1}{LC} \Rightarrow \omega_n = \sqrt{\frac{1}{(.001)(100 \times 10^{-9})}} = 100,000 \text{ rad/s}$$

$$2\zeta\omega_n = \frac{R}{L} \Rightarrow \zeta = \frac{10}{2(.001)(100,000)} = 0.05$$

$$M_p \approx 1 - \frac{\zeta}{0.6} = 0.91$$

$$y_p = y_{ss} + y_{ss} \cdot M_p = 3V + (3V)(0.91) = \underline{5.73V}$$

$$t_r \approx \frac{1.8}{\omega_n} = \underline{1.8 \times 10^{-5} \text{ sec}}$$

These values are very close to the simulation values, considering the approximations used!

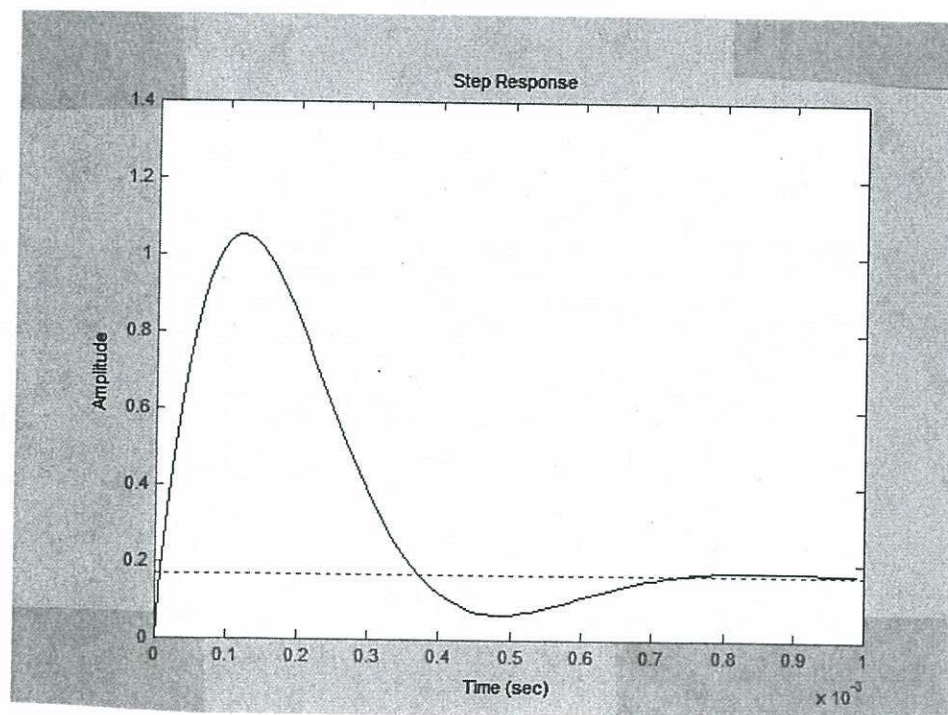
9.5 The state variable model for this circuit was created in exercise 2 of chapter 9.2. We just need to plug in numbers and run the simulation.

(b)

Matlab commands:

```
R1 = 10;  
R2 = 100;  
R3 = 1.1;  
L = 1e-3;  
C = 10e-6;  
  
A = [-1/R3/L 1/L; -1/C -1/C*(1/R1+1/R2)];  
b = [0; 1/(R1*C)];  
c = 2*[0 1]; % Scale to account for input = 2*u0(t)  
d = 0;  
  
sys = ss(A,b,c,d);  
step(sys)
```

Figure:



10.1

$$2 \frac{di}{dt} + 10i = 10v$$

$$v(t) = 3 e^{j5t}, \quad i(t) = \underline{I} e^{j5t}$$

$$2 [(j5) \underline{I} e^{j5t}] + 10 [\underline{I} e^{j5t}] = 30 e^{j5t}$$

$$\underline{I} = \frac{30}{10 + j10} = \frac{30}{10\sqrt{2}} \angle -45^\circ$$

$$\underline{i(t)} = \underline{\frac{30}{10\sqrt{2}} \cos(5t - 45^\circ)}$$

10.2

Governing differential equation:

$$3 \frac{di(t)}{dt} + 6 i(t) = v_s(t)$$

Represent inputs & outputs as complex exponentials:

$$v_s(t) = 5 e^{j30^\circ} e^{j4t}$$

$$i(t) = \underline{I} e^{j4t} \Rightarrow \frac{di(t)}{dt} = (j4) \underline{I} e^{j4t}$$

Substitute into governing diff. eqn:

$$3 [(j4) \underline{I} e^{j4t}] + 6 [\underline{I} e^{j4t}] = 5 e^{j30^\circ} e^{j4t}$$

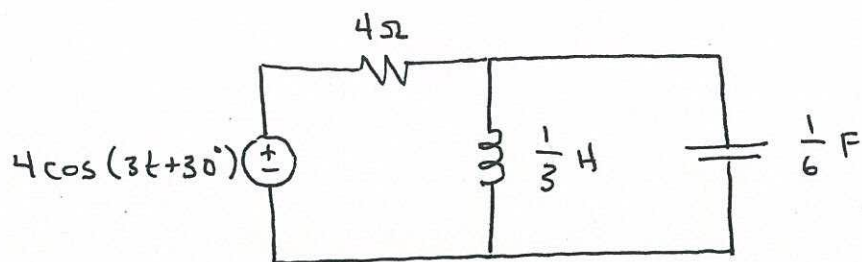
$$[j12 + 6] \underline{I} = 5 e^{j30^\circ}$$

$$\underline{I} = \frac{5 e^{j30^\circ}}{j12 + 6} = \frac{5 \angle 30^\circ}{6\sqrt{5} \angle +63^\circ}$$

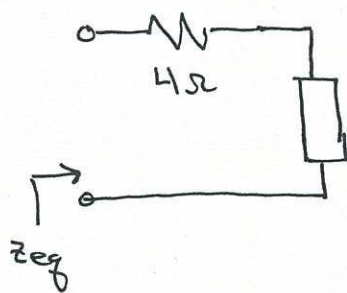
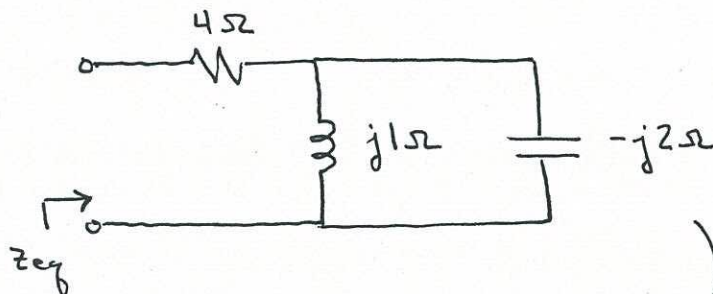
$$\underline{I} = \frac{\sqrt{5}}{6} \angle -33.4^\circ$$

$$i(t) = \frac{\sqrt{5}}{6} \cos(4t - 33.4^\circ)$$

10.3



(a)



$$z = \frac{(j1)(-j2)}{j1 - j2} = \frac{2}{-j} = j2$$

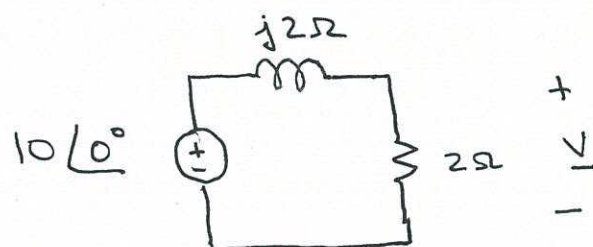
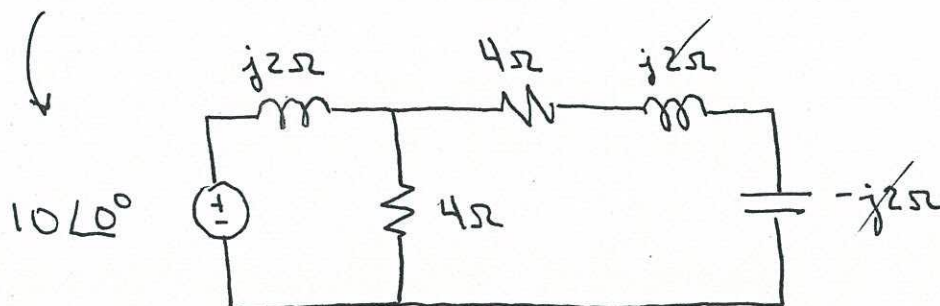
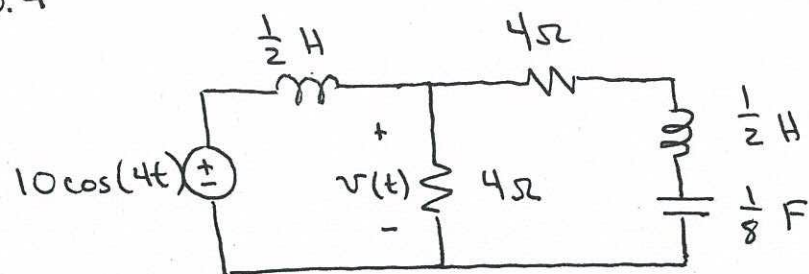
$$\underline{z_{eq}} = (4 + j2)\Omega$$

$$(b) \quad \underline{I_c} = \underbrace{\underline{V_s} \left(\frac{1}{4 + j2} \right)}_{\underline{I_s}} \underbrace{\left(\frac{j1}{j1 - j2} \right)}_{\text{current divider}} = 4 \angle 30^\circ \left(\frac{1}{4.47 \angle 26.6^\circ} \right) (-1)$$

$$\underline{I_c} = \frac{-4 \angle 30^\circ}{4.47 \angle 26.6^\circ} = -0.894 \angle 3.4^\circ$$

$$\underline{i_c(t)} = \underline{-0.894 \cos(3t + 3.4^\circ)}$$

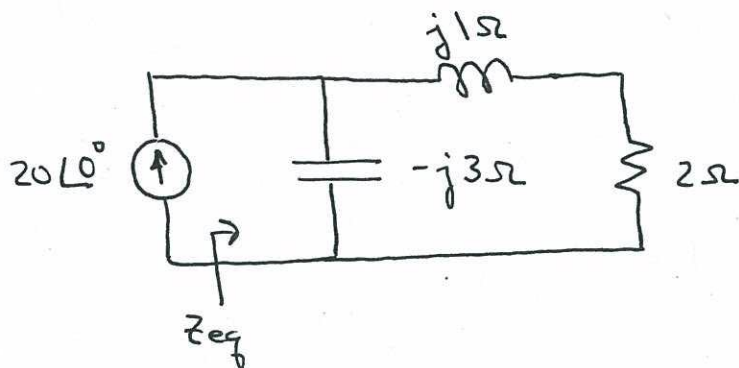
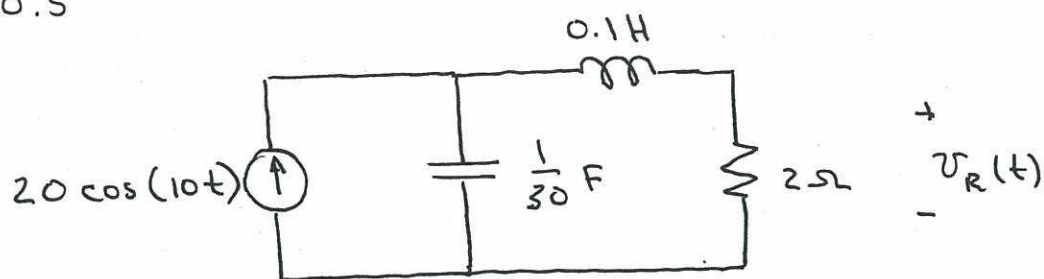
10.4



$$\underline{V} = \frac{2}{2+j2} (10 \angle 0^\circ) = \frac{20 \angle 0^\circ}{2\sqrt{2} \angle 45^\circ} = \frac{10}{\sqrt{2}} \angle -45^\circ$$

$$\underline{v(t)} = \underline{\underline{\frac{10}{\sqrt{2}} \cos(4t - 45^\circ)}}$$

10.5



(a)

$$Z_{eq} = \frac{-j3\Omega (2+j1)\Omega}{(2+j1)\Omega - j3\Omega} = \frac{(3-j6)\Omega}{(2-j2)} = \frac{3\sqrt{5} \angle -63^\circ}{2\sqrt{2} \angle -45^\circ}$$

$$\underline{Z_{eq} = \frac{3\sqrt{5}}{2\sqrt{2}} \angle -18^\circ}$$

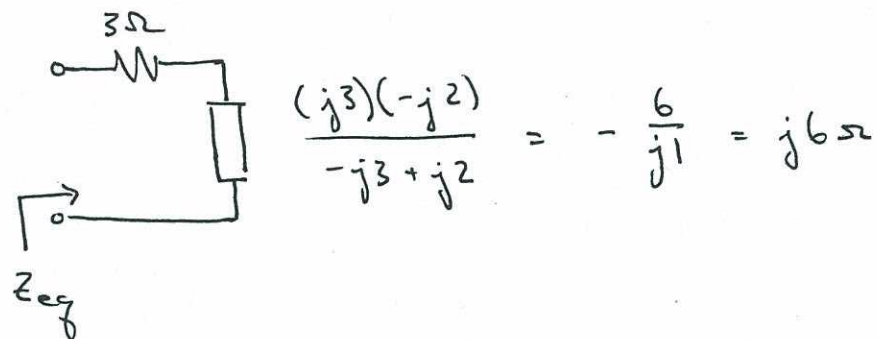
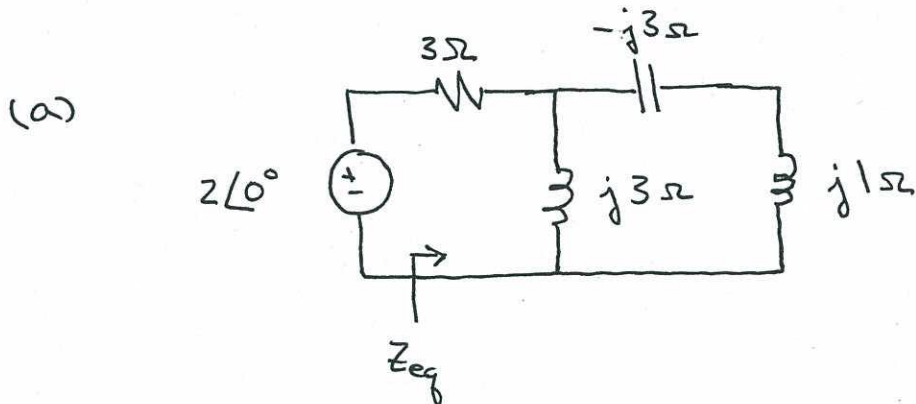
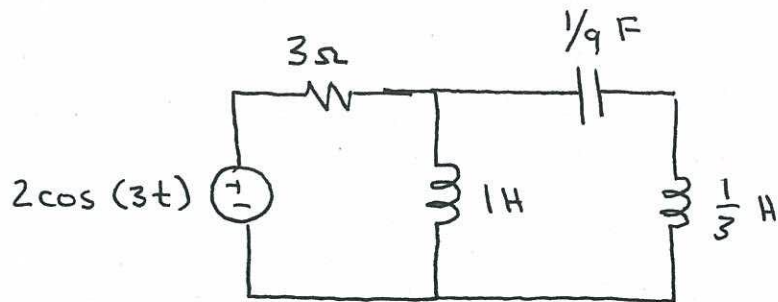
(b) Voltage divider

$$\underline{V_R} = (20 \angle 0^\circ) \underbrace{\left[\frac{-j3 (2+j1)}{2-j2} \right]}_{\text{voltage across current source}} \underbrace{\left(\frac{2}{2+j1} \right)}_{\text{voltage divider}}$$

$$\underline{V_R} = 20 \angle 0^\circ \left[\frac{-j6}{2-j2} \right] = 20 \angle 0^\circ \left[\frac{6 \angle -90^\circ}{2\sqrt{2} \angle -45^\circ} \right] = \frac{60}{\sqrt{2}} \angle -45^\circ$$

$$\underline{v_R(t) = \frac{60}{\sqrt{2}} \cos(10t - 45^\circ)}$$

10.6



$$\underline{Z_{eq}} = (3 + j6) \Omega$$

(b)

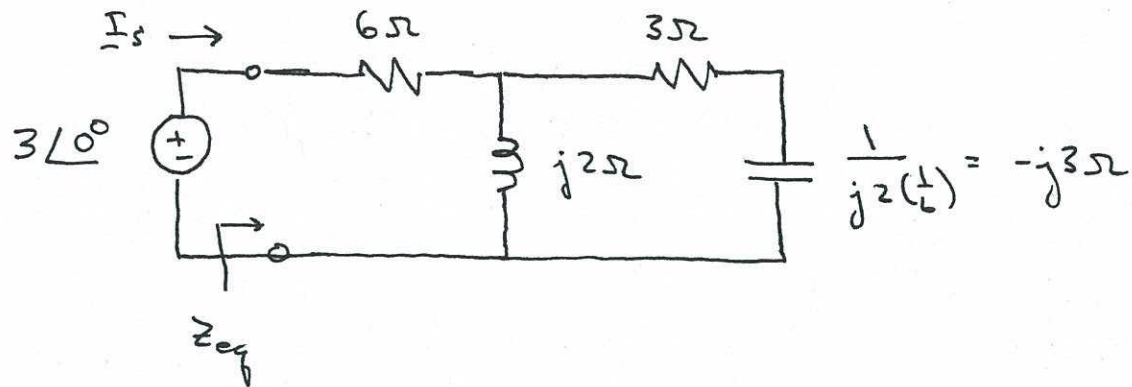
$$\underline{I_s} = \frac{2 \angle 0^\circ}{Z_{eq}} = \frac{2 \angle 0^\circ}{3 + j6} = \frac{2 \angle 0^\circ}{\sqrt{45} \angle 60^\circ}$$

$$\underline{I_s} = \frac{2}{3\sqrt{5}} \angle -60^\circ$$

$$\underline{i_s(t)} = \underline{\underline{\frac{2}{3\sqrt{5}} \cos(3t - 60^\circ)}}$$

10.7

(a) Frequency-domain circuit:



$$Z_{eq} = 6\Omega + \frac{(j2\Omega)(3-j3)\Omega}{j2\Omega + (3-j3)\Omega} = 6\Omega + \frac{j6\Omega + 6\Omega}{(3-j)\Omega}$$

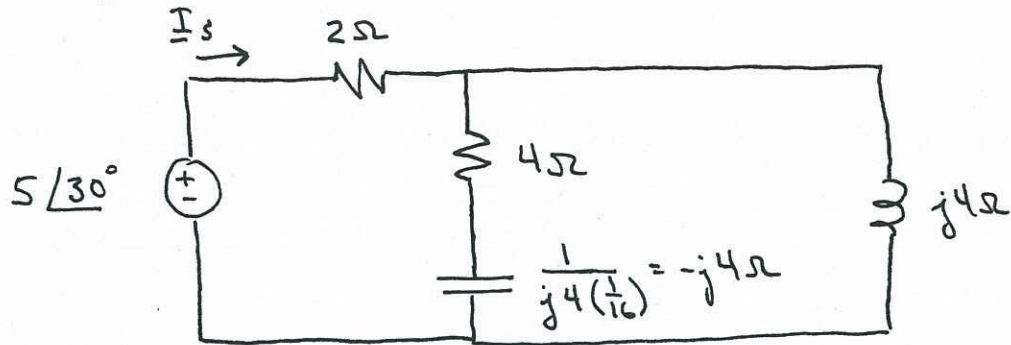
$$Z_{eq} = \frac{18-j6}{3-j} + \frac{6+j6}{3-j} = \boxed{\frac{24}{(3-j)} \Omega = Z_{eq}}$$

$$(b) \quad \underline{I_s} = \frac{\underline{V_s}}{Z_{eq}} = \frac{3\angle 0^\circ}{\frac{24}{3-j}} = \frac{3\sqrt{10}}{24} e^{-j18.4^\circ}$$

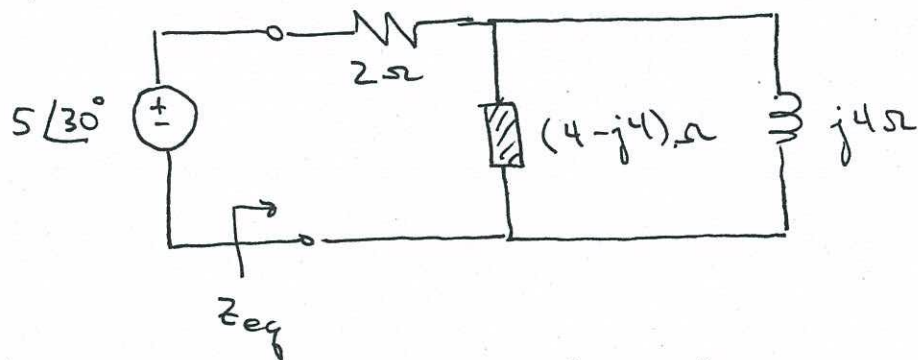
$$i_s(t) = \frac{\sqrt{10}}{8} \cos(2t - 18.4^\circ)$$

10.8

Frequency domain circuit:



(a) Equivalent circuit:



$$Z_{eq} = 2\Omega + \frac{(j4\Omega)(4-j4)\Omega}{j4\Omega + (4-j4)\Omega} = 2\Omega + \frac{j16 + 16}{4}\Omega$$

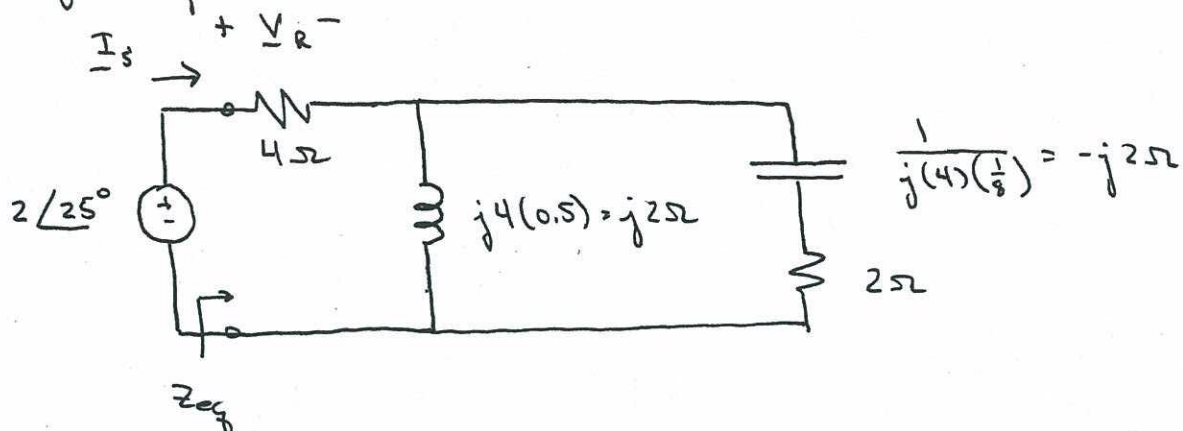
$$Z_{eq} = (j4 + 6)\Omega$$

$$(b) \quad \underline{I}_s = \frac{5\angle 30^\circ}{(6+j4)\Omega} = \frac{5\angle 30^\circ}{2\sqrt{13}\angle 33.7^\circ} = \frac{5}{2\sqrt{13}}\angle -3.7^\circ$$

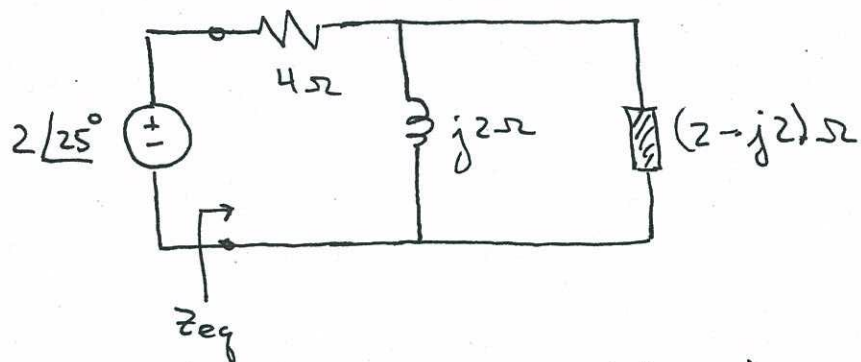
$$i_s(t) = \frac{5}{2\sqrt{13}} \cos(4t - 3.7^\circ)$$

10.9

Frequency domain circuit:



(a) Equivalent circuit:



$$Z_{eq} = 4\Omega + \frac{(j2)(2-j2)}{j2+2-j2} \Omega = (j2+6)\Omega$$

$$Z_{eq} = (j2+6)\Omega = 2\sqrt{10} \angle 18.4^\circ \Omega$$

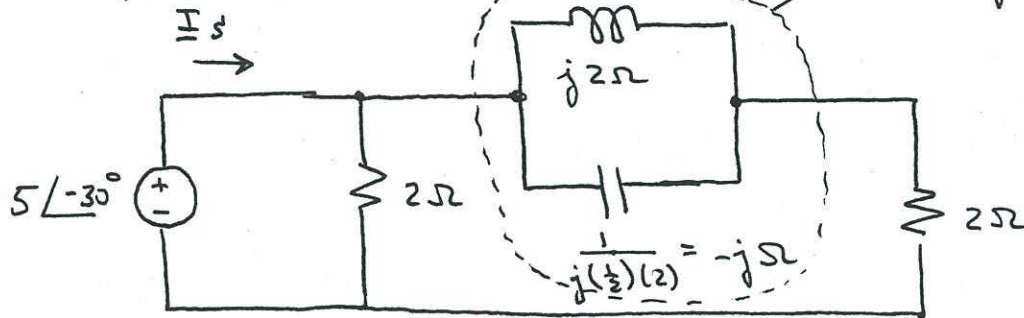
$$(b) \quad \underline{V}_R = (4\Omega)(\underline{I}_s) = (4\Omega) \left(\frac{2\angle 25^\circ}{2\sqrt{10} \angle 18.4^\circ} \right) = \frac{8\angle 25^\circ}{2\sqrt{10} \angle 18.4^\circ}$$

$$\underline{V}_R = \frac{4}{\sqrt{10}} \angle 6.57^\circ$$

$$v_R(t) = \frac{4}{\sqrt{10}} \cos(4t + 6.57^\circ)$$

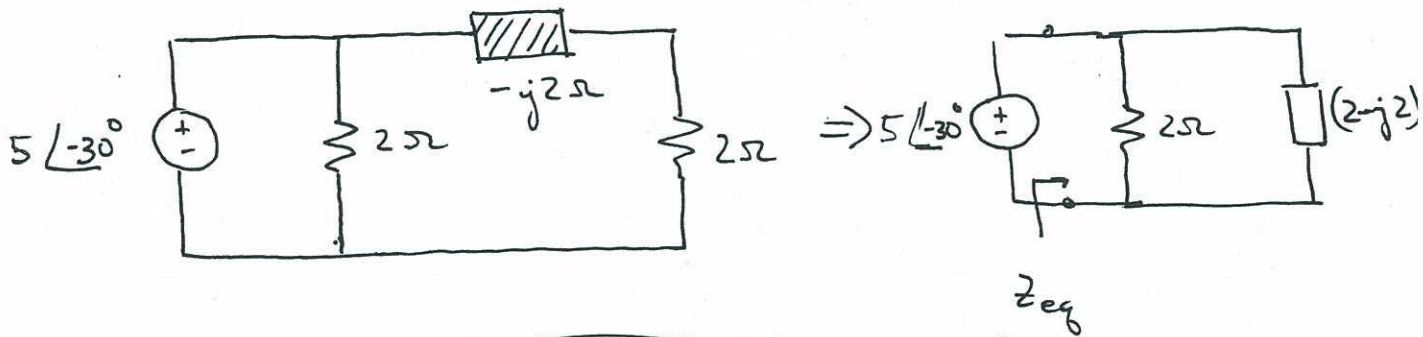
10.10

Frequency-domain circuit:



$$Z_{eq} = \frac{(-j)(j2)}{j2 - j} = \frac{2}{j} = -j2$$

Equivalent circuit:



$$Z_{eq} = \frac{(2\Omega)(2-j2)\Omega}{2\Omega + (2-j2)\Omega} = \frac{(4-j4)\Omega}{(4-j2)}$$

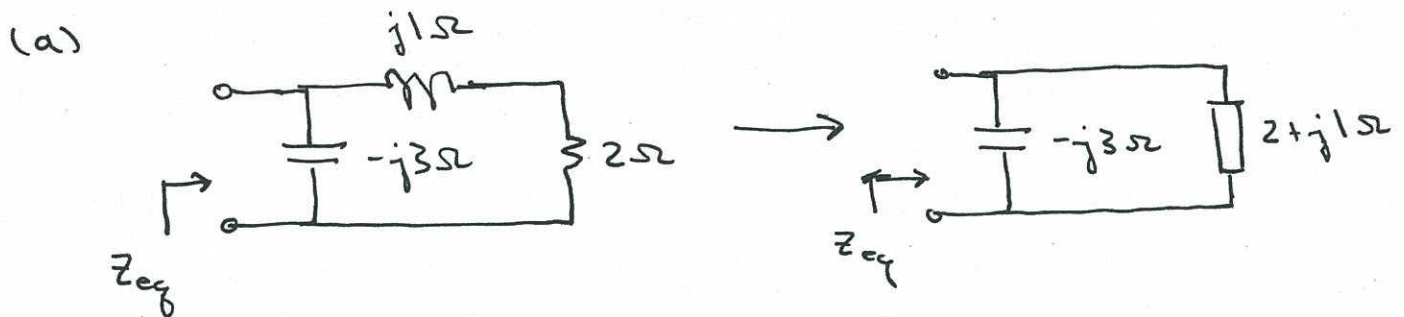
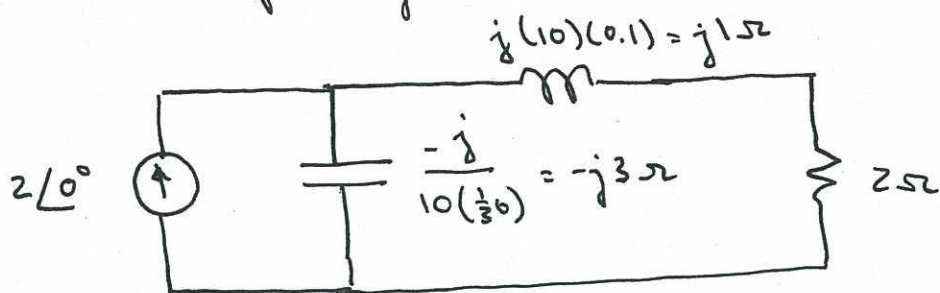
$$Z_{eq} = \frac{2\sqrt{2} \angle -45^\circ}{\sqrt{5} \angle -26.6^\circ} = \frac{2\sqrt{2}}{\sqrt{5}} \angle -18.4^\circ$$

$$\underline{I}_s = \frac{5 \angle -30^\circ}{Z_{eq}} = \frac{5 \angle -30^\circ}{\frac{2\sqrt{2}}{\sqrt{5}} \angle -18.4^\circ} = \frac{5\sqrt{5}}{2\sqrt{2}} \angle -11.6^\circ$$

$$i_s(t) = \frac{5\sqrt{5}}{2\sqrt{2}} \cos(2t - 11.6^\circ)$$

10.11

Frequency domain circuit:



$$Z_{eq} = \frac{(-j3\Omega)(2+j1\Omega)}{-j3\Omega + 2+j1\Omega} = \frac{3-j6}{2+j2} \Omega$$

(b) Probably easiest to use current divider:

$$\underline{I}_R = 2\angle 0^\circ \left[\frac{-j3\Omega}{-j3\Omega + 2\Omega + j1\Omega} \right] = 2\angle 0^\circ \left(\frac{-j3}{2+j2} \right)$$

$$\underline{I}_R = \frac{(2\angle 0^\circ)(3\angle -90^\circ)}{2\sqrt{2}\angle 45^\circ} = \frac{3}{\sqrt{2}} \angle -135^\circ$$

$$\underline{V}_R = (2\Omega) \underline{I}_R \Rightarrow \underline{V}_R = \frac{6}{\sqrt{2}} \angle -135^\circ$$

$$\underline{v}_R(t) = \underline{\underline{\frac{6}{\sqrt{2}} \cos(10t - 135^\circ)}}$$

11.1

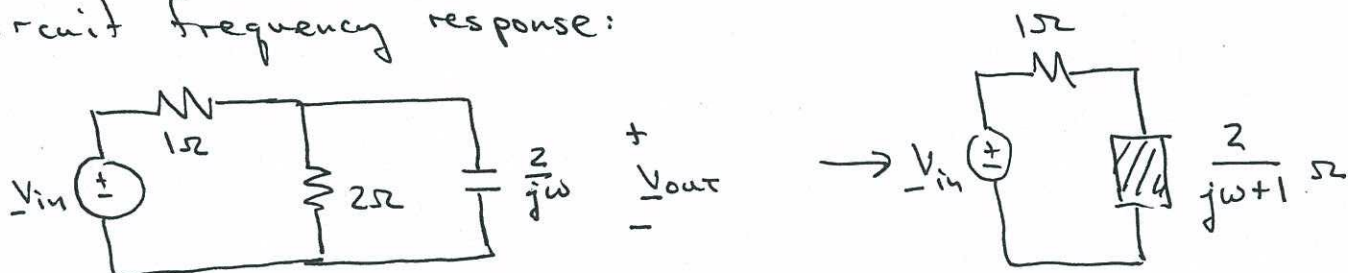
Input voltage: $V_{in}(t) = 3 \cos(2t - 20^\circ) - 2 \cos(3t + 30^\circ) + \cos(4t)$

$$\omega = 2 \Rightarrow \underline{V}_{in} = 3 \angle -20^\circ$$

$$\omega = 3 \Rightarrow \underline{V}_{in} = 2 \angle 30^\circ$$

$$\omega = 4 \Rightarrow \underline{V}_{in} = 1 \angle 0^\circ$$

Circuit frequency response:



$$\underline{V}_{out} = \underline{V}_{in} \left[\frac{\frac{2}{j\omega+1}}{1 + \frac{2}{j\omega+1}} \right] = \underline{V}_{in} \left[\frac{2}{j\omega+3} \right]$$

$$\omega = 2 \Rightarrow \frac{2}{j\omega+3} = 0.555 \angle -33.7^\circ$$

$$\omega = 3 \Rightarrow \frac{2}{j\omega+3} = 0.471 \angle -45^\circ$$

$$\omega = 4 \Rightarrow \frac{2}{j\omega+3} = 0.40 \angle -53.1^\circ$$

$$\text{Output: } \underline{V}_{out} = \underline{V}_{in} \left[\frac{2}{j\omega+3} \right]$$

$$\omega = 2 \Rightarrow \underline{V}_{out} = (3 \angle -20^\circ)(0.555 \angle -33.7^\circ) = 1.66 \angle -53.7^\circ$$

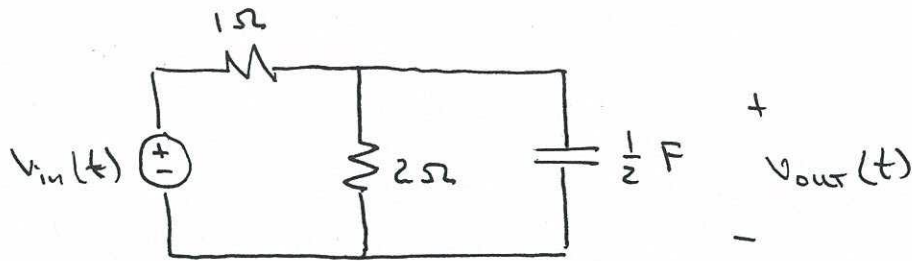
$$\omega = 3 \Rightarrow \underline{V}_{out} = (2 \angle 30^\circ)(0.471 \angle -45^\circ) = 0.94 \angle -15^\circ$$

$$\omega = 4 \Rightarrow \underline{V}_{out} = (1 \angle 0^\circ)(0.4 \angle -53.1^\circ) = 0.4 \angle -53.1^\circ$$

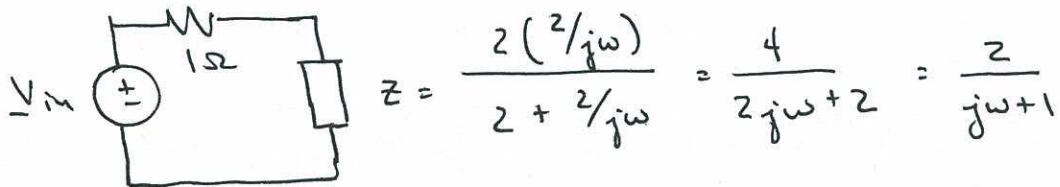
$$V_{out}(t) = 1.66 \cos(2t - 53.7^\circ) - 0.94 \cos(3t - 15^\circ) + 0.4 \cos(4t - 53.1^\circ)$$

$$i_{n(11)} - V_{out}(t) = 1.66 \cos(2t - 53.7^\circ) - 0.94 \cos(3t - 15^\circ) + 0.4 \cos(4t - 53.1^\circ)$$

11.2



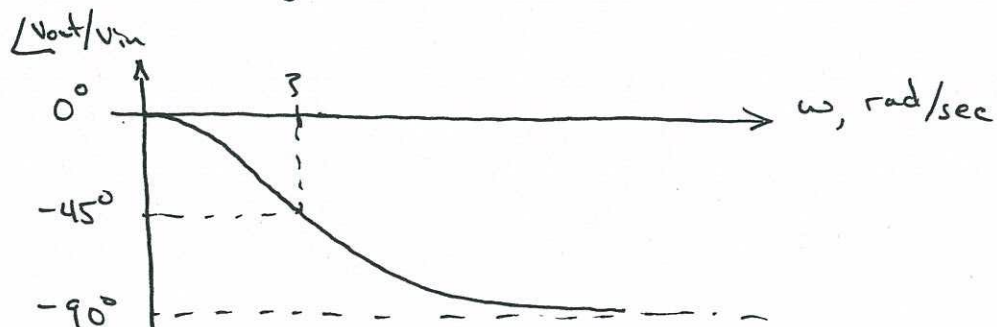
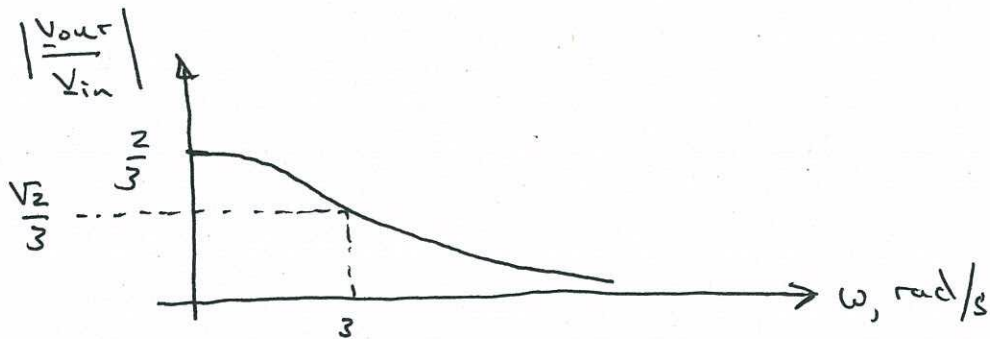
Frequency domain circuit:



$$V_{out} = V_{in} \left[\frac{\frac{2}{j\omega + 1}}{1 + \frac{2}{j\omega + 1}} \right] \Rightarrow \frac{V_{out}}{V_{in}} = \frac{2}{j\omega + 3}$$

$$\text{DC gain} = \frac{2}{j0 + 3} = \frac{2}{3}$$

$$\omega_c = 3 \text{ rad/sec}$$



11.2 (cont'd)

$$2 \frac{di}{dt} + 10i = 10v$$

Re-write in "standard" form:

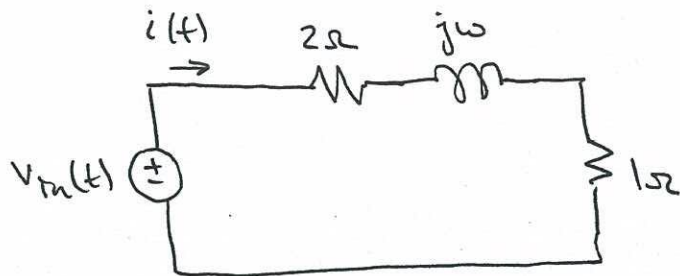
$$\frac{di}{dt} + 5i = 5v$$

↑

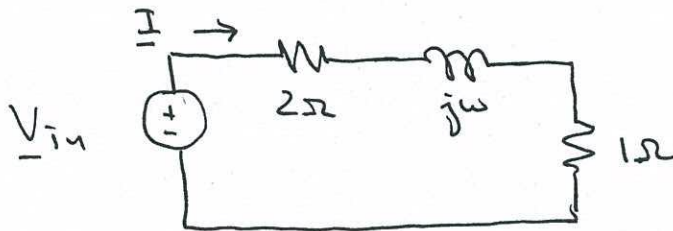
$$\frac{1}{\tau} = \omega_c$$

$$\underline{\underline{\omega_c = 5 \text{ rad/sec}}}$$

11.3



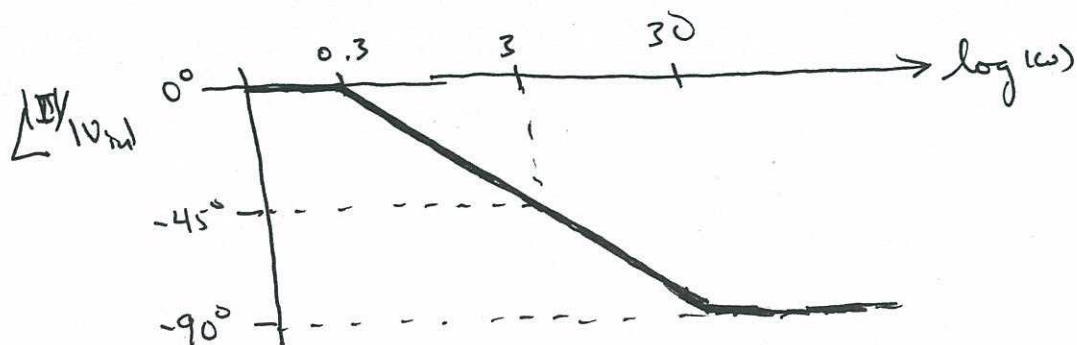
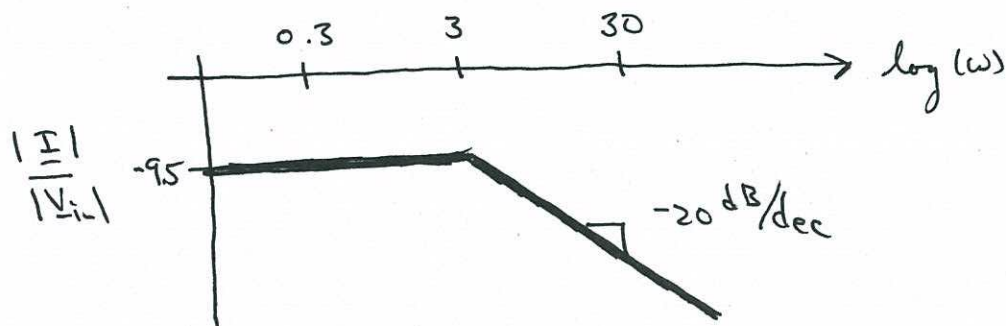
Frequency domain



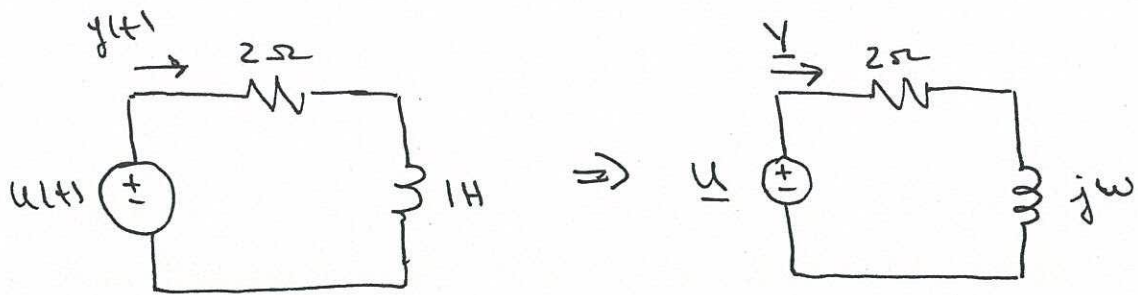
$$\underline{I} = \underline{V}_{in} \left[\frac{1}{j\omega + 3} \right] \Rightarrow \frac{\underline{I}}{\underline{V}_{in}} = \frac{1}{j\omega + 3}$$

$$\left| \frac{\underline{I}}{\underline{V}_{in}} \right| = \frac{1}{\sqrt{\omega^2 + 9}}, \quad \angle \frac{\underline{I}}{\underline{V}_{in}} = -\tan^{-1} \left(\frac{\omega}{3} \right)$$

$$\omega_c = 3 \text{ rad/sec}, \quad \text{DC gain} = \frac{1}{3} = -9.54 \text{ dB}$$



11.4

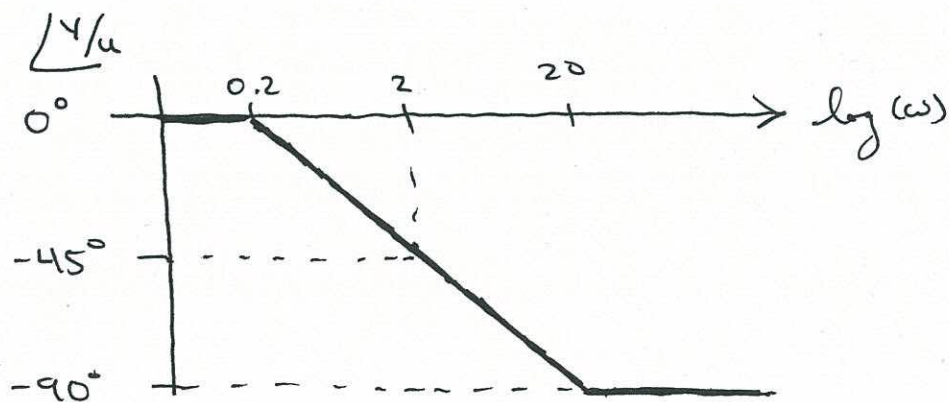
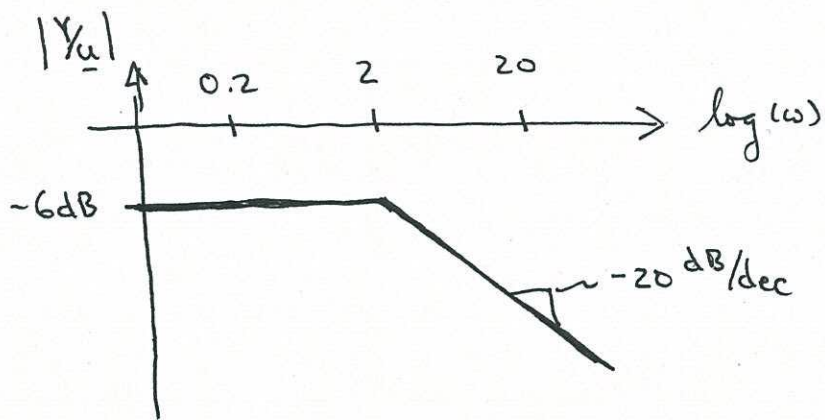


$$\underline{Y} = \frac{\underline{u}}{2 + j\omega} \Rightarrow \frac{\underline{Y}}{\underline{u}} = \frac{1}{2 + j\omega}$$

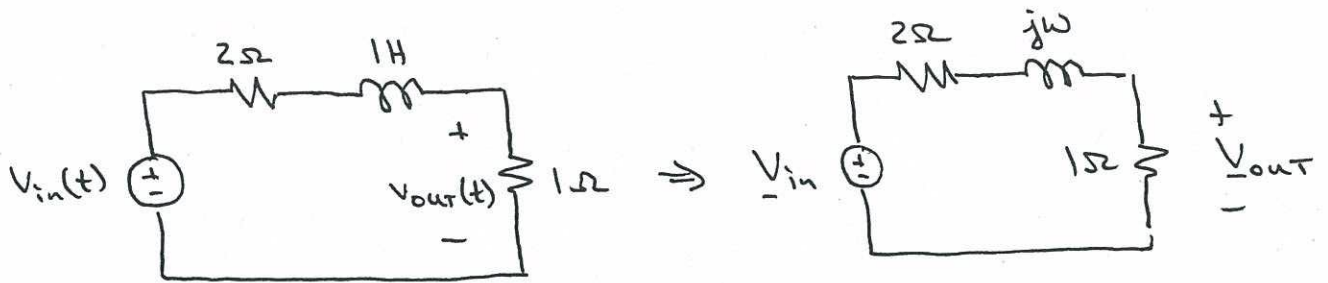
$$\left| \frac{\underline{Y}}{\underline{u}} \right| = \frac{1}{\sqrt{\omega^2 + 4}}, \quad \angle \frac{\underline{Y}}{\underline{u}} = -\tan^{-1}\left(\frac{\omega}{2}\right)$$

$$\text{DC gain} = \frac{1}{2} = -6 \text{ dB}$$

$$\omega_c = 2 \text{ rad/sec}$$



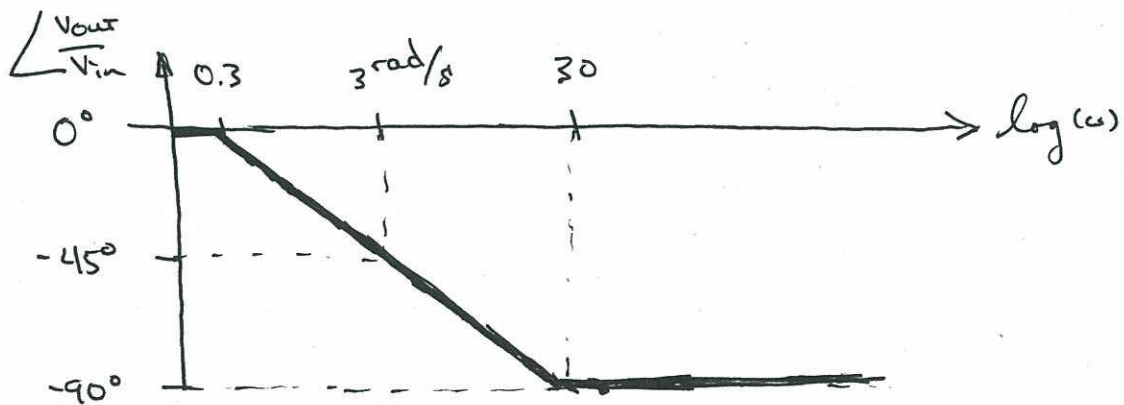
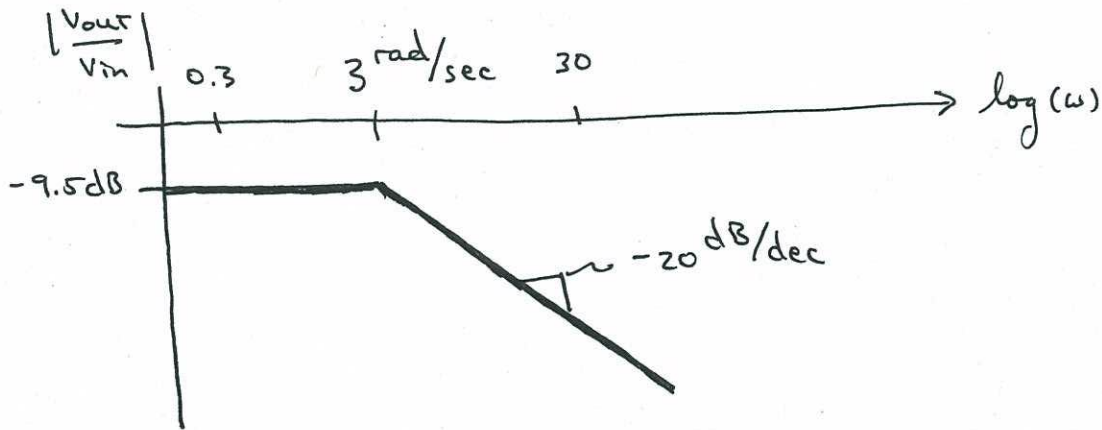
11.5



$$V_{out} = V_{in} \left[\frac{1\Omega}{2\Omega + j\omega + 1\Omega} \right] = \frac{1}{j\omega + 3} V_{in}$$

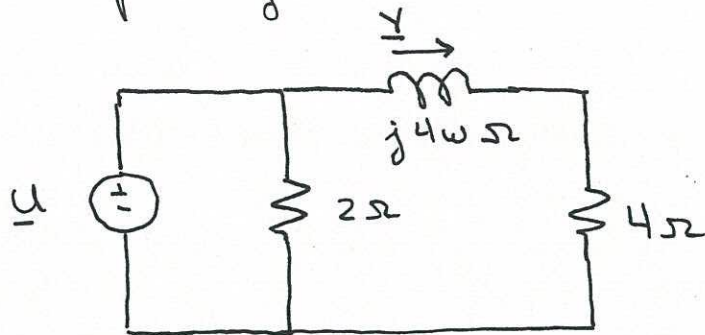
$$\text{DC gain} = \frac{1}{3} \Rightarrow \text{DC gain}_{dB} = 20 \log_{10}\left(\frac{1}{3}\right) = -9.54 \text{ dB}$$

$$\omega_c = 3 \text{ rad/sec}$$



11.6

Frequency domain circuit:

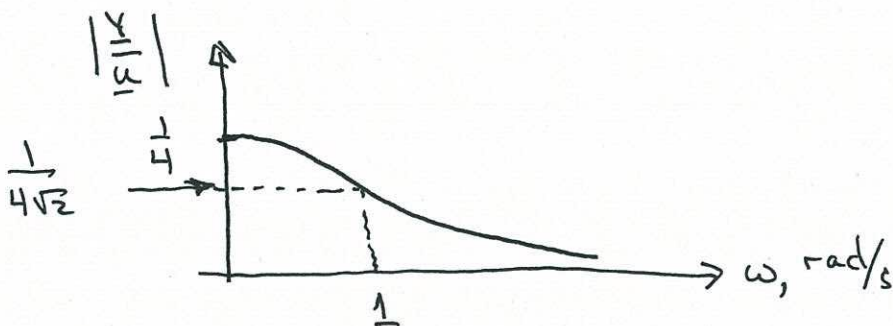


$$(a) \quad \underline{Y} = \underline{U} \left[\frac{1}{4 + j4\omega} \right] \Rightarrow \boxed{\frac{\underline{Y}}{\underline{U}} = \frac{1}{4 + j4\omega}}$$

$$(b) \quad \left| \frac{\underline{Y}}{\underline{U}} \right| = \left| \frac{1}{4 + j4\omega} \right| = \frac{1}{\sqrt{16 + 16\omega^2}} = \frac{1}{4\sqrt{1 + \omega^2}}$$

$$\text{DC gain} \Rightarrow \left| \frac{\underline{Y}}{\underline{U}} \right|_{\omega=0} = \frac{1}{4}$$

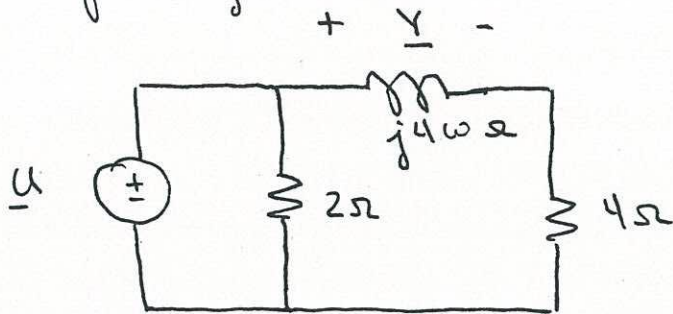
$$\omega_c = 1 \text{ rad/s}$$



(c) Low pass. Gain at low frequencies is non-zero. High frequencies are "stopped" by the circuit.

11.7

Frequency domain circuit:



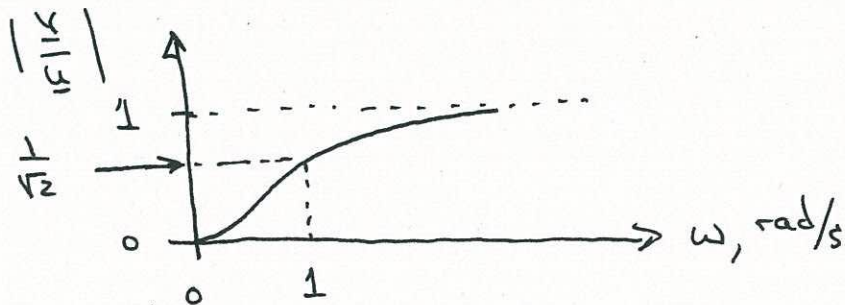
$$(a) \quad \underline{Y} = \underline{U} \left[\frac{j4\omega}{4 + j4\omega} \right] \Rightarrow \boxed{\frac{\underline{Y}}{\underline{U}} = \frac{j4\omega}{4 + j4\omega}}$$

$$(b) \quad \left| \frac{\underline{Y}}{\underline{U}} \right| = \left| \frac{j4\omega}{4 + j4\omega} \right| = \left| \frac{j\omega}{1 + j\omega} \right|$$

$$\left| \frac{\underline{Y}}{\underline{U}} \right| = \frac{\omega}{\sqrt{1 + \omega^2}}$$

$$\text{DC gain} = \left| \frac{\underline{Y}}{\underline{U}} \right|_{\omega=0} = 0$$

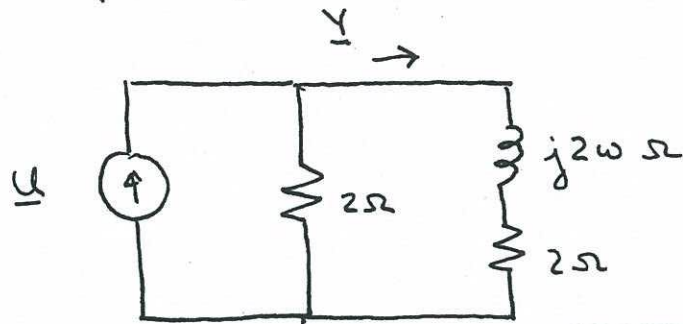
$$\omega_c = 1 \text{ rad/s}$$



(c) High-pass. Low frequencies are "stopped" by the filter.

11.8

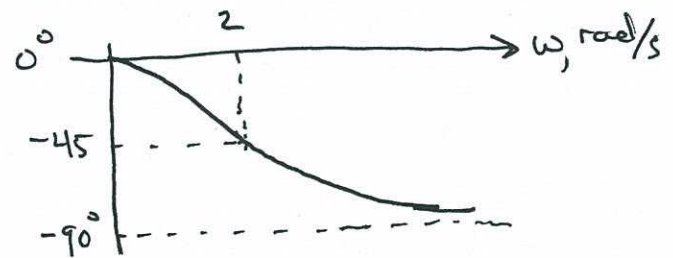
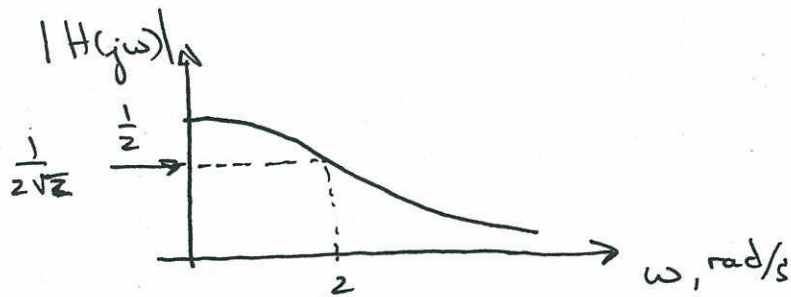
Frequency domain circuit:



$$(a) \quad \underline{Y} = \underline{U} \left[\frac{2}{4 + j2\omega} \right] \Rightarrow \boxed{H(j\omega) = \frac{\underline{Y}}{\underline{U}} = \frac{2}{4 + j2\omega}}$$

$$(b) \quad |H(j\omega)| = \frac{2}{\sqrt{16 + 4\omega^2}} = \frac{1}{\sqrt{4 + \omega^2}} \Rightarrow \begin{aligned} \text{DC gain} &= \frac{1}{2} \\ \omega_c &= 2 \text{ rad/sec} \end{aligned}$$

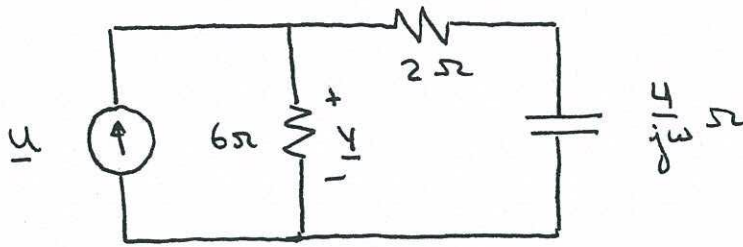
$$\angle H(j\omega) = 0^\circ - \tan^{-1}\left(\frac{\omega}{2}\right) \Rightarrow \begin{aligned} \angle H(j0) &= 0^\circ \\ \angle H(j\infty) &= -90^\circ \end{aligned}$$



(c) Low pass. High frequencies are "stopped" by the circuit.

11.9

Frequency domain circuit:

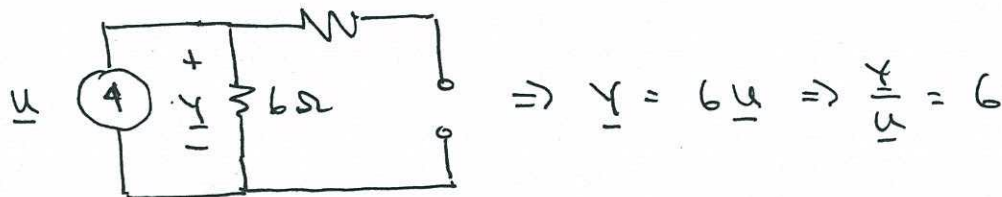


$$(a) \quad \underline{Y} = \underline{U} \underbrace{\left[\frac{2 + \frac{4}{j\omega}}{8 + \frac{4}{j\omega}} \right]}_{\substack{\text{current} \\ \text{divider}}} \times (6\Omega) \Rightarrow \boxed{\frac{\underline{Y}}{\underline{U}} = \frac{j12\omega + 24}{j8\omega + 4}}$$

current through
6Ω resistor.

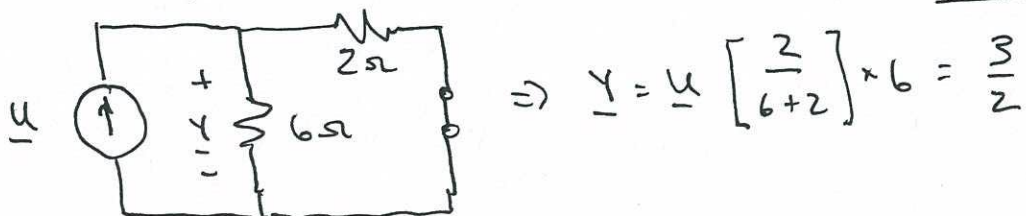
(b) Frequency response, $\omega \rightarrow 0$: $\frac{\underline{Y}}{\underline{U}} = \frac{24}{4} = 6$

circuit, $\omega \rightarrow 0$:



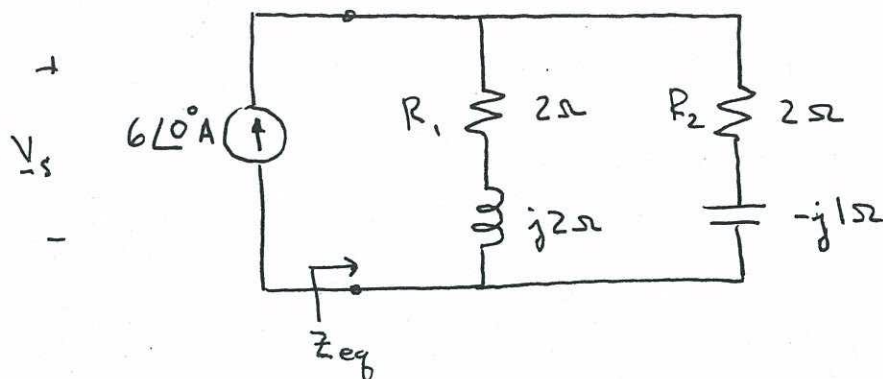
AGREE

Frequency response, $\omega \rightarrow \infty \Rightarrow \frac{\underline{Y}}{\underline{U}} \approx \frac{j12\omega}{j8\omega} = \frac{3}{2}$

circuit, $\omega \rightarrow \infty$ 

AGREE

12.1



$$Z_{eq} = \frac{(2+j2)(2-j)}{2+j2+2-j} = \frac{6+j2}{4+j} = \frac{6.32 \angle 18.4^\circ}{4.12 \angle 14^\circ}$$

$$V_s = I_s \cdot Z_{eq} = 6 \angle 0^\circ \left[\frac{6.32 \angle 18.4^\circ}{4.12 \angle 14^\circ} \right] = 9.2 \angle 4.39^\circ \text{ V}$$

- Power supplied by source:

$$P_s = \frac{1}{2} (9.2)(6) \cos(4.39^\circ - 0^\circ) = \underline{\underline{27.52 \text{ W}}}$$

- Power absorbed by R_1 :

$$I_1 = \frac{6 \angle 0^\circ (2-j)}{4+j} = 3.26 \angle -40.6^\circ \text{ A} \Rightarrow P_{R1} = \frac{1}{2} (3.26^2) (2\Omega)$$

$$\underline{\underline{P_{R1} = 10.63 \text{ W}}}$$

- Power absorbed by R_2 :

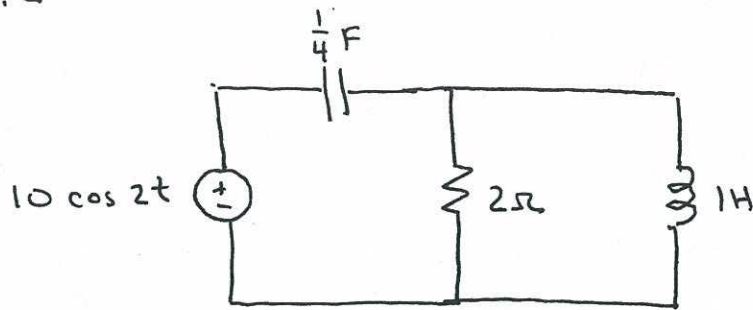
$$I_2 = \frac{6 \angle 0^\circ (2+j2)}{4+j1} = 4.12 \angle 30.96^\circ \text{ A} \Rightarrow P_{R2} = \frac{1}{2} (4.12)^2 (2\Omega)$$

$$\underline{\underline{P_{R2} = 16.9 \text{ W}}}$$

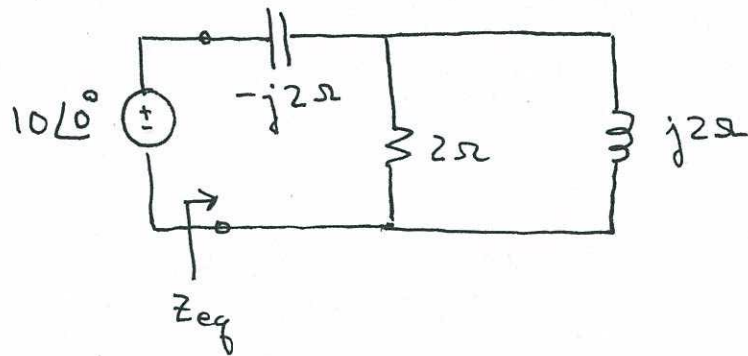
- No average power is absorbed or generated by the inductor or capacitor.

- Note: $P_{R1} + P_{R2} = P_s \quad \checkmark$

12.2



Frequency domain circuit:

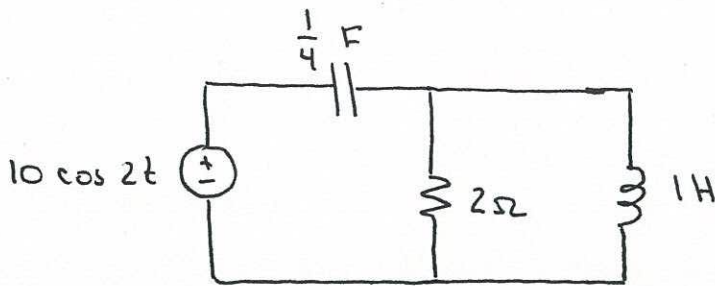


$$Z_{eq} = -j2\Omega + \frac{(2\Omega)(j2\Omega)}{(2+j2)\Omega} = (1-j)\Omega$$

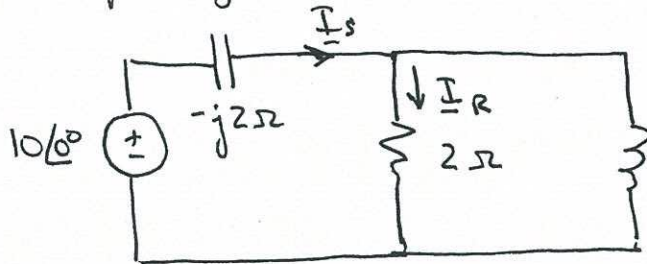
$$\underline{I}_s = \frac{10\angle 0^\circ}{1-j} = \frac{10}{\sqrt{2}} \angle 45^\circ$$

$$P_s = \frac{10 \left(\frac{10}{\sqrt{2}} \right)}{2} \cos(0^\circ - 45^\circ) \Rightarrow \underline{\underline{P_s = 25 \text{ W}}}$$

12.3



Frequency domain circuit:



From previous problem, $\underline{I}_s = \frac{10\angle 0^\circ}{1-j}$

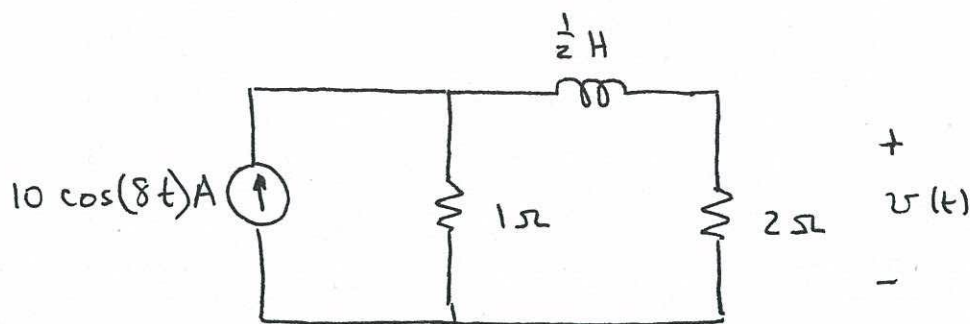
$$\underline{I}_R = \underbrace{\underline{I}_s \left(\frac{j2\Omega}{(2+j2)\Omega} \right)}_{\text{current divider}} = \underbrace{\frac{10\angle 0^\circ}{1-j}}_{\underline{I}_s} \left(\frac{j}{1+j} \right) = \underline{j\frac{10}{2}}$$

$$\underline{I}_R = j5 \text{ A} \Rightarrow \underline{I}_R = 5\angle 90^\circ$$

$$P_R = \frac{\underline{I}_R^2}{2} \cdot R = \frac{5^2}{2} (2\Omega) \Rightarrow \underline{\underline{P_R = 25 \text{ W}}}$$

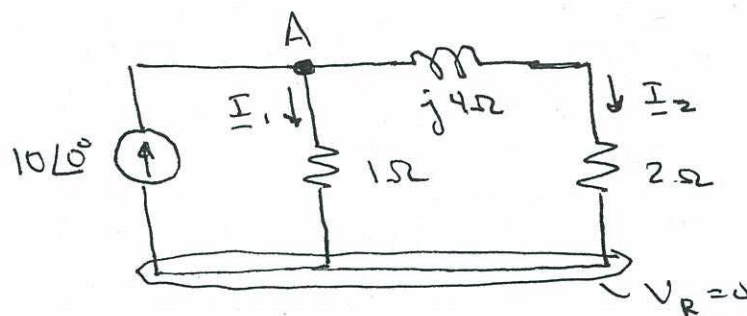
Note: We calculated the average power delivered by the source in the last problem. The power delivered by the source is equal to the power absorbed by the resistor.

12.4



(a) I'll determine $v(t)$ using several different approaches, to illustrate the flexibility of impedance analysis techniques.

Approach 1: Nodal Analysis



KCL at A:

$$10 \angle 0^\circ = \underline{I}_1 + \underline{I}_2 = \frac{V_A}{1 \Omega} + \frac{V_A}{(2 + j4) \Omega}$$

$$10 \angle 0^\circ = \frac{(2 + j4) \underline{V}_A + \underline{V}_A}{2 + j4} = \frac{3 + j4}{2 + j4} \underline{V}_A$$

$$\underline{V}_A = \left[\frac{2 + j4}{3 + j4} \right] \cdot 10 \angle 0^\circ$$

voltage divider to get $\underline{V}$:

$$\underline{V} = \underline{V}_A \left(\frac{2}{2 + j4} \right) = 10 \angle 0^\circ \left[\frac{2 + j4}{3 + j4} \right] \left[\frac{2}{2 + j4} \right]$$

↓ (continued)

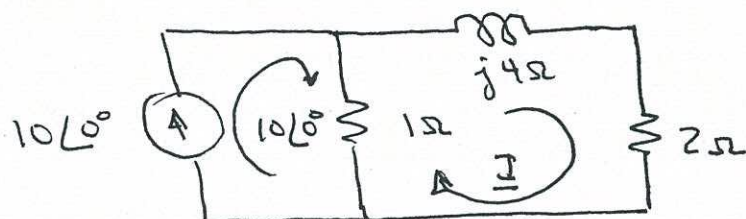
12.4 (cont'd)

Approach 1 (cont'd)

$$\underline{V} = 10 \angle 0^\circ \left[\frac{2}{3+j4} \right] = \frac{20 \angle 0^\circ}{\sqrt{3^2+4^2} \angle \tan^{-1}(\frac{4}{3})}$$

$$\underline{V} = 4 \angle 53.3^\circ \Rightarrow \underline{v(t)} = \underline{4 \cos(8t - 53.3^\circ) \text{ V}}$$

Approach 2: Mesh analysis



$$1 \Omega [\underline{I} - 10 \angle 0^\circ] + j4 \Omega (\underline{I}) + 2 \Omega \underline{I} = 0$$

$$\underline{I} (j4 + 3) = 10 \angle 0^\circ$$

$$\underline{I} = \frac{10 \angle 0^\circ}{3+j4} = \frac{10 \angle 0^\circ}{\sqrt{3^2+4^2} \angle \tan^{-1}(\frac{4}{3})} = \frac{10 \angle 0^\circ}{5 \angle 53.13^\circ}$$

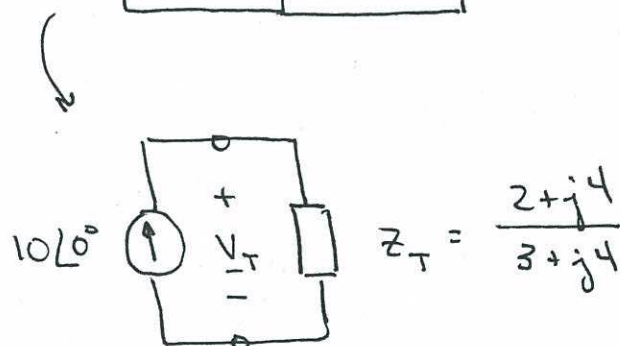
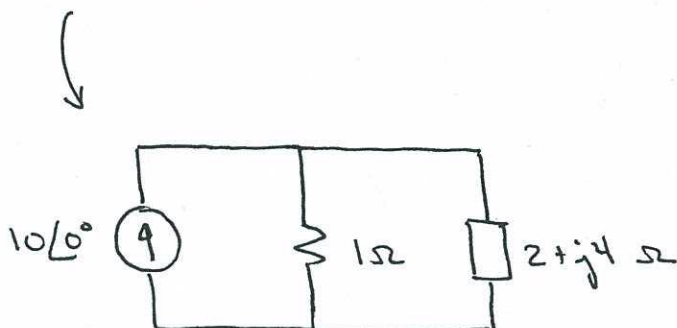
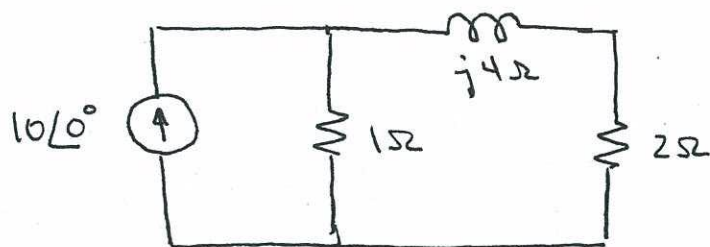
$$\underline{I} = 2 \angle -53.13^\circ \text{ A}$$

$$\underline{V} = (2 \Omega) \underline{I} = 4 \angle -53.13^\circ$$

$$\underline{v(t)} = \underline{4 \cos(8t - 53.13^\circ)}$$

12.4 (cont'd again...)

Approach 3: circuit reduction



$$\underline{V}_T = \underline{I}_s Z_T = 10\angle 0^\circ \left[\frac{2+j4}{3+j4} \right]$$

Voltage divider to get $\underline{V}$:

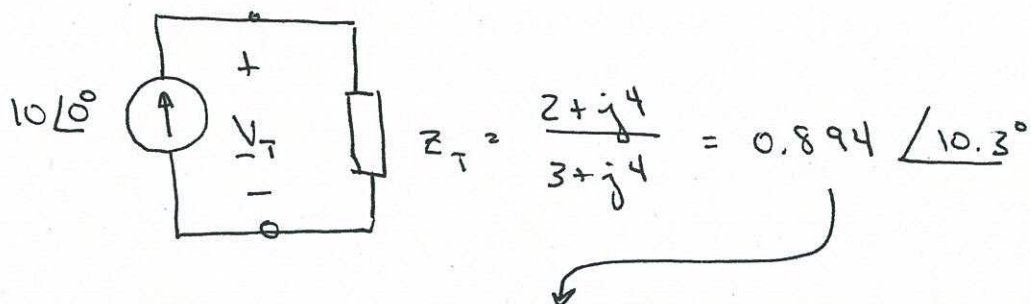
$$\underline{V} = \underline{V}_T \left(\frac{2}{2+j4} \right) = 10\angle 0^\circ \left[\frac{2+j4}{3+j4} \right] \left[\frac{2}{2+j4} \right]$$

$$\underline{V} = 4\angle -53.13^\circ$$

$$\underline{v(t)} = \underline{4 \cos(8t - 53.13^\circ)}$$

12.4 (still cont'd...)

(b) Use the equivalent circuit determined in Approach 3 above:



$$V_T = (10\angle 0^\circ)(0.894 \angle 10.3^\circ) = 8.944 \angle 10.29^\circ$$

$$P = \frac{V_m I_m}{2} \cos(\theta_r - \theta_i) = \frac{(10)(8.94)}{2} \cos(10.29^\circ - 0^\circ)$$

$$\underline{\underline{P = 44 \text{ W}}}$$

(c) Complex power delivered by source:

$$S = \frac{\underline{V}^* \underline{I}^*}{2} = \frac{(10\angle 0^\circ)(8.94 \angle 10.29^\circ)}{2} = 44.7 \angle 10.29^\circ$$

$$\text{Rectangular coordinates: } \underline{\underline{S = 44 + j8 \text{ VA}}}$$

(d) Inductor power:

$$\text{From approach 2 above, } \underline{I}_1 = 2 \angle -53.13^\circ \text{ A}$$

$$Q = \frac{I^2}{2} [I_m(Z)] = \frac{2^2}{2} (4) = 8 \text{ VAR}$$

12.5

$$V_{RMS} = \frac{150,000 \text{ W}}{(0.7)(240 \text{ A})}$$

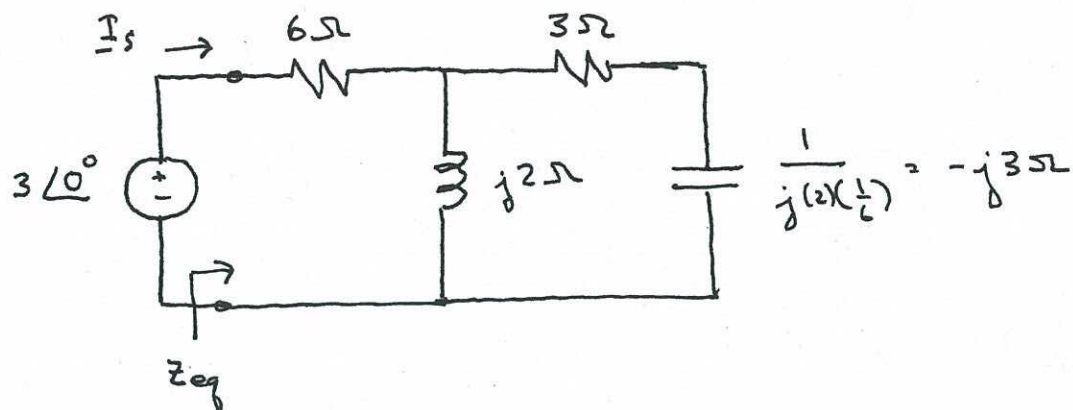
$$\underline{\underline{V_{RMS} = 892.9 \text{ V}}}$$

12.6

$$\theta = \cos^{-1}(0.8) = \underline{\underline{36.87^\circ}} \quad (\text{current } \underline{\text{lags}} \text{ voltage for inductive load})$$

12.7

Frequency domain circuit



(a) $S = \frac{V \cdot I_s^*}{2} \Rightarrow$ need to determine current out of source:

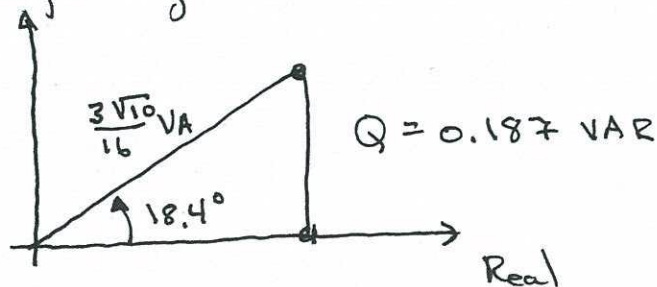
$$Z_{eq} = 6\Omega + \frac{(j2\Omega)(3-j3)\Omega}{j2\Omega + 3\Omega - j3\Omega} = 6\Omega + \frac{j6\Omega + 6\Omega}{(3-j)\Omega}$$

$$Z_{eq} = \frac{24}{3-j}$$

$$\underline{I}_s = \frac{3\angle 0^\circ}{Z_{eq}} = \frac{3\angle 0^\circ}{\frac{24}{3-j}} = \frac{3\sqrt{10}}{24} \angle -18.4^\circ$$

$$S = \frac{(3\angle 0^\circ) \left(\frac{3\sqrt{10}}{24} \angle +18.4^\circ \right)}{2} = \underline{\underline{\frac{3\sqrt{10}}{16} \angle 18.4^\circ}}$$

(b) Imaginary



$$P = 0.563 \text{ W}$$

12.7 continued

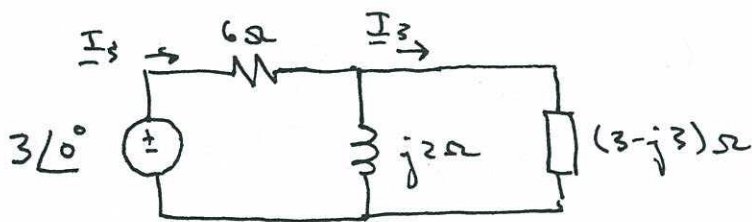
(c) Average power delivered by the source is the real part of the complex power (shown on part (b)).

$$P = \frac{3\sqrt{10}}{16} \cos(0 + 18.4^\circ) = \underline{\underline{0.563 \text{ W}}}$$

(d) The average power delivered by the source is dissipated in the resistors. Thus

$$P_{6\Omega} + P_{3\Omega} = P = \underline{\underline{0.563 \text{ W}}}$$

(e) Need to find current through the 3Ω resistor (or voltage difference across it) to get $P_{3\Omega}$. I'll use circuit reduction...

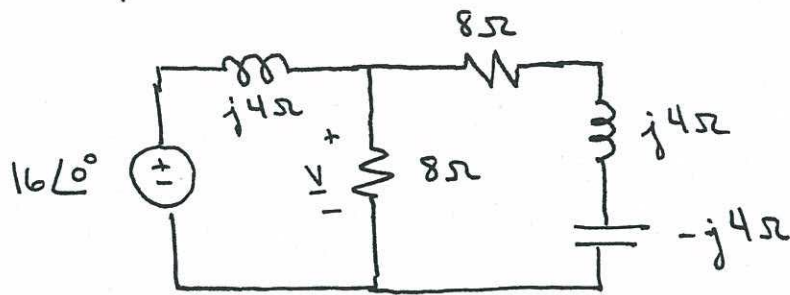


$$\begin{aligned} \underline{I}_3 &= \underline{I}_s \left[\frac{j2\Omega}{3-j3\Omega} \right] \\ &= \frac{3\angle 0^\circ (3-j)}{24} \left[\frac{j^2}{3-j} \right] \\ &= 0.25j = \frac{1}{4} \angle 90^\circ \end{aligned}$$

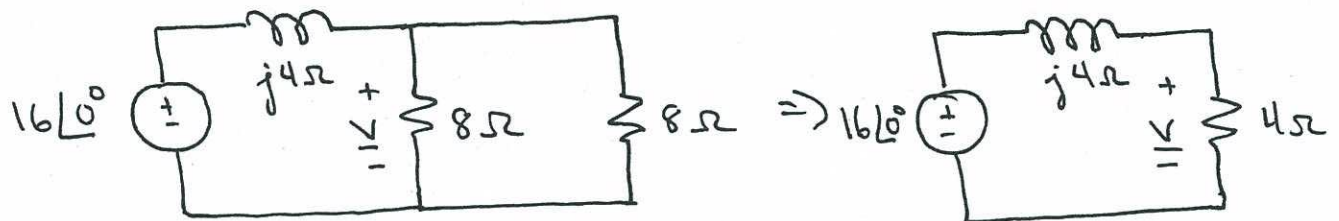
$$P_{3\Omega} = |\underline{I}_3|^2 \cdot (3\Omega) = \underline{\underline{\frac{3}{16} \text{ W}}}$$

12.8

Frequency domain circuit:

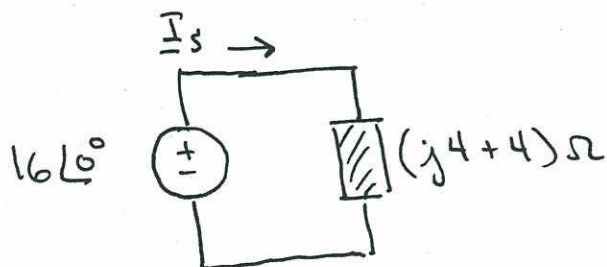


(a) Reduced circuit:



$$\underline{V} = 16\angle 0^\circ \left[\frac{4\Omega}{(4+j4)\Omega} \right] = \frac{16\angle 0^\circ}{\sqrt{2}\angle 45^\circ} = \frac{16}{\sqrt{2}} \angle -45^\circ$$

$$v(t) = \frac{16}{\sqrt{2}} \cos(2t - 45^\circ)$$

(b) Find source current, $\underline{I}_s$ 

$$\underline{I}_s = \frac{16\angle 0^\circ}{4+j4} = \frac{4}{\sqrt{2}} \angle -45^\circ$$

$$S' = \frac{\underline{V} \cdot \underline{I}^*}{2} = \frac{(16\angle 0^\circ) \left(\frac{4}{\sqrt{2}} \angle 45^\circ \right)}{2} = \frac{32}{\sqrt{2}} \angle 45^\circ$$

$$S = 16\sqrt{2} \angle 45^\circ$$

12.8 (cont'd)

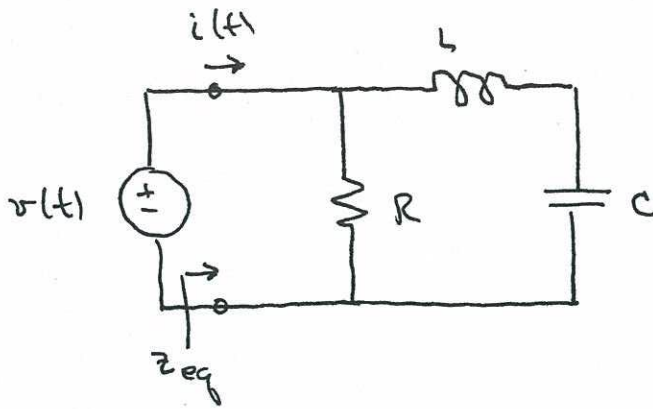
$$(c) \quad P = \operatorname{Re}\{s\} = |s| \cos(\theta_v - \theta_i)$$

$$\boxed{P = 16 \text{ W}}$$

$$(d) \quad Q = \operatorname{Im}\{s\} = |s| \sin(\theta_v - \theta_i)$$

$$\boxed{Q = 16 \text{ VAR}}$$

12.9



$$v(t) = 120 \cos(100t + 65^\circ) \Rightarrow \underline{V} = 120 \angle 65^\circ$$

$$i(t) = 2 \cos(100t + 30^\circ) \Rightarrow \underline{I} = 2 \angle 30^\circ$$

$$(a) \quad S' = \frac{\underline{V} \underline{I}^*}{2} = \frac{(120 \angle 65^\circ)(2 \angle -30^\circ)}{2} = \underline{\underline{120 \angle 35^\circ}}$$

(b) The power dissipated by the resistor is the same as the real power delivered by the source. Therefore,

$$P = |S| \cos(\theta_v - \theta_i) = 120 \cos(35^\circ)$$

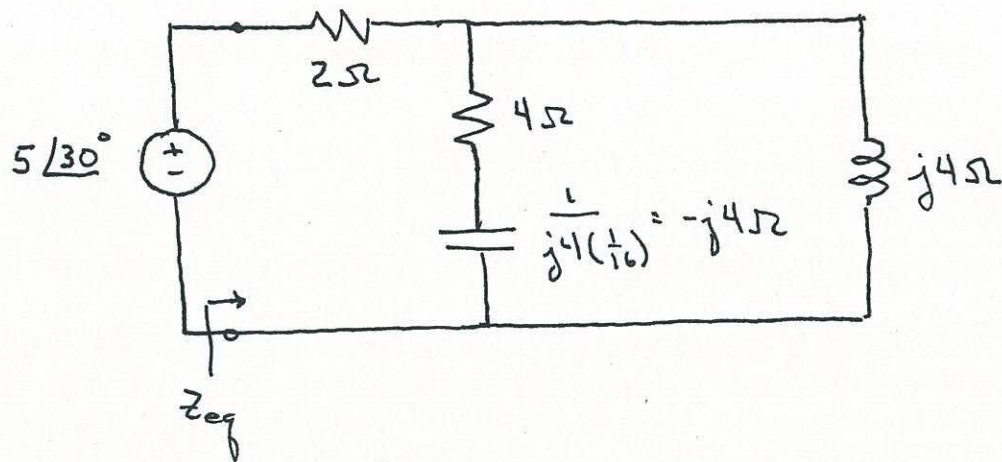
$$\boxed{P = 98.3 \text{ W}}$$

$$(c) \quad \underline{Z}_{eq} = \frac{\underline{V}}{\underline{I}} = \frac{120 \angle 65^\circ}{2 \angle 30^\circ}$$

$$\boxed{\underline{Z}_{eq} = 60 \angle 35^\circ}$$

12.10

Frequency domain circuit



(a) Circuit reduction:



$$z_{eq} = 2\Omega + \frac{j16 + 16}{4}\Omega$$

$$= (2 + j4 + 4)\Omega$$

$$z_{eq} = (6 + j4)\Omega$$

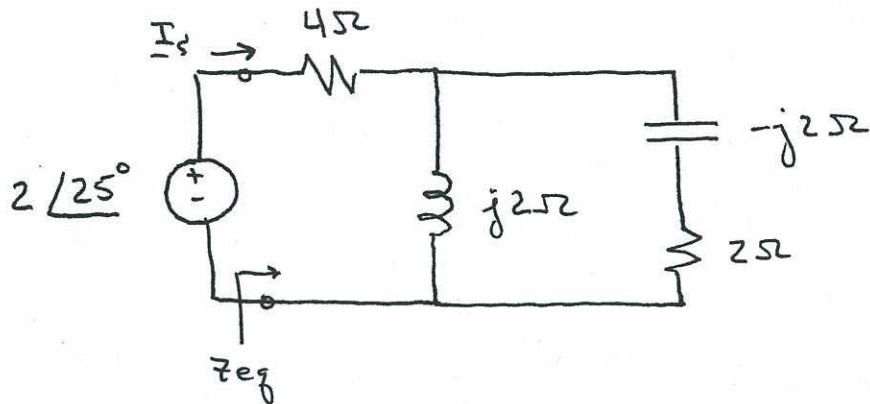
$$(b) \quad \underline{I}_s = \frac{5\angle 30^\circ}{6 + j4\Omega} = \frac{5}{2\sqrt{13}} \angle -3.7^\circ$$

$$S = \frac{\underline{V}_s \underline{I}_s^*}{2} = \frac{(5\angle 30^\circ) \left(\frac{5}{2\sqrt{13}} \angle +3.7^\circ \right)}{2} =$$

$$S = \frac{25}{2\sqrt{13}} \angle 33.7^\circ$$

12.11

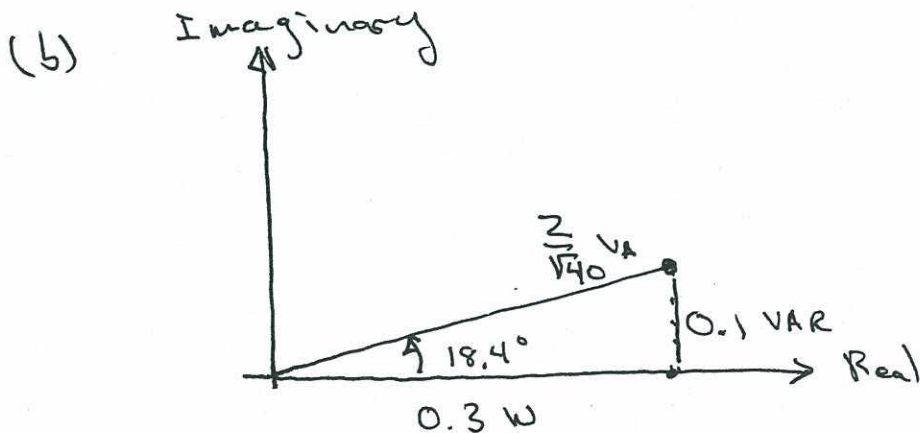
Frequency domain circuit



$$(a) \quad Z_{eq} = 4\Omega + \frac{(j2)(2-j2)}{j2+2-j2} = 4 + j2 + 2 = (6 + j2)\Omega$$

$$\underline{I}_s = \frac{2\angle 25^\circ}{6+j2} = \frac{2\angle 25^\circ}{\sqrt{40}\angle 18.4^\circ} = \frac{2}{\sqrt{40}}\angle 6.57^\circ$$

$$S = \frac{(2\angle 25^\circ)\left(\frac{2}{\sqrt{40}}\angle -6.57^\circ\right)}{2} \Rightarrow \boxed{\frac{2}{\sqrt{40}}\angle 18.43^\circ = S}$$



$$P = \frac{2}{\sqrt{40}} \cos(18.43^\circ) = 0.3 \text{ W}$$

$$Q = \frac{2}{\sqrt{40}} \sin(18.43^\circ) = 0.1$$